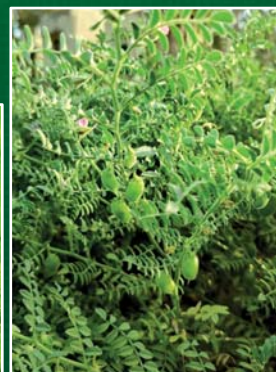
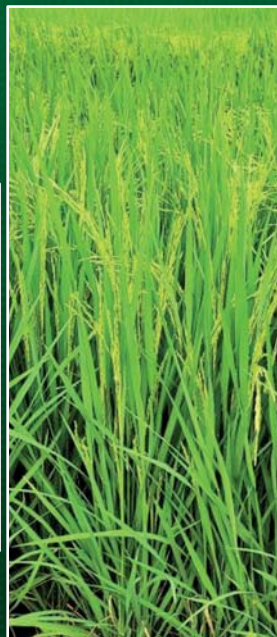


वार्षिक प्रतिवेदन ANNUAL REPORT 2021



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ICAR-NATIONAL INSTITUTE FOR PLANT BIOTECHNOLOGY

Lal Bahadur Shastri Centre, Pusa Campus, New Delhi-110012 (India)



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आमुख

भा. कृ.अ.प.—राष्ट्रीय पादप जैवप्रौद्योगिकी संस्थान (भा.कृ.अ.प.—एनआईपीबी) पादप आण्विक जीवविज्ञान एवं फसल जैवप्रौद्योगिकी क्षेत्र में अनुसंधान, शिक्षण एवं प्रशिक्षण में रत एक प्रमुख अनुसंधान संस्थान है। एनआईपीबी मुख्यतः पांच अधिदेशित फसलों नामतः चावल, गेहूं, चना, अरहर और सरसों से संबंधित मौलिक, कार्यनीतिपरक एवं व्यावहारिक अनुसंधान के लिए कार्य करता है। यह संस्थान राष्ट्रीय महत्व के अन्य कृषि एवं बागवानी फसलों के प्रक्षेत्र में सहयोगी अनुसंधान भी करता है। एनआईपीबी में किए जा रहे अनुसंधानों में फसलों में सस्यविज्ञानी गुणों, जैविक—अजैविक प्रतिबल सहिष्णुता में सुधार के लिए पराजीनी तथा जीनोम संपादन के विकास के साथ-साथ संरचनात्मक एवं कार्यात्मक जीनोमिक्स और आण्विक मानचित्रण पर भी अनुसंधान किया जाता है। रिपोर्टाधीन अवधि के दौरान संस्थान द्वारा प्राप्त की गई उपलब्धियां निम्नानुसार हैं:

वर्ष 2021 में भारतीय चाय (*कैमेलिया एसेमिका* किस्म मास्टर्स सीवी. टीवी-1) के गुणसूत्र स्तर की जीनोम असेम्बली (2.93Gb; 97.66% कवरेज) को सृजित कर 15 स्यूडोमोलेक्यूल्स पर एंकर किया गया। उड़द की एलबीजी 20 किस्म का जीनोम अनुक्रमित किया गया तथा 11 स्यूडोमोलेक्यूल्स में लगभग 524.23 Mb

जीनोम असेम्बली संयोजित की गई। चना सुधार की दिशा में निरंतर प्रयास करते हुए भा.कृ.अ.प.—भारतीय कृषि अनुसंधान संस्थान (भा.कृ.अ.प.—भा.कृ.अ.सं.) द्वारा चने की एक सूखा सहिष्णु किस्म BG4005 जारी की गई जिसमें भा.कृ.अ.प.—एनआईपीबी का मुख्य सहयोग था। BG4005 (Pusa Chickpea 4005), पूसा 362 की एक सूखा—सहिष्णु किस्म है। अरहर में *कैजानस प्लेटीकार्पस*, *सी. स्काराबीओडेस* और *सी. एक्यूटीफोलियस* जैसे वन्य संबंधियों का उपयोग फली भेदक प्रतिरोध तथा उच्च ताप के प्रति सहिष्णुता की यांत्रिकी को समझने के लिए किया जा रहा है। हमारे अध्ययन में, *सी. स्काराबीओडेस* की सर्वाधिक ताप सहिष्णु जाति के रूप में पहचान की गई थी। गेहूं में नाइट्रोजन उपयोग की दक्षता (एनयूई), सूखा और उच्च तापमान प्रतिबल सहिष्णुता पर किए गए अध्ययनों से संस्थान में विशेष रूप से नाइट्रोजन स्वांगीकरण, नाइट्रेट ट्रांसपोर्टर और इसके साथ ही उच्च तापमान प्रतिबल सहिष्णुता को रेखांकित करने वाली आण्विक यांत्रिकियों से संबंधित रुचिकर परिणाम प्राप्त हुए हैं।

संस्थान में ब्रेसिका में CRISPR/Cas मध्यित नॉकआउट प्रक्रिया को उपयुक्ततम बनाया गया है जिसका उपयोग ब्रेसिका में उपज को बढ़ाने, तेल की



गुणवत्ता संबंधी गुणों में सुधार करने तथा अगुणित इंड्यूसर वंशक्रम विकसित करने के लिए नए लक्ष्य जीनों के सत्यापन के लिए किया जाएगा। *ब्रेसिका* की वन्य प्रजातियां, कीट तथा रोगजनक प्रतिरोध के लिए जीनों का एक समृद्ध स्रोत हैं। विगत वर्ष में हमारे संस्थान ने ब्रेसिका जर्मप्लाज्म को स्क्रीन कर कई ऐसी प्रजातियों को चिन्हित किया गया जिनमें सफेद रतुआ, अल्टरनेरिया रोग, तना गलन और ओरोबैंके संक्रमण के विरुद्ध प्रतिरोध क्षमता पाई गई। *एस. स्केरोटियोरम* संक्रमण तथा ओरोबैंके प्रकोप की यांत्रिकीय अंतरदृष्टि प्राप्त करने के लिए सैक्रेटोम और ट्रांसक्रिप्टोमिक विश्लेषण किए गए हैं।

चावल में पूसा 1509 x NKSUR 70 से व्युत्पन्न समष्टि में अवायवीय अंकुरण (Anaerobic Germination) के लिए QTL मानचित्रित किया गया तथा गुणसूत्र 7 पर एक प्रमुख QTL की पहचान की गई। गहरे जल के अंतर्गत तने के लंबा होने में नॉनकोडिंग RNAs की भूमिका का अध्ययन किया गया। इसके अतिरिक्त अंतरगांठ के लंबा होने से संबंधित 18 eQTL के एक हैप्लोटाईप नेटवर्क तथा अवायवीय अंकुरण (Anaerobic Germination) सहिष्णुता से संबंधित जीनों का असम की चावल की 94 गहरी जल की भूप्रजातियों में अन्वेषण किया गया। चावल में एक नए अजैविक प्रतिबल प्रदान करने वाले जीन 'OsCHI2' की पहचान की गई। *AtRd29A:OsCHI2* की अतिअभिव्यक्ति करने वाले चावल के पराजीनी वंशक्रमों में सूखा, लवण तथा शीत संबंधित प्रतिबलों के प्रति बढ़ी हुई सहिष्णुता प्रदर्शित हुई। पुष्पगुच्छ प्रध्वंस के दौरान चावल-मैग्नापोर्थ ओराइजी के तुलनात्मक ट्रांसक्रिप्टोम विश्लेषण का टेपे (प्रतिरोधी) तथा एचपी 2216 (संवेदी) जीनप्ररूपों में विश्लेषण किया गया। संस्थान में कार्यात्मक जीनोमिक्स विश्लेषण पर फसल सुधार के लिए वन्य चावल तथा ईएमएस प्रेरित उत्परिवर्तकों का एक विशाल संकलन अनुरक्षित किया गया है। इस वर्ष अनुरक्षण तथा रखरखाव के लिए वन्य चावल की

255 प्रविष्टियां एनबीपीजीआर को प्रस्तुत की गई। अधिदेशित फसलों के अतिरिक्त रागी जैसी फसलों में नए जीनों की पहचान के लिए इनका उपयोग किया जा रहा है। इस वर्ष आम की लोकप्रिय किस्मों में सुगंध और स्वाद के विश्लेषण के लिए एक नये शोध की शुरुआत की गई है।

भा.कृ.अ.प.—एनआईपीबी, पादप आण्विक जीवविज्ञान एवं जैवप्रौद्योगिकी (एमबीबी) के क्षेत्र में मानव संसाधन विकास में अपना सक्रिय योगदान देता रहा है। इस वर्ष छह पीएच.डी. और 8 एम.एससी. छात्रों को क्रमशः डॉक्टरल और स्नातकोत्तर उपाधियां प्रदान की गईं। ये उपाधियां भा.कृ.अ.प.—भा.कृ.अ.सं. के 59वें दीक्षांत समारोह में प्रदान की गईं। इसके अतिरिक्त भा.कृ.अ.प.—एनआईपीबी अनुसूचित जाति उप परियोजना (एससीएसपी), पोषण वाटिका महाअभियान, वृक्षारोपण अभियान, हिंदी चेतना मास, सतर्कता जागरूकता सप्ताह तथा स्वच्छता अभियान जैसी विस्तार एवं संस्थान इतर गतिविधियों में सक्रिय रूप से भाग लेता आ रहा है। हमने भारत के छह राज्यों (उत्तर प्रदेश, उत्तराखण्ड, मध्य प्रदेश, राजस्थान, बिहार और ओडिशा) में 14 एससीएसपी कार्यक्रम आयोजित किए हैं जिनसे 2500 से अधिक किसान लाभान्वित हुए हैं।

मैं 'हिन्दी चेतना मास' तथा 'स्वच्छ भारत अभियान' सहित संस्थान की विभिन्न गतिविधियों में योगदान करने के लिए भा.कृ.अ.प.—एनआईपीबी के समस्त वैज्ञानिक तथा तकनीकी, प्रशासनिक स्टाफ, छात्रों, अनुसंधान अध्येताओं और संविदा पर सहायी स्टाफ को हृदय से धन्यवाद देता हूँ। मैं डॉ. मोनिका दलाल, डॉ. प्रशांत दाश, डॉ. रोहिणी श्रीवत्स, डॉ. अमिता मिश्रा सेवंथी, डॉ. अमोल कुमार सोलंके, डॉ. नवीन गुप्ता, डॉ. एम.एस. निम्मी, डॉ. दीपक बिष्ट और डॉ. आर.एस. निरंजन को इस रिपोर्ट के संकलन में उनकी सहायता के लिए विशेष रूप से धन्यवाद देता हूँ।

मैं डॉ. टी. महापात्र, सचिव, डेयर एवं महानिदेशक, भा.कृ.अ.प., डॉ. टी.आर. शर्मा, उप महानिदेशक (फसल विज्ञान) और डॉ. डी.के. यादव, सहायक महानिदेशक (बीज), भा.कृ.अ.प. का संस्थान की समस्त गतिविधियों में निरंतर सहायता व मार्गदर्शन करने के लिए हृदय से आभार व्यक्त करता हूँ।

अजित

दिनांक: 18.4.2022

(अजित कुमार शसानी)

PREFACE

The ICAR-National Institute for Plant Biotechnology (ICAR-NIPB) is a premier research institute engaged in research, teaching and training in the area of plant molecular biology and crop biotechnology. NIPB primarily works on five mandate crops viz. rice, wheat, chickpea, pigeonpea and mustard in the areas of basic, strategic and applied research. The institute also undertakes collaborative research in other agricultural and horticultural crops of national importance. Research at NIPB involves structural and functional genomics, molecular mapping, development of transgenics and genome editing for improving the agronomic traits, biotic -abiotic stress tolerance and nutritional quality in crops. The achievements of the institute during the reporting period are as follows

In the year 2021, a chromosome-level reference genome assembly (2.93 Gb; 97.66% Coverage) of Indian tea (*Camellia assamica* var. Masters cv. TV-1) was generated and 99.40% of super-scaffold level was anchored into 15 pseudomolecules. The genome of black gram cv. LBG 20 was sequenced and about 524.23

Mb of genome assembly was organized into 11 pseudo molecules. With continuing efforts towards chickpea improvement, a drought-tolerant variety, BG 4005, was released by ICAR-Indian Agricultural Research Institute (ICAR-IARI), in which ICAR-NIPB is a Collaborator. BG 4005 (Pusa Chickpea 4005) is a drought-tolerant introgression line of Pusa 362, developed through marker-assisted backcross breeding. In pigeonpea, wild relatives like *Cajanus platycarpus*, *C. scarabaeoides* and *C. acutifolius* are being used to understand the mechanism of pod borer resistance and heat stress tolerance. In our study, *C. scarabaeoides* was identified as the most thermotolerant species. In wheat, studies on nitrogen use efficiency (NUE), drought and high temperature stress tolerance have provided interesting leads at the institute especially on N assimilation, nitrate transporters as well as molecular mechanisms underlying high temperature stress tolerance.

The institute has optimized the process of CRISPR/Cas mediated knockouts in Brassica which will be further used for validating novel target genes for enhancing yield, improving oil quality traits and developing haploid



inducer lines in *Brassica*. The wild relatives of brassica are a rich reservoir of genes for insect and pathogen resistance. Brassica germplasm and/or introgression lines were screened for resistance to white rust, *Alternaria* leaf spot disease and *Orobanche* infestation. To get the mechanistic insight in to the *S. sclerotiorum* infection and *Orobanche* infestation secretome and transcriptomic analyses have been carried out.

In rice, QTLs for anaerobic germination were mapped in a population derived from Pusa1509

x NKSUR70 and a major QTL on chromosome 7 which explained about 27% of the phenotypic variation was identified. To study the role of non coding RNAs in stem elongation under deep water, miRNAs and lncRNAs were sequenced from internode and node tissues and validated. Further, a haplotype network of 18 expression quantitative trait loci (eQTL) related to internode elongation and an anaerobic germination tolerance related genes was investigated among 94 deepwater rice land races of Assam. A novel abiotic stress responsive gene *OsCHI2* was identified in rice. The rice transgenic lines over expressing *AtRd29A:OsCHI2* showed enhanced tolerance to drought, salt and cold stresses. A comparative transcriptome analysis of Rice-Magnaporthe *Oryzae* interaction during Panicle Blast was analysed in Tetep (resistant) and HP2216 (susceptible) genotypes. The institute also maintains a huge collection of wild rice and EMS induced mutants for functional genomic analysis and crop improvement. This year, 255 wild rice accessions were submitted to NBPGR

for maintenance and upkeep. In addition to mandate crops, crops such as finger millet are also being exploited for identification of novel genes. This year a new program on analysis of aroma and flavour in popular mango varieties has been initiated.

ICAR-NIPB is involved in Human Resource Development in the area of Plant Molecular Biology and Biotechnology (MBB). 6 Ph.D. and 8 M.Sc. students were awarded doctoral and master's degrees, respectively, in the 59th Convocation of ICAR-IARI. Additionally, ICAR-NIPB has been actively participating in extension and outreach activities like Scheduled Castes Sub Plan (SCSP), Poshan Vatika Mahaabhiyan, Tree Plantation Abhiyan, Hindi Chetna Maas, Vigilance Awareness week and Swachhata abhiyan. We have organised 14 SCSP programmes across 6 states of India (Uttar Pradesh, Uttarakhand, Madhya Pradesh, Rajasthan, Bihar and Odisha) in which more than 2500 farmers benefited.

I would like to extend my sincere thanks to all scientific, technical, administrative staff, students, research fellows and contractual supporting staff of ICAR-NIPB for their contributions in various institute activities including "Hindi Chetna Maas" and "Swachh Bharat Abhiyan". Special thanks are due to Dr. Monika Dalal, Dr. Prasanta Dash, Dr. Rohini Sreevathsa, Dr. Amitha Mithra Sevanthi, Dr. Amolkumar Solanke, Dr. Naveen Gupta, Dr. M. S. Nimmy, Dr. Deepak Bisht and Dr. R. S. Niranjana for their help in compilation of this report.

I am grateful to Dr. T. Mohapatra, Secretary, DARE and Director General, ICAR, Dr. T. R. Sharma, Deputy Director General (Crop Sciences) and Dr. D.K. Yadava, Assistant Director General (Seeds), ICAR, for their constant support and help in overall functioning of the institute.

Date: 18.04.2022



(Ajit K Shasany)

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Executive Summary

विशिष्ट सारांश

राष्ट्रीय पादप जैवप्रौद्योगिकी संस्थान (एनआईपीबी) भा.कृ.अ.प. की छत्रछाया में कार्यरत अनुसंधान संस्थान है जिसका अधिदेश पादप जैवप्रौद्योगिकी के क्षेत्र में आधुनिक जैवप्रौद्योगिकी से संबंधित युक्तियों और तकनीकों का विकास एवं उपयोग करना है। भा.कृ.अ.प.—एनआईपीबी को भारतीय कृषि अनुसंधान संस्थान के स्नातकोत्तर छात्रों को शिक्षा प्रदान करने के लिए आण्विक जीवविज्ञान एवं जैवप्रौद्योगिकी प्रभाग के रूप में सेवा करने का कार्य भी सौंपा गया है। साथ ही, यहां एनएआरईएस स्टाफ और छात्रों को संस्थान स्तर पर उपरोक्त विषय में प्रशिक्षण भी प्रदान किया जाता है। भारत सरकार के अनुसूचित जाति उप योजना (एससीएसपी) और 'मेरा गांव मेरा गौरव' (एमजीएमजी) कार्यक्रमों के माध्यम से यह संस्थान किसानों की भी सेवा कर रहा है। संस्थान मुख्यतः पांच प्रमुख फसलों नामतः चावल, गेहूँ, अरहर, चना और सरसों पर कार्य करता है तथा भा.कृ.अ.प. के अंतर्गत पादप जैवप्रौद्योगिकी का प्रमुख संस्थान होने के कारण अन्य राष्ट्रीय व अंतरराष्ट्रीय संस्थानों के सहयोग के माध्यम से अन्य कृषि एवं बागवानी फसलों पर भी कार्य करता है, जिसके अंतर्गत उन फसलों की प्रमुख समस्याओं के जैवप्रौद्योगिकीय हल तलाश करने तथा जीनोमिक संसाधन विकसित करने का कार्य किया जाता है। वर्ष 2021 के दौरान संस्थान द्वारा प्राप्त की गई प्रमुख उपलब्धियों का विवरण संक्षेप में निम्नानुसार है:

जीनोमिक्स, भा.कृ.अ.प.—एनआईपीबी के प्रमुख स्तंभों में से एक है। रिपोर्टाधीन वर्ष के दौरान उड़द की LBG20 किस्म का जीनोम असेम्बल किया गया तथा उड़द के लगभग 558.22 Mb में से 524.23 Mb जीनोम असेम्बली को 11

गुणसूत्रों में संयोजित किए गए। भारतीय चाय (*कैमेलिया एसेमिका* किस्म मास्टर्स सीवी. टीवी-1) पर अपने कार्य का विस्तार करते हुए हमने पहले उपांग जीनोम अनुक्रमित किया था। तत्पश्चात पिछले वर्ष विकासात्मक अध्ययन के आधार पर हमने भारतीय चाय के सम्पूर्ण जीनोम को 2.93GB में असेम्बल कर 15 स्यूडोमोलेक्यूलस एंकर किया। एकल प्रति जीन आधारित विकासात्मक अध्ययन तथा सम्पूर्ण जीनोम डुप्लीकेशन (डब्ल्यूजीडी) विश्लेषण से लगभग दो और 5.5 एमवाईए पर क्रमशः सीएसए और सीएसएस से ट्यूवी-1 के विपथित होने की पुष्टि हुई है। चाय प्रजनन कार्यक्रमों में तेजी लाने के लिए चाय के 369 जीनप्ररूपों के सार्वजनिक क्षेत्र में उपलब्ध सम्पूर्ण—जीनोम पुनः—अनुक्रमण आंकड़ों का उपयोग करते हुए चाय का प्रथम पीढ़ी के हैप्लोटाइप मानचित्र (Tea HapMap 1) सृजित किया गया। पिछले वर्ष हमने ग्वार की गुणसूत्र स्तर की असेम्बली के बारे में रिपोर्ट किया था। इस वर्ष हमने फ्लेवोनॉइड और फिनाइल प्रोपेनॉइड उपापचयी मार्ग से संबंधित कुल 29 और 27 जीनों की पहचान की है। यद्यपि ये जीन ग्वार के सभी 7 गुणसूत्रों में वितरित थे तथापि, गुणसूत्र 1 में फिनाइल प्रोपेनाइड जीनों की तथा गुणसूत्र 1 और 5 में फ्लेवोनॉइड जैव संश्लेषण जीनों की सर्वाधिक संख्या थी। अरहर जो हमारे देश की एक प्रमुख दलहनी फसल है, उसकी खेती को तीन प्रमुख पर्यावरणीय कारकों, *हेलिकोवर्पा आर्मीजेरा* (फली बेधक), माहुओं का संक्रमण तथा ताप प्रतिबल प्रभावित करती है। इन मुद्दों से निपटने के लिए संस्थान में अनेक युक्तियां अपनाई गई हैं। रिपोर्टाधीन वर्ष में अरहर के वन्य संबंधी, *कैजानस प्लेटीकार्पस* में *एच. आर्मीजेरा* द्वारा निरंतर शाकभोजिता के अंतर्गत गतिज ट्रांसक्रिप्टोम के आधार पर CHI4 जीन की पहचान की गई थी।

अंततः इस जीन को तम्बाकू के पौधे में स्थानांतरण कर उसका लक्षण—वर्णन और सत्यापन किया गया। यह जीन स्वपात्रे पत्ती जैवमूल्यांकन के आधार पर *एच. आर्मीजेरा* के विरुद्ध प्रभावी पाया गया था जिससे फलीबेधक के नियंत्रण में इसकी सक्षम उपयोगिता का सुझाव मिला। फसल सुरक्षा के लिए पादप लेक्टिन, प्रोटीएज निरोधकों, एल्फा—एमाइलेज निरोधकों तथा काइटिनेज के कीटनाशी गुणों का उपयोग किया जा सकता है। ब्रैसिका में प्रतिरोध के परिदृश्य को व्यापक बनाने के लिए चने के लैक्टिन तथा उड़द के प्रोटीएज निरोधक जीन से युक्त दो पराजीनी ब्रैसिका वंशक्रमों का संकरीकरण कराया गया तथा संततियों का स्वनिषेचन कराया गया और पराजीनियों के लिए छंटाई की गई। सक्षम पीड़क नियंत्रण लक्ष्यों जैसे *हेलिकोवर्पा आर्मीजेरा* से प्राप्त काइटिन डीएसिटाइलेज तथा *चिलो सुप्रेसेलिस* से प्राप्त आर्जीनीन काइनेज का होमोलॉजी मॉडलिंग सम्पन्न किया गया। *कैजानस* प्रजातियों में ताप प्रतिबल अनुक्रिया में भिन्नता का अध्ययन किया गया। जैवरासायनिक मूल्यांकन के द्वारा तुलनात्मक ट्रांसक्रिप्टोम विश्लेषण से यह स्पष्ट हुआ कि *सी. स्केराबीयोडेस* प्रजाति में सर्वाधिक ताप सहिष्णुता थी। इसके पश्चात् *सी. कैजानीफोलियस*, *सी. कैजान* और *सी. एक्यूटिफोलियस* में सहिष्णुता पायी गई।

भारत चने का एक प्रमुख उत्पादक एवं उपभोक्ता देश है जिसका चने के कुल वैश्विक उत्पादन में लगभग 70% योगदान है। सूखा चने की उत्पादकता के मार्ग में आने वाली प्रमुख बाधाओं में से एक है। इस दिशा में भा.कृ.अ. सं. द्वारा भा.कृ.अ.प.—एनआईपीबी के सहयोग से एक सूखा सहिष्णु किस्म BG 4005 जारी की गई है। BG 4005 (पूसा चिकपी 4005), पूसा 362 का एक सूखा—सहिष्णु समाहन

वंशक्रम है, जिसे मार्कर-सहायी प्रतीप संकर प्रजनन के माध्यम से विकसित किया गया है। चने के जीनप्ररूपों के तुलनात्मक विश्लेषण से सूखा-सहिष्णु जीनप्ररूप के जड़ ऊतक में तीन गुने से अधिक अपरेगुलेशन के साथ तीन miRNAs, miR167, miR168 और miR171 ज्ञात किए गए। ये सूखा-अनुक्रियाशील miRNAs चने में सूखा प्रतिबल सहिष्णुता के पश्च-ट्रांसक्रिप्शनल जीन विनियमन की नई अंतरदृष्टि उपलब्ध करा सकते हैं, ऐसी संभावना है।

जड़ गांठ सूत्रकृमि चने के प्रमुख पीड़कों में से एक है तथा सूत्रकृमि संक्रमण को रोकने के लिए उचित प्रमोटर्स का HD-RNAi में उपयोग करते हुए सूत्रकृमि परजीविता के प्रमुख जीनों को लक्षित करना एक सर्वाधिक प्रभावी युक्ति पाया गया है। हमने पूर्व में सूत्रकृमि प्रभावक जीन *Mi-msp2* के dsRNA कांस्ट्रक्ट की अभिव्यक्ति के लिए *एरेबिडॉप्सिस* से परॉक्सीडेज पिटिका-विशिष्ट प्रमोटर, *pAt2g18140* का सत्यापन किया था। जब संक्रमण कराया गया तो *CaMV35S::Mi-msp2* वंशक्रमों की तुलना में *pAt2g18140::Mi-msp2* वंशक्रमों में अंडों की संख्या में कमी पाई गई थी। इसके साथ ही पिटिका विशिष्ट प्रमोटर वाले RNAi वंशक्रमों में अगली पीढ़ी की प्रजनन दर में अंततः कमी के साथ-साथ मादा के आकार में भी कमी देखी गई।

संस्थान में गेहूं की फसल की नाइट्रोजन उपयोग की दक्षता (एनयूई) तथा उच्च तापमान प्रतिबल सहिष्णुता कार्य के प्रमुख क्षेत्र हैं। गेहूं और उनकी संततियों में नाइट्रोजन स्वांगीकरण जीनों (*GS2* और *Fd-GOGAT*) के तुलनात्मक विश्लेषण में सीमित अनुक्रम विचलन प्रदर्शित हुआ तथापि, गेहूं और उनकी संततियों में उनकी अभिव्यक्ति और अंततः पड़ने वाले प्रभाव भिन्न-भिन्न थे। गेहूं के NRT2 (उच्च संलयता युक्त नाइट्रेट ट्रांसपोर्टर कुल) और NAR2 (अनेक गतिविधियों के लिए लेकिन

सभी NRT2 के लिए नहीं लेकिन गौण प्रोटीन के लिए वांछित) जीनों का विश्लेषण नाइट्रेट को सीमित करने वाली दशा के अंतर्गत विभिन्न प्रमुख ऊतकों में गेहूं की फसल की वृद्धि की तीन प्रमुख अवस्थाओं में विश्लेषण किया गया। इसके अतिरिक्त समान बाह्य दशा के अंतर्गत ¹⁵N इनपलक्स का अध्ययन किया गया, ताकि चार चुने गए जीनप्ररूपों में रेखांकित नाइट्रेट परिवहन क्षमता को समझा जा सके। हमारे परिणामों से नाइट्रेट सीमित करने वाली दशा के अंतर्गत गेहूं में ¹⁵N इनपलक्स (नाइट्रेट अंतर्ग्रहण क्षमता) में आनुवंशिक भिन्नता के साथ-साथ ऊतक विशिष्ट *TaNRT2* जीन की ऊतक विशिष्ट अभिव्यक्ति की अवस्था का संकेत मिला। गेहूं की नाइट्रेट ट्रांसपोर्टर जीनों के साथ उच्च संलयता का कार्यात्मक लक्षण-वर्णन करने के लिए उनमें से कुछ को कार्यात्मक सम्पूरण के लिए *एरेबिडॉप्सिस* के *atnrt2.1* उत्परिवर्तक में रूपांतरित किया जा चुका है।

गेहूं में उच्च-तापमान प्रतिबल सहिष्णुता की आण्विक यांत्रिकियों को सुस्पष्ट करने की दिशा में ताप प्रेरणशील विनिमायक जीन MBF1 (बहुप्रोटीन सेतुकारी कारक I) का सम्पूर्ण जीनोम विश्लेषण किया गया। इसके अतिरिक्त ताप प्रतिबल सहिष्णुता लाने में गेहूं से लवण प्रतिबल जड़ प्रोटीन आरएस1 की भूमिका को समझने के लिए तम्बाकू के पराजीनी पौधे सृजित किए गए। GTPase कुल के केन्द्रक प्रोटीन जीनों से संबंधित आरएस के जीनोमव्यापी विश्लेषण चपाती गेहूं में किए गए तथा विकास की विभिन्न अवस्थाओं और अजैविक प्रतिबल दशाओं के दौरान उनकी अभिव्यक्ति प्रोफाइल का विश्लेषण किया गया।

पिछले दो दशकों से यह संस्थान सरसों की उत्पादकता बढ़ाने के लिए निरंतर अनुसंधान करता आ रहा है। भारतीय सरसों में इस प्रौद्योगिकी के उपयोग के लिए वैक्टर कांस्ट्रक्ट के विकास के साथ-साथ जीनोम संपादन की CRISPR/Cas प्रणाली को उपयुक्ततम बनाया गया

है। *ब्रैसिका जंसिया* और *बी. ओल्लेरेंसिया* में अगुणित प्रेरक वंशक्रमों के विकास हेतु इन दोनों जातियों में CENH3 पैरालॉग को लक्षित करते हुए CRISPR/Cas9 वैक्टर को डिजाइन किया गया। चूंकि CENH3 नॉकआउट भ्रूण घातक है इसलिए इन CENH3 नॉकआउट वंशक्रमों को बचाने के लिए जीएफपी-टेलस्वैप वैक्टर डिजाइन किया गया। हमने इन दो वैक्टर को *बी. जंसिया* की किस्मों वरुना और आरएमएल 198 तथा *बी. ओल्लेरेंसिया* किस्म बोटाइटिस की वाणिज्यिक किस्म पूसा मेघना में सह-रूपांतरित किया है। हमने CRISPR/Cas9 और जीएफपी-टेलस्वैप कांस्ट्रक्ट के एकीकरण के लिए पीसीआर विश्लेषण में सकारात्मकता प्रदर्शित करने वाले *बी. जंसिया* के पौधे की भी पहचान की है।

भा.कृ.अ.प.-एनआईपीबी में हम *ब्रैसिका* के ऐसे वन्य संबंधियों का उपयोग कर रहे हैं जो कीट और रोगजनक प्रतिरोध के लिए जीनों का समृद्ध स्रोत हैं और इसके लिए विभिन्न युक्तियों को आजमाया जा रहा है। *आल्टर्नरिया ब्रैसिकी* द्वारा भारतीय सरसों में होने वाला पत्ती धब्बा रोग न केवल भारत में वरन पूरे विश्व में सरसों की वार्षिक उपज में उल्लेखनीय क्षति पहुंचाता है। हमने *आल्टर्नरिया ब्लाइट* की प्रतिरोधी वन्य जातियों का उपयोग करके *बी. जंसिया* के समाहन वंशक्रम (आईएल) विकसित किए हैं। आईएल में सम्बद्ध गुणों से संबंधित जीनों और पथों की पहचान के लिए स्थायी प्रतिरोधी तथा संवेदी आईएल में पत्ती ट्रांसक्रिप्टोम और जीबीएस सूचना सृजित की गई है। *सिनेपिस एल्बा* और *कैमेलिना सेटाइवा* में प्रतिरक्षा यांत्रिकियों को समझने के लिए भिन्न रूप से प्रेरित प्रोटीनों (पीएमई, सीएसएमटी, एचडीएलपी, एक अज्ञात जीन और Cs-काइनेज) के ट्रांसक्रिप्शनल विश्लेषण से *बी. जंसिया*, *एस. एल्बा* और *सी. सेटाइवा* से इनकी पहचान पूरी की गई है। पुनर्संश्लेषण युक्ति के अंतर्गत अगेती पीढ़ियों में *ब्रैसिका*

जंसिया के 37 वंशक्रमों और 90 प्रगत पीढ़ियां विकसित की गई हैं। इनका अनेक वर्षों/अनेक स्थलों पर परीक्षण किया गया था और 15 आल्टर्नरिया ब्रैसिकी प्रतिरोधी आईएल पहचाने गए।

सरसों में सफेद रतुआ प्रतिरोध विकसित करने हेतु सरसों की खेती वाले प्रमुख हॉट-स्पॉट स्थलों से एल्ब्यूगो कैडिडा के 15 विलगक एकत्र किए गए। कुल 721 ब्रैसिका जननद्रव्यों/प्रविष्टियों/समाहित वंशक्रमों/पुनःसंश्लेषी वंशक्रमों को एल्ब्यूगो कैडिडा के नौ विद्यमान विलगकों से कृत्रिम रूप से संक्रमित कराया गया और उन पर रोग प्रतिक्रिया का अध्ययन किया गया। वन्य संबंधियों नामतः बी. कैरिनाटा, क्रैम्बे एबिसिनिका, एरुका सेटाइवा और इसके साथ-साथ 4 समाहित एवं 4 पुनःसंश्लेषित ब्रैसिका वंशक्रमों की नियंत्रित पर्यावरण में कृत्रिम संरोपित दशाओं के अंतर्गत रोगजनक के 3-7 विलगकों के प्रति रोगरोधी प्रतिक्रिया वाले आशाजनक वंशक्रमों की पहचान की गई है। इसके साथ ही खेत की दशा के अंतर्गत बी. जंसिया की प्रतिरोधी किस्मों की भी पहचान की गई है।

पिछले वर्ष हमने भारतीय सरसों में एस. स्क्लेरोटियोरम संक्रमण के विरुद्ध सैक्रेटोम लाइब्रेरी के बारे में रिपोर्ट किया था। इस वर्ष हमने पाया कि सभी पूर्वानुमानित इफेक्टर प्रोटीन एस. स्क्लेरोटियोरम माईसिलियम में अभिव्यक्त हुए थे। हमने रोगजनन की कार्यात्मक प्रासंगिकता के आधार पर 27 एनोटेड इफेक्टर से 13 इफेक्टर प्रत्याशियों का चयन किया है। इसके अतिरिक्त हमने बी. जंसिया के सहिष्णु (आरएच1222-28) तथा संवेदी (एनआरसीपीबी101) वंशक्रमों के समय आधारित आरएनए-क्रम विश्लेषण के माध्यम से अपेक्षाकृत भिन्न रूप से व्यक्त जीन (RDGEs) प्राप्त किए हैं जो अनेक पोषक प्रतिरक्षा संबंधी जीनों से संबंधित हैं।

ओरोबैंके के लिए प्रतिरोधी/सहिष्णु जीनप्ररूप की पहचान के लिए हमने हरियाणा के हॉट-स्पॉट क्षेत्रों में 216 ब्रैसिका जननद्रव्यों की छंटाई की है। ओरोबैंके जाति की पहचान के लिए विशिष्ट आण्विक मार्करों का उपयोग किया गया था। विभिन्न पथों के लिए उत्तरदायी विभिन्न जीनों को पहचानने के लिए ओरोबैंके के विभिन्न ऊतकों का ट्रांसक्रिप्टोम विश्लेषण किया गया।

चावल में, चावल का प्रध्वंस और जीवाण्विक अंगमारी, पत्ती अंगमारी तथा प्रमुख अजैविक प्रतिबल जैसे सूखा, लवणता तथा उच्च तापमान प्रतिबल भा.कृ.अ.प.-एनआईपीबी में कार्य के प्रमुख क्षेत्र हैं। चावल के प्रध्वंस रोग के अनेक प्रकारों में से पुष्पगुच्छ प्रध्वंस सर्वाधिक विनाशकारी है। संस्थान में प्रतिरोधी किस्म टेपे और संवेदी किस्म एचपी2216 का उपयोग करके संक्रमण के क्रमशः 48, 72 और 96 घंटे बाद आरएनए-अनुक्रमण का उपयोग करते हुए चावल में प्रथम पुष्पगुच्छ प्रध्वंस ट्रांसक्रिप्टोम रिपोर्ट किया गया था। हमारे अध्ययनों से यह प्रदर्शित हुआ कि पुष्पगुच्छ प्रध्वंस प्रतिरोध एक जटिल प्रक्रिया है जो आरओएस उत्पादन तथा वीऑक्सीकरण, एमएपीके एवं एलआरआर सिग्नलिंग, प्रति सूक्ष्मजैविक यौगिकों तथा द्वितीयक चायपचयजों के एकत्रित होने तथा कोशिका भित्ति को सशक्त बनाते हुए प्रारंभिक प्रतिरक्षा अनुक्रिया के परिणामस्वरूप होती है, ताकि कवकों के प्रवेश और प्रसार से बचा जा सके। खनिज सिलिकॉन (Si) पादप प्रतिरक्षा में महत्वपूर्ण भूमिका निभाने के लिए जाना जाता है। मैग्नेपोर्थे संक्रमण के दौरान चावल के पौधों पर Si अनुप्रयोग के प्रभाव को समझने के लिए चावल की वाणिज्यिक किस्मों टेपे और एचपी2216 में Si ट्रांसपोर्टर जीनों पर अध्ययन किया गया। कैल्सियम सिलिकेट संबंधी उपचारों में संक्रमण की उपस्थिति में Si ट्रांसपोर्टर, *OsLsi1* और *OsLsi3* का उल्लेखनीय अपरेगुलेशन हुआ। इस प्रकार, जैविक प्रतिबल दशा के अंतर्गत पौधों

के सिलिकेट स्तरों में परिवर्तन लाए गए। प्रयोगशाला में किए गए एक अध्ययन में आच्छद अंगमारी रोग के विरुद्ध पोषक द्वारा उत्पन्न की गई RNAi बाधा का उपयोग करके चावल के पराजीनी पौधे विकसित किए गए।

जेंथोमोनास ओराइजी पीवी ओराइजी के एक भारतीय प्रभेद जो चावल में जीवाण्विक अंगमारी का कारक रोगजनक है, तीन प्रमुख ट्रांसक्रिप्शन एक्टिवेटर लाइक इफेक्टर (TALE) का उपयोग किया जाता है, ताकि *xa13* मध्यत प्रतिरोध से बचा जा सके। सभी तीन प्रमुख टीएल इफेक्टर में एबरेट रिपीट से प्राप्त किए गए चयन दबाव के अंतर्गत विकास की एक अतिरिक्त परत देखी गई। सभी तीनों TALEs के कार्यात्मक लक्षण-वर्णन से इंडिका तथा जैपोनिका दोनों किस्मों में रोग के प्रति व्यक्तिगत योगदान के बारे में ज्ञात किया गया।

वैश्विक अभिव्यक्ति आंकड़ों से पहचाने गए चावल के एक नए अजैविक प्रतिबल अनुक्रियाशील, LOC_Os06g10210 (OsCHI2) को प्रतिबल उत्प्रेरणशील *AtRd29A* प्रमोटर के नियंत्रण के अंतर्गत नगीना 22 (एन22) से विलगित किया गया और वाणिज्यिक किस्म पूसा सुगंध-2 (पीएस-2) में रूपांतरित किया गया। दो स्वतंत्र पराजीनी घटनाओं से यह प्रदर्शित हुआ कि वे सूखा, शीत तथा लवणता प्रतिबलों के अंतर्गत वन्य प्रकार के पीएस2 पौधों की तुलना में अनेक भौतिक/रासायनिक प्राचलों के संदर्भ में श्रेष्ठ थे। *OsCHI2* पराजीनी वंशक्रमों तथा एन22 ने फ्लेवोनाइड जैव संश्लेषण पथ के संरचनात्मक जीनों की ट्रांसक्रिप्टकी की प्रचुरता को बढ़ाया है जिसके कारण वन्य प्रकार के पीएस 2 पौधों की तुलना में अजैविक प्रतिबल की दशाओं के अंतर्गत फ्लेवोनाइड का उच्चतर उत्पादन हुआ है। हमने चावल में एक आरंभिक ऊष्मा-उत्प्रेरणशील जीन PUB63 जो प्लांट U बॉक्स लिगेज़ का सदस्य है, के बारे में भी अन्वेषण किया है तथा

इसे क्लोन भी किया है, ताकि यह प्रदर्शित हो सके कि विकसित किए गए पराजीनी वंशक्रम ताप प्रतिबल उपचार के प्रति अधिक सहिष्णु हैं।

गहरे जल के चावल में तने के लंबायमान होने पर किए गए अध्ययन में miRNAs और lncRNAs की पहचान की है। ये गहरे जल प्रतिबल उपचार के 14 दिन बाद सबसे ऊपरी प्रथम अंतरगांठ तथा गांठ के ऊतकों से एकत्र किए गए थे। ceRNA के तुलनात्मक अध्ययन से यह स्पष्ट हुआ कि 69 और 27 lncRNAs क्रमशः अंतरगांठ तथा गांठ miRNAs के साथ एंडोजीनस टारगेट मिमिक्स (eTM) के रूप में कार्य कर रहे थे। ऐसे एक अणु का तम्बाकू की पत्तियों में परिवर्तनशील अभिव्यक्ति के माध्यम से सत्यापन किया गया।

एक अन्य अध्ययन में अंतरगांठ के लंबायमान होने से संबंधित 18 अभिव्यक्ति मात्रात्मक गुण स्थलों (eQTL) तथा एक अवायवीय अंकुरण सहिष्णु संबंधित जीन के हैप्लोटाइप नेटवर्क का अन्वेषण असम की चावल की 94 गहरे जल की भूप्रजातियों में किया गया। *tak* रिसेप्टर-जैसे *काइनेज 1* तथा स्वभोजी से संबंधित जीन *18b* (गुणसूत्र 1 पर) की भी जांच की गई। प्रमुख भारतीय उपराऊं और तराऊं चावल के 24 जीनप्ररूपों में गहरी जड़ें बनाने वाले (डीआरओ1) जीन के युग्मविकल्पी खनन से संबंधित एक अध्ययन के द्वारा पूरी सामग्री का 16 प्रमुख संरचना वेरियंट में वर्गीकरण किया गया जो सम्पूर्ण कार्यात्मक युग्मविकल्पी (सट्टी, आईआर 36 तथा डीटीवाई 3.1 डोनर, आईआर81896-बी-बी-195) के भिन्न आकार का था, ताकि ट्रंकेटिड कार्यहीन युग्मविकल्पी (पीएमके 2, आईआर 64, आईआर 20 और स्वर्णा) सृजित किए जा सकें। एक अन्य अध्ययन में चावल में अवायवीय अंकुरण सहिष्णुता के लिए गुणसूत्र-7 पर एक प्रमुख क्यूटीएल की पहचान की गई।

दस विभिन्न Poaceae जातियों में *MIR397* और *MIR408* की अनुक्रिया में भिन्नता ज्ञात करने के लिए विभिन्न ऊतकों में *MIR397* और *MIR408* के अभिव्यक्ति प्रोफाइल सृजित किए गए। *MIR397* और *MIR408* की सर्वाधिक प्रचुरता जननात्मक ऊतकों जैसे पताका पत्ती में देखी गई, जबकि वानस्पतिक ऊतकों जैसे जड़ ऊतकों में, *MIR397* के मामले में न्यूनतम अभिव्यक्ति देखी गई, जबकि *MIR408* के मामले में तना ऊतकों में इसकी न्यूनतम अभिव्यक्ति देखी गई। *MIR397* की असामान्य उच्च अभिव्यक्ति चावल की वन्य जातियों में सीमित रही तथा इसे चावल के कृष्य वंशक्रमों तथा पोएसी कुल के अन्य सदस्यों में नहीं पाया गया। इस संस्थान में कार्यात्मक जीनोमिक्स विश्लेषण तथा फसल सुधार के लिए वन्य चावल तथा ईएमएस प्रेरित उत्परिवर्तकों का विशाल संकलन है। इस वर्ष अनुसंधान तथा रखरखाव के लिए चावल की 255 वन्य प्रविष्टियां एनबीपीजीआर में प्रस्तुत की गईं।

इन अधिदेशित फसलों में हुए विकासों के अतिरिक्त रागी में तम्बाकू के पराजीनों की अति अभिव्यक्ति सृजित करके, DREB2A जीन का लक्षण-वर्णन किया गया जिससे बढ़ी हुई आरओएस स्केवेंजिंग यांत्रिकी के माध्यम से प्राप्त प्रतिबल के प्रति बढ़ी हुई सहिष्णुता प्रदर्शित हुई तथा ताप प्रतिबल दशा के अंतर्गत वन्य प्रकार के पौधों की तुलना में घटा हुआ इलेक्ट्रोलाइट लिकेज, H₂O₂ और मैलनडाइएल्डीहाइड अंश प्रदर्शित हुआ।

भारतीय आम की चुनी हुई किस्मों के सुगंध प्रोफाइलिंग पर आधारित गैस क्रोमेटोग्राफी-मास स्पेक्ट्रोस्कोपी-हैंड स्पेस (जीसी-एमएस-एचएस) से लिमोनीन जैसे यौगिकों की उच्च प्रचुरता प्रदर्शित हुई। ऊतक संवर्धन के द्वारा उगाए गए पौधों (एनएससी-टीसीपी) की राष्ट्रीय प्रमाणीकरण प्रणाली के अंतर्गत संस्थान में इस वर्ष citrus के पौधों के आनुवंशिक फीडेलिटी परीक्षण के लिए मानक

परिचालनशील प्रोटोकॉल विकसित किया गया। हमने आनुवंशिक फीडेलिटी के लिए 8 ISCR मार्कर प्रस्तावित किए हैं जिनमें से दो मार्करों में, इस प्रोटोकॉल का उपयोग करके, मातृ पौधे तथा संतति में भिन्नता प्रदर्शित हुई है।

भा.कृ.अ.प.-एनआईपीबी अपने स्थापना काल से ही पादप आण्विक जीवविज्ञान तथा जैवप्रौद्योगिकी (एमबीबी) के क्षेत्र में मानव संसाधन विकास के कार्य में सक्रिय भूमिका निभा रहा है। वर्तमान में 50 पीएच.डी. और 17 एम.एससी. छात्र आण्विक जीवविज्ञान एवं जैवप्रौद्योगिकी विषय में पंजीकृत हैं। वर्ष 2021 के दौरान भा.कृ.अ. प.-भारतीय कृषि अनुसंधान संस्थान के 59वें दीक्षांत समारोह में छह पीएच.डी. और 8 एम.एससी.छात्रों को क्रमशः डॉक्टरल तथा स्नातकोत्तर उपाधियां प्रदान की गईं।

‘ऊतक संवर्धन द्वारा उगाए गए पौधों का आनुवंशिक फीडेलिटी परीक्षण एवं विषाणु अनुसूचीकरण’ पर एक डीबीटी-बीसीआईएल प्रायोजित प्रशिक्षण कार्यक्रम ऑन-लाइन मोड में आयोजित किया गया। भा.कृ. अ.प.-एनआईपीबी द्वारा भारत के छह राज्यों (उत्तर प्रदेश, उत्तराखण्ड, मध्य प्रदेश, राजस्थान, बिहार और ओडिशा) में 14 एससीएसपी कार्यक्रम आयोजित किए गए। इन कार्यक्रमों से 2500 से अधिक किसानों को लाभ हुआ। संस्थान द्वारा भा.कृ.अ.प.-एनआईपीबी में एक कृषक-वैज्ञानिक परिचर्चा आयोजित की गई तथा संस्थान ने भा.कृ.अ.प.-भारतीय कृषि अनुसंधान संस्थान द्वारा आयोजित कृषि मेले में अपनी प्रौद्योगिकियों तथा अनुसंधान गतिविधियों का प्रदर्शन किया। इसके अतिरिक्त कोविड बचाव के दिशानिर्देशों का अनुपालन करते हुए संस्थान में अनेक गतिविधियों जैसे पोषण वाटिका महाअभियान, वृक्षारोपण अभियान, हिंदी चेतना मास, सतर्कता जागरूकता सप्ताह, स्वच्छता अभियान आदि भी आयोजित की गईं।

Executive Summary

National Institute for Plant Biotechnology (NIPB) is a research institution under the umbrella of ICAR, with a mandate for development and applications of modern biotechnology tools and techniques in the area of plant biotechnology. ICAR-NIPB also has the responsibility of serving as the Division of Molecular Biology and Biotechnology for teaching and training of post graduate students of Indian Agricultural Research Institute, besides its institute level training programmes for NARES staff and students. Through Scheduled Castes Sub Plan (SCSP) and Mera Gaon Mera Gaurav (MGMG) programmes of the Government of India, the institute also serves the farmers. The institute mainly works on five major crops, namely rice, wheat, pigeonpea, chickpea and Indian mustard; and being the premiere institute of plant biotechnology under ICAR, it works on other agricultural and horticultural crops through strong collaborations with national and international institutions for developing genomic resources and finding biotechnological solutions to the major problems in those crops. Salient achievements of the institute during the year 2021 are summarized below.

Genomics is one of the strong pillars of ICAR-NIPB. In the reporting year, the black gram genome of cv. LBG 20 was assembled and organized in 3758 scaffolds with high N50 values of 48.01 Mb. Eventually; about 524.23 Mb out of 558.22 Mb of black gram genome assembly was organized into 11 pseudomolecules/chromosomes. Expanding our work on Indian tea (*Camellia assamica* var. Masters cv. TV-1) in which we earlier sequenced the organelle genome, we have now generated a chromosome-level reference genome assembly (2.93 Gb) covering 97.66% of the predicted 3.0 Gb tea genome, anchored to 15 pseudomolecules. The single-copy gene-based evolutionary study and the whole genome duplication (WGD) analysis have confirmed the divergence of TV-1 from CSA and CSS approximately at around 2 and 5.5 MYA, respectively. To speed up tea breeding programs, a first-generation haplotype map of tea (Tea HapMap 1) was generated using publicly available whole-genome re-sequencing data of 369 tea genotypes. Last year, we reported chromosome level assembly of clusterbean. This year, we identified a total of 29 and 27 genes related with flavonoid

and phenylpropanoid biosynthetic pathway. Though these genes were distributed on all 7 chromosomes of cluster bean, chromosome 1 had the most number of phenylpropanoid genes and chromosomes 1 and 5 had the maximum number of flavonoid biosynthesis genes.

Cultivation of pigeonpea, a major legume crop of the country, is challenged by three major environmental factors, infestation by *Helicoverpa armigera* (pod borer), aphids and heat stress. To address these issues, the institute employs several approaches. In the reporting year, a *CHI4* chitinase gene was identified in pigeonpea wild relative, *Cajanus platycarpus* based on a dynamic transcriptome under continued herbivory by *H. armigera*. Subsequently the gene was cloned, characterized and validated *in planta* in tobacco. The gene was found to be effective against *H. armigera* based on the *in vitro* leaf bioassay suggesting it to be of potential utility in the control of pod borer. Insecticidal property of plant lectins, protease inhibitors, α -amylase inhibitors, and chitinases can also be exploited for crop protection. To broaden

the resistance spectrum of the brassica, two transgenic brassica lines harbouring chickpea lectin and urd bean protease inhibitor gene respectively, were crossed and progenies were selfed and screened for presence of transgenes. Homology modelling of potential pest control targets such as chitin deacetylase from *Helicoverpa armigera* and arginine kinase from *Chilo suppressalis* was carried out. The variation in heat stress response was studied in *Cajanus* species. Comparative transcriptome analyses accompanied by biochemical assays revealed that species, *C. scarabaeoides* was the most thermotolerant followed by *C. cajanifolius*, *C. cajan*, and *C. acutifolius*.

India is the major chickpea-producing and consuming country, contributing almost 70% of the world's total production. Drought is one of the major constraint for chickpea productivity. Towards this, a drought-tolerant variety, BG 4005, was released by IARI, in which ICAR-NIPB is a Collaborator. BG 4005 (Pusa Chickpea 4005) is a drought-tolerant introgression line of Pusa 362, developed through Marker-assisted backcross breeding. A comparative analysis in chickpea genotypes revealed three miRNAs, miR167, miR168 and miR171 with more than 3-fold upregulation in root tissue of drought-tolerant genotype. These drought-responsive miRNAs are likely to provide novel insights into post-

transcriptional gene regulation of drought stress tolerance in chickpea.

Root-knot nematodes is one of the major pests of chickpea and targeting major nematode parasitism genes via HD-RNAi using appropriate promoters is an effective approaches to curb nematode infection. We employed a previously validated peroxidase gall specific promoter, pAt2g18140, from *Arabidopsis* to express the dsRNA construct of the nematode effector gene *Mi-msp2*. When subjected to infection, the reductions in the numbers of egg masses in the pAt2g18140::Mi-msp2 lines was more than the CaMV35S::Mi-msp2 lines. A reduction in female size and a subsequent reduction in next-generation fecundity was also observed in the gall specific promoter driven RNAi lines.

At the institute, nitrogen use efficiency (NUE), drought and high temperature stress tolerance are the major areas of work in wheat. Comparative analysis of N assimilation genes (GS2 and Fd-GOGAT) in cultivated wheat and their progenitors showed limited sequence divergence; however, their expression and subsequent effects were different in cultivated wheat and their progenitors. The expression of NRT2 (high affinity nitrate transporter family) and NAR2 (nitrate assimilation related; accessory protein required for the activity of many but not all NRT2 protein) genes of bread

wheat was analysed at three key growth stages of wheat in different major tissues under nitrate limiting condition. Furthermore, ¹⁵N influx under similar external condition was studied to understand the underlying nitrate transport capacity in four selected genotypes. Our results indicated stage and tissue specific expression of TaNRT2 genes under nitrate-limiting condition, and genetic variation in ¹⁵N influx (nitrate uptake capacity) in wheat. To functionally characterize high affinity nitrate transporter genes of wheat, few of them have been transformed into *atnrt2.1* mutant of *Arabidopsis* for functional complementation.

Towards deciphering the molecular mechanisms of high-temperature stress tolerance in wheat, whole-genome analysis of heat-inducible regulatory gene MBF1 (multiprotein bridging factor 1) was carried out. Further, to understand the role of Salt Stress Root Protein RS1 from wheat (TaSSRP) in imparting heat stress tolerance, transgenic tobacco plants were generated. Genome-wide analysis of RAS related nuclear protein genes of GTPase family was carried out in bread wheat and their expression profile was analysed during different developmental stages and abiotic stress conditions.

Over the past two decades this centre has been continuously undertaking research

for improvement of mustard productivity. CRISPR/Cas system of genome editing has been optimized along with development of vector constructs for utilization of this technology in Indian mustard. For developing haploid inducer lines in *Brassica juncea* and *B. oleracea*, CRISPR/Cas9 vector targeting *CENH3* paralog(s) in both these species were designed. Since *CENH3* knockout is embryo lethal, a GFP-tailswap vector was designed to rescue these *CENH3* knockout lines. We have co-transformed these two vectors in *B. juncea* cvs. Varuna and RLM-198 and *B. oleracea* var. botrytis cv. Pusa Meghna. We have identified *B. juncea* plantlets showing positive in PCR analysis for integration of CRISPR/Cas9 and GFP-tailswap constructs.

At ICAR-NIPB we are exploiting the wild relatives of Brassica which are rich reservoirs of genes for insect and pathogen resistance through different approaches. *Alternaria* leaf spot disease of Indian mustard caused by *Alternaria brassicae* inflicts a significant annual yield loss in India and across the world. We have developed *B. juncea* introgression lines (ILs) using a wild species resistant to *Alternaria* blight. For identification of genes and pathways associated with the trait in ILs, leaf transcriptome and GBS information have been generated in the stable resistant and susceptible ILs. To understand the mechanisms

of defense in *Sinapis alba* and *Camelina sativa*, transcriptional analysis of differentially induced proteins (PME, CSMT, HDLP, one unknown gene, and Cs-Kinase) earlier identified from *B. juncea*, *S. alba*, and *C. sativa* was done. In yet another approach, 37 resynthesized *Brassica juncea* lines in early generations and 90 from advanced generations, developed in the institute programme, were tested in multi-year/ multi-location and 15 *Alternaria brassicae* resistant ILs were identified.

Towards developing white rust resistance in mustard, 15 isolates of *Albugo candida* from major Indian mustard growing/hot spot locations were collected. A total of 721 Brassica germplasm/accessions/introgressed lines/resynthesized lines were artificially inoculated with nine prevalent isolates of *A. candida* and their disease reaction was studied. Promising lines showing immune reaction to 3-7 isolates of the pathogen have been identified from wild relatives, *B. carinata*, *Crambe abyssinica*, *Eruca sativa*, in addition to 4 introgressed and 4 resynthesized Brassica lines under environmentally controlled artificially inoculated conditions. *B. juncea* accessions resistant under field condition have also been identified.

We reported secretome library against *S. sclerotiorum* infection in Indian mustard last year.

This year, we found that 100% of the predicted secreted effector proteins were expressed at both the genomic and transcript level in the *in vitro* grown *S. sclerotiorum* mycelium. We have shortlisted 13 effector candidates from the 27 annotated effectors based on their functional relevance to pathogenesis. Further, the relative differentially expressed genes (RDGEs) obtained through time course RNA-Seq analysis of the tolerant (RH1222-28) and susceptible (NRCPB101) *B. juncea* lines have revealed several host defense-related genes.

To identify the resistant/tolerant genotype for *Orobanche*, we screened 216 Brassica germplasm in hotspot regions of Haryana. To identify the *Orobanche* species, specific molecular markers were employed. Transcriptome analysis of various tissues of *Orobanche* was done to identify different genes responsible for various pathways.

In rice, major diseases namely rice blast, bacterial blight, sheath blight and major abiotic stresses namely, drought, salinity and heat stress are the major areas of work at ICAR-NIPB. Among the many manifestations of rice blast disease, panicle blast is the most devastating. The institute reported the first panicle blast transcriptome in rice using blast resistant cultivar Tetep, and susceptible cultivar HP2216 by using RNA-Seq after 48, 72 and 96 h

post infection. Our study showed that panicle blast resistance is a complex phenomenon contributed by early defense response through ROS production and detoxification, MAPK and LRR signalling, accumulation of antimicrobial compounds and secondary metabolites, and cell wall strengthening to prevent the entry and spread of the fungi. The mineral silicon (Si) is known to play an important role in plant defence. To understand the effect of Si application on rice plants during *Magnaporthe* infection, study on Si transporter genes were carried out in cvs. Tetep, and HP2216. In the calcium silicate treated plants, there was significant upregulation of Si transporters, *OsLsi1* and *OsLsi3*, in the presence of infection, thus bringing about changes in silicate levels of plant under biotic stress. In one of the studies in the lab, transgenic rice plants were developed using host delivered RNAi interference against sheath blight disease.

An Indian strain of *Xanthomonas oryzae* pv *oryzae*, the causal pathogen of bacterial blight in rice, deploys three major Transcription Activator Like effectors (TALE) to defy the *xa13* mediated resistance. An additional layer of evolution adapted under selection pressure was seen in the form of aberrant repeats in all the three major TAL effectors. Functional characterization of all the three TALEs revealed them to be individually

contributing to disease in both indica and japonica varieties.

A novel abiotic stress responsive gene in rice, LOC_Os06g10210 (*OsCHI2*), identified from the global expression data, was isolated from cv. Nagina 22 (N22) and transformed into cv. Pusa Sugandh 2 (PS2) under the control of stress inducible *AtRd29A* promoter. Two independent transgenic events showed that they were superior in several physio-chemical parameters as compared to wildtype PS2 plants under drought, cold and salinity stresses. *OsCHI2* transgenic lines and N22 had enhanced transcript abundance of structural genes of flavonoids biosynthesis pathway leading to higher production of flavonoids under abiotic stresses than the wild type PS2 plants. We have also investigated a member of Plant U Box ligases, PUB63 - an early heat-inducible gene in rice and cloned it to demonstrate that the developed transgenic lines are more tolerant to heat stress treatment.

To study stem elongation in deepwater rice, miRNAs and lncRNAs were identified from the uppermost 1st internode and node tissues were collected after 14 days of deepwater stress treatment. Competitive endogenous RNA (ceRNA) study revealed that 69 and 27 lncRNAs were acting as endogenous target mimics (eTM) with the internode and node

miRNAs, respectively. One such module was validated through transient expression in tobacco leaves.

In another study, a haplotype network of 18 expression quantitative trait loci (eQTL) related to internode elongation and an anaerobic germination tolerance related gene were investigated among 94 deepwater rice landraces of Assam. Haplotype variation in *tak receptor-like kinase 1* and *autophagy associated gene 18b* (on chromosome 1) was also examined. An allele mining study of *Deeper Rooting 1* (*DRO1*) gene in 24 major Indian upland and lowland rice genotypes classified the material into 16 primary structure variants ranging from complete functional allele (Satti, IR36 and DTY 3.1 donor, IR81896-B-B-195) to truncated non-functional alleles (PMK2, IR64, IR20 and Swarna). In yet another study, a major QTL for anaerobic germination tolerance in rice was identified on Chromosome 7.

To determine the variation in the response of *MIR397* and *MIR408* in ten different poaceae species, expression profile of *MIR397* and *MIR408* were generated in different tissues. Highest abundance of *MIR397* and *MIR408* was observed in the reproductive tissue i.e., flag leaf and lowest expression was observed in vegetative tissues such as root tissue in case of *MIR397* and stem tissues in case of *MIR408*.

The unusual high expression of *MIR397* was restricted to wild species of rice and was not observed among cultivated rice lines and other poaceae members. The institute also maintains a huge collection of wild rice and EMS induced mutants for functional genomic analysis and crop improvement. This year, 255 wild rice accessions were submitted to NBPGR for maintenance and upkeep.

Besides these developments in the mandate crops, *DREB2A* gene from finger millet was characterized by generating overexpression tobacco transgenics, which showed enhanced tolerance to heat stress through enhanced ROS scavenging mechanism and showed decreased electrolyte leakage, H_2O_2 and malondialdehyde content than wildtype plants under heat stress condition.

A Gas Chromatography-Mass Spectroscopy-Head Space (GC-MS-HS) based aroma

profiling of selected Indian mango varieties demonstrated compounds of high abundance like limonene. Under National Certification System of Tissue culture Raised Plant (NCS-TCP), this year the institute developed standard operating protocol for genetic fidelity testing of citrus plant. We proposed 8 ISSR markers for the genetic fidelity, out of which two markers showed variation in mother plant and progeny using this protocol.

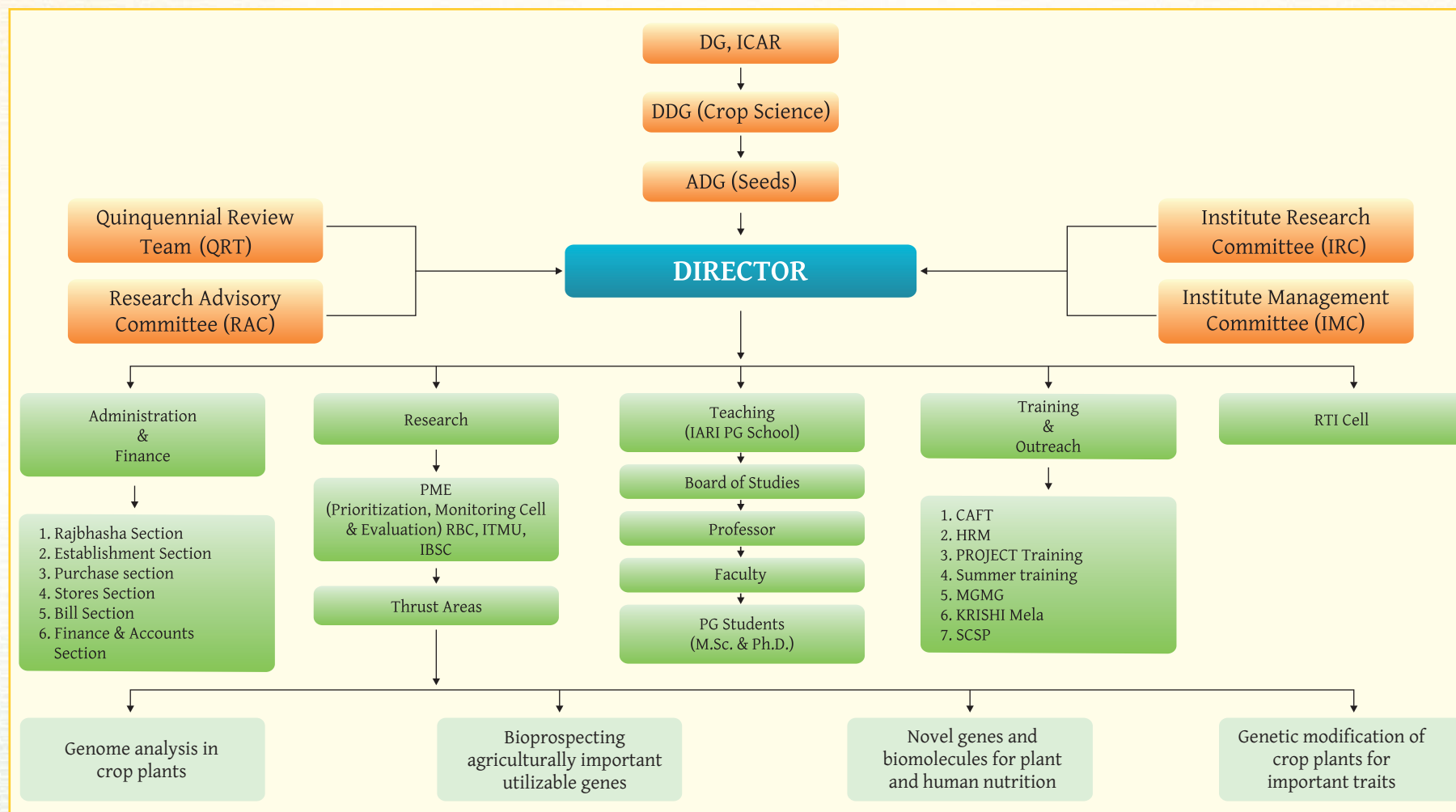
ICAR - NIPB has been actively engaged in Human Resource Development in the area of Plant Molecular Biology and Biotechnology (MBB) since its inception. Currently, 50 Ph. D. and 17 M. Sc. students are registered in the discipline of Molecular Biology and Biotechnology. In the year 2021, six Ph. D. and eight M. Sc. students were awarded with Doctoral and Master's degrees, respectively, in the 59th Convocation of ICAR-IARI.

A DBT-BCIL sponsored Training program for "Genetic Fidelity Testing and Virus Indexing of Tissue Culture-Raised Plants" was conducted in online mode. ICAR - NIPB organized 14 SCSP programmes across six states of India (Uttar Pradesh, Uttarakhand, Madhya Pradesh, Rajasthan, Bihar and Odisha). More than 2500 farmers benefited with these programmes. The institute conducted Farmers-Scientist interaction at ICAR-NIPB and also showcased its technologies and research activities in Krishi mela organized by ICAR-IARI. In addition, several activities such as Poshan Vatika Mahaabhiyan, Tree Plantation Abhiyan, Hindi Chetna Maas, Vigilance Awareness week, Swachchata abhiyan were celebrated following COVID prevention guidelines.

About the Institute

- NIPB AND THE MANDATE
- FINANCIAL STATEMENT
- PERSONNEL
- STAFF POSITION
- RESOURCE GENERATION

ORGANOGRAM OF ICAR-NIPB



ICAR-NIPB

National Institute for Plant Biotechnology (NIPB) is the premiere research institution of the Indian Council of Agricultural research (ICAR), engaged in plant molecular biology and biotechnology research. The Biotechnology Centre, established in 1985, as part of the Indian Agricultural Research Institute (IARI), was upgraded to National Research Centre on Plant Biotechnology in the year 1993, with a vision to impart the biotechnology advantage to the National Agricultural Research System (NARS). In February 2019, NRCPB was elevated to the level of National Institute for Plant Biotechnology (ICAR-NIPB). This brings us much greater national responsibility to conduct basic and applied research, and human resource development in plant biotechnology. NIPB has acquired an excellent infrastructure in terms of equipment and other physical facilities and also a high degree of scientific competence. Research at NIPB involves structural and functional genomics, molecular mapping and marker development, transgenics and genome editing for improving the agronomic

traits, biotic -abiotic stress tolerance and nutritional quality in crops. The institute mainly works on five major crops, namely rice, wheat, pigeonpea, chickpea and Indian mustard. Being the premiere institute of plant biotechnology of ICAR, NIPB works on other agricultural and horticultural crops through strong collaborations with national and international institutions for developing genomic resources and finding biotechnological solutions to the major problems in these crops. In addition to research, the institute contributes significantly to development of human resource by offering regular M.Sc. and Ph.D. programmes in partnership with PG School, IARI. The institute also imparts short and long term trainings to students and scientists from other universities and institutions.

Mandate

1. To undertake basic plant molecular biology research for understanding molecular mechanisms underlying basic biological processes
2. To develop capabilities of devising tools and techniques of biotechnology and genetic engineering for crop improvement
3. To use the knowledge gained and technologies developed for advancing agriculture development
4. To serve as a national lead centre for plant molecular biology and biotechnology research and to create trained manpower in the areas of plant biotechnology and genetic engineering

Staff Strength of the Institute

Staff	Sanctioned	Filled	Vacant
Scientific	36+1	34+1	02
Technical	14	08	06
Administrative	21	13	08
Skilled Supporting Staff	02	-	02
Total Strength	73+1	55+1	18

Financial Statement 2021-22

(Rs. In lakhs)

Institute Grant		
	Allocation	Utilization
Capital	129.44	117.56
Revenue		
Establishment	1136.09	1135.96
Pension & other retirement benefits	62.79	62.69
Travelling Allowances	9.50	9.44
Research and Operational Expenses	403.50	103.22
Administrative Expenses	433.20	432
Miscellaneous Expenses	7.30	7.18
Total	2182.32	2168.62

Resource Generation

Sales of Farm Produce	0
License Fee	0
Leave Salary and Pension Contribution	0
Interest Earned on Short Term Deposits	10.87
Income Generated from Internal Resource Generation	17.10
Miscellaneous Receipts	23.86
Total	51.83

Network projects on transgenic crops (NPTC)

	Allocation	Utilization
Capital	61	32.93
Travelling Allowances	0	0
Research and Operational Expenses	155	154.3
Total	216	187.23

Fund Received through Externally Funded Projects

Externally Funded Projects	612.20
Consultancy Projects	0.00
Total	612.20

Scientific Staff

Director
Dr. Ajit Kumar Shasany
National Professor B. P. Pal Chair
Dr. Nagendra Kumar Singh

Principal Scientist	Senior Scientist
Dr. Sarvjeet Kaur Dr. Anita Grover Dr. Rekha Kansal Dr. Sanjay Singh Dr. Jasdeep Chatrath Padaria Dr. Ram Charan Bhattacharya Dr. Debasis Pattanayak Dr. Pranab Kumar Mandal Dr. Pradeep Kumar Jain Dr. Kishor Gaikwad Dr. Sharmistha Barthakur Dr. Kanika Dr. Monika Dalal Dr. Tapan Kumar Mondal Dr. Ashish Kumar Dr. Prasanta Kumar Dash Dr. Rhitu Rai Dr. Vandana Rai Dr. Subodh Kumar Sinha Dr. Rohini Sreevathsa Dr. Anil Kumar Singh	Dr. S.V. Amitha Charu Rama Mithra Dr. Amolkumar U. Solanke Dr. Navin Chandra Gupta Scientist (S. S.) Dr. Nimmy M.S. Dr. Ramavatar Nagar Dr. Mahesh Rao Dr. Anshul Watts Dr. Deepak Singh Bisht Scientist Dr. A. Dinabandhu Dr. Sandhya Sharma Dr. Joshitha Vijayan Dr. Shbana Begam Dr. Yuvaraj Iyyapan

Chief Technical Officer	Technical Officer	Technical Assistant
Mrs. Seema Dargan Dr. Rohit Chamola Dr. Pankaj Kumar	Dr. Rampal Singh Niranjan	Mr. Deepak Kumar Rathore Ms. Rita Ms. Megha Mr. Anshul Kumar Verma

Administrative Staff

Senior Administrative Officer	Sr Finance & Account Officer	Assistant Administrative Officer
Mr. Sumit Singh	Mr. Rahul Kumar	Mrs. Sangeeta Jain Mr. Vipin Kumar
Personal Secretary	Assistant	Lower Division Clerk
Mr. Anoj Kumar Jain	Mr. Mohit Sikka Mr. Sudarshan Kumar Jha Ms. Nidhi Nailwal Mrs. Rekha Chauhan	Mr. Rajesh Kumar Pal Mr. Kunal Maan Mrs. Priyanka Nair Mr. Mittravesh Choudhari

Research Achievements

1. GENOME ANALYSIS IN CROP PLANTS
2. BIOPROSPECTING AGRICULTURALLY IMPORTANT UTILIZABLE GENES
3. NOVEL GENES AND BIOMOLECULES FOR PLANT AND HUMAN NUTRITION
4. GENETIC MODIFICATION OF CROP PLANTS FOR IMPORTANT TRAITS
5. NATIONAL PROFESSOR-B. P. PAL CHAIR

1. GENOME ANALYSIS IN CROP PLANTS



Functional Genomics of heat stress tolerance in Pigeon pea for identification of genes and pathways

Ramakrishna, Sandhya Sharma and Kishor Gaikwad*

Pigeonpea is one of the climate-resilient grain legumes. Though the optimum temperature for cultivated pigeonpea is ~ 25-35 °C, its wild relatives can grow in temperatures ranging between 18 and 45 °C. To gain insight into molecular mechanisms responsible for the heat stress tolerance in pigeonpea, one pigeon pea cultivar (*Cajanus cajan*) and two wild relatives, *Cajanus acutifolius*, and *Cajanus scarabaeoides* which differ in heat tolerance were subjected to heat stress at 42 ± 2 °C for 30 min and 3 h and transcriptome analysis was carried out. A total of 9521, 12,447, and 5282 transcripts were differentially expressed in *C. cajan*, *C. acutifolius*, and *C. scarabaeoides*, respectively (Fig.1.1). We observed that a significant number of genes undergo alternative splicing in a species-specific pattern during heat stress. The histochemical assay showed more lipid peroxidation in *C.*

acutifolius compared to other *Cajanus* species inferring the presence of higher quantities of polyunsaturated fatty acids in the plasma

membrane which might have led to severe damage of membrane-bound organelles and subsequently high electrolyte leakage during

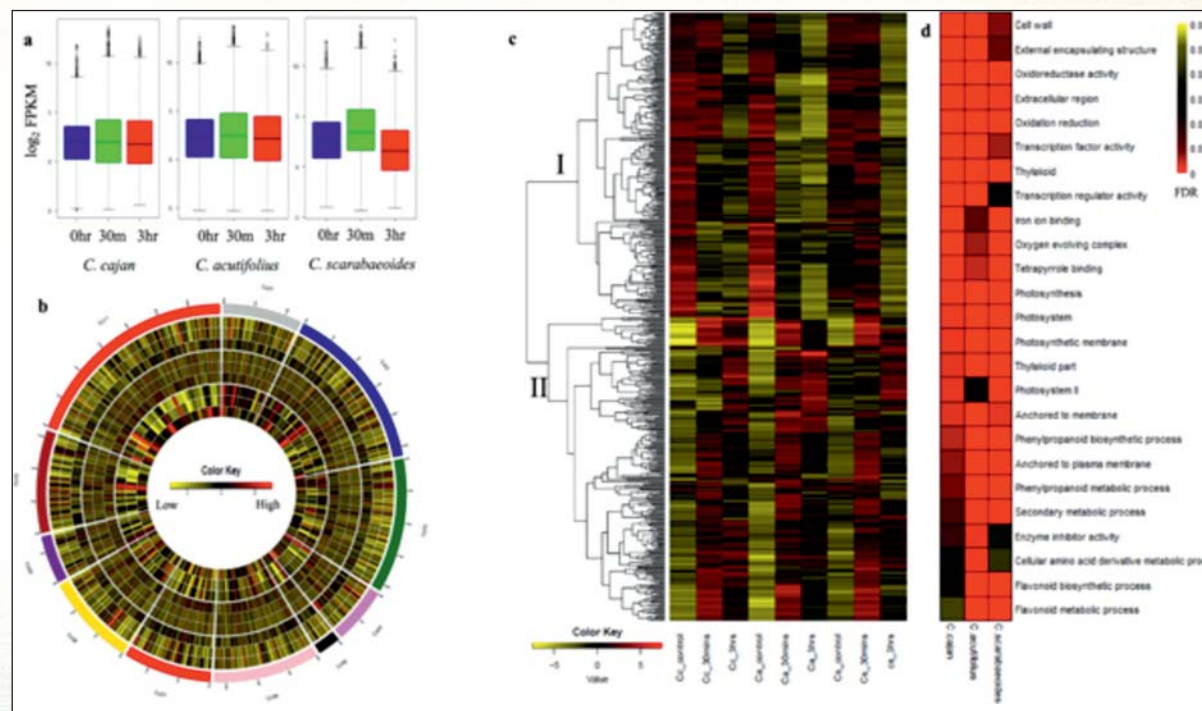


Fig.1.1: Expression profile of pigeonpea and its wild relatives during heat stress. a) Box plot showing expression levels of DEGs (log₂ FPKM) in pigeonpea and its wild relatives during heat. b) Circos plot showing global transcriptome expression of pigeonpea and CWRs during heat stress. The nine heatmap tracks (from inside-to-outside) representing expression levels of DEGs (log₂ FPKM) at Control, 30 min and 3 h of heat stress in *C. scarabaeoides* (316 DEGs), *C. acutifolius* (2913 DEGs) and *C. cajan* (1143 DEGs), respectively. c) Heatmap showing expression profile (log₂ FPKM) of 563 differentially expressed transcripts in all three pigeonpea species, cluster-I comprise down-regulated and cluster-II comprise up-regulated genes during heat stress. d) Heat map showing 25 highly enriched GOBPs in pigeonpea and CWRs during heat stress. The FDR value is the significance of the GOBPs enriched by DEGs in each *Cajanus* species

heat stress (Fig. 1.2). Our study revealed that *C. scarabaeoides* was the most thermotolerant followed by *C. cajanifolius*, *C. cajan*, and *C. acutifolius*. This study paves the way for the identification of candidate genes, which can be useful for the development of thermo-tolerant pigeon pea cultivars.

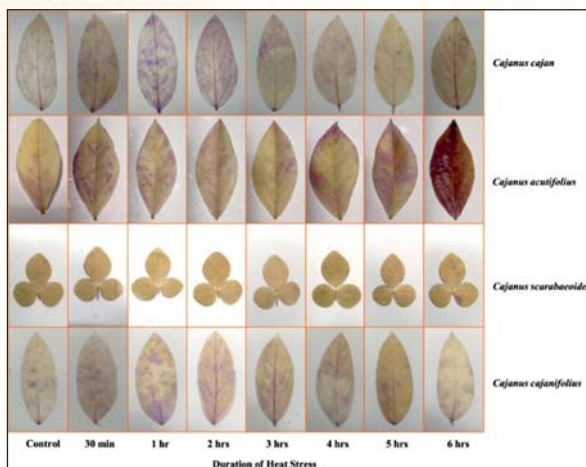


Fig.1.2: Histochemical assay of lipid peroxidation during heat stress in pigeonpea and CWRs with Schiff's reagent

Decoding the genome of Black gram

Kishor Gaikwad*

High quality DNA was isolated from black gram variety LBG 20, a popular variety suitable for all seasons and regions. It is also early maturing and high yielding in nature. Multiple genomic libraries of sizes ranging from 300-700 bp were prepared (Illumina) for deep sequencing on a X10 platform (Illumina). Similarly, another run of 10X Genomics was initiated for generating

data which will be used for scaffolding. The current data is being assembled and till now 558.22 Mb of black gram genome is organized in 3758 scaffolds with high N50 values of 48.01 Mb. About 524.23 Mb out of 558.22 Mb of black gram genome assembly is organized into 11 pseudo molecules/chromosomes.



Tissue specific expression analysis of MIR397 and MIR408 in ten different poaceae species

Payal Gupta and Prasanta K Dash*

MicroRNAs (MIR) are reported to play regulatory role in improvement of agronomic traits in crops. The over-expression of *OsMIR397* improved yield by promoting panicle branching and increasing grain size. Similarly, constitutive expression of *MIR408* improved biomass and seed yield in *Arabidopsis*. A regulatory microRNA in wheat *Tae-MIR408* is reported to be involved in regulating heading-time in wheat. To determine the variation in the responsiveness of *MIR397* and *MIR408* in ten different poaceae species, expression profile of *MIR397* and *MIR408* were generated in different tissues such as seedling, stem, root, leaf, flag leaf and panicle. Highest abundance of *MIR397* and *MIR408* was observed in the reproductive tissue i.e., flag leaf and lowest expression was

observed in vegetative tissues such as root tissue in case of *MIR397* and stem tissues in case of *MIR408*. The unusual high expression of *MIR397* was restricted to wild species of rice and was not observed among cultivated rice lines and other poaceae members (Fig. 1.3).

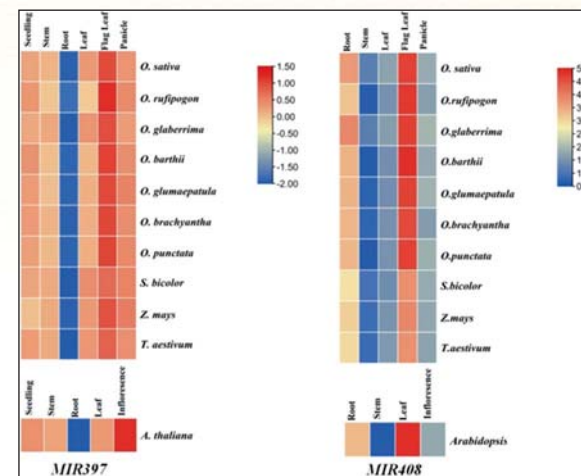


Fig. 1.3: Differentially expressed *MIR397* and *MIR408* across seven *Oryza* spp.; three Poaceae and one non-Poaceae spp. Expression pattern was obtained by qRT-PCR. Each block shows $\log_2$ -fold expression in tissue specific manner



Characterization of heat stress responsive genes in wheat (*Triticum aestivum* L.)

Jasdeep Chatrath Padaria* and Mawuli Azmeti

Terminal heat stress that is high temperature stress at flowering and grain filling stage is the

major constraint for wheat (*Triticum aestivum* L.) production. It is therefore imperative to understand the molecular networks involved in the plant heat stress response (HSR), and using the information generated to breed climate resilient wheat crop. Towards this, expression analysis of a novel gene, *TaSSRP* (Salt Stress Root Protein RS1) identified in transcriptome data generated in our laboratory, was analyzed under heat stress at post anthesis stage in 15 genotypes through qRT-PCR (Fig. 1.4). The gene showed differential expression in response to heat stress among the genotypes analysed. For detailed functional characterization, *TaSSRP* was isolated from heat tolerant wheat genotype Raj 3765 and cloned and sequenced. Tobacco transgenic plants harbouring *TaSSRP* were developed via *Agrobacterium* mediated transformation. Putative transgenic plants

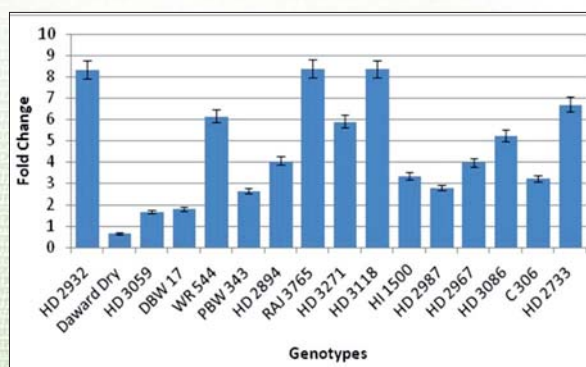


Fig 1.4: Expression analysis of *SSRP* (Salt Stress Root Protein RS1) under heat stress by qRT-PCR in wheat genotypes.

were screened by PCR and PCR positive transgenic plants were transferred to pots (Fig. 1.5).

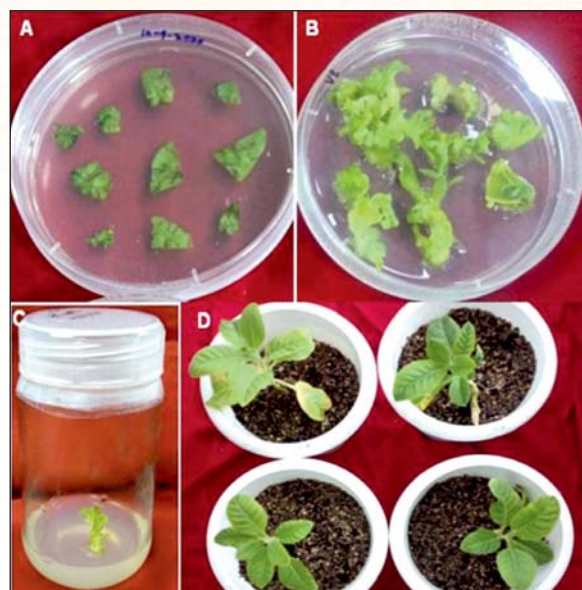


Fig 1.5: Different stages of transformation of *Nicotiana tabacum*: A) Leaf discs on co-cultivation media, B) Explants at callus stage, C) Putative transgenics on selection media, D) Transfer of PCR positive transgenics to pots



Molecular modeling of chitin deacetylase from *Helicoverpa armigera*

Yuvaraj Iyyappan* and Sarvjeet Kaur

Helicoverpa armigera, the most impactful pest of agriculture causes greatest damage

to cotton, tomatoes, maize, chickpeas and tobacco. The insecticides such as organophosphates, pyrethroids etc. are the major methods for *H. armigera* control. The extensive usage of these pesticides is an environmental concern and it has led to the development of pesticide resistance in this pest. It is therefore necessary to develop novel insecticides to manage the population of *H. armigera*. Chitin deacetylase (CDA) is a key enzyme involved in the modification of chitin and plays critical roles in moulting and pupation. Defects in the CDA genes lead to stunted insect development and even death. Therefore, CDA can be used as a potential pest control target. However, there are no effective pesticides known to target CDA. The enzyme chitin deacetylase from *H. Armigera* has been cloned in the Pet28a vector for expression of the protein. The structure of chitin deacetylase from *H. armigera* has been modeled through homology modeling (Fig. 1.6).

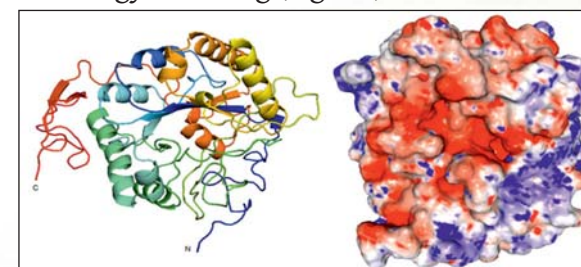


Fig. 1.6: Modeled Structure of chitin deacetylase from *Helicoverpa armigera* and the electrostatic potential surface of the structure.

Molecular modeling of Arginine kinase from *Chilo suppressalis*

Yuvaraj Iyyapan* Sarvajeet Kaur

The rice stem borer, *Chilo suppressalis*, is an important rice pest in India, China, Sri Lanka, eastern Asia, Japan, Taiwan and Malaysia and has evolved resistance to several classes of insecticides. It is necessary to develop novel insecticides to manage the population of this pest. Arginine kinase is a phosphagen kinase protein that plays a

central role in invertebrates by catalyzing the reversible transfer of a phosphoryl group. Defects in Arginine kinase gene has lead to larval mortality and delay the larval pupation and significantly retarded the larval development. The fact that arginine kinase is completely absent in mammals makes this protein a favorable target for rational drug design for the development of novel insecticides. The enzyme Arginine kinase from *chilo suppressalis* has been cloned in Pet28a vector for expression of the protein and the

structure of the protein has been modeled through homology modeling (Fig.1.7).

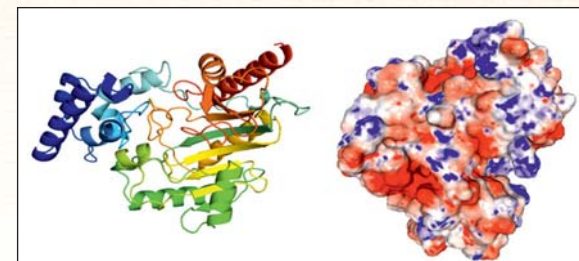


Fig. 1.7: Modeled Structure of Arginine kinase from *Chilo suppressalis* and the electrostatic potential surface of the structure.

2. BIOPROSPECTING AGRICULTURALLY IMPORTANT UTILIZABLE GENES



Chromosome level reference assembly of Indian Tea (*Camellia assamica* var. Masters cv. TV-1) genome

Hukam C. Rawal, Megha Rohilla, Abhishek Mazumder, Tilak Raj Sharma, Nagendra Kumar Singh, Tapan Kumar Mondal*

The genome size of an Indian tea (*Camellia assamica* var. Masters cv. TV-1) was determined as 3.0 Gb using a FACS cell sorting analysis. A chromosome-level reference genome assembly (2.93 Gb; 97.66% Coverage) was generated anchoring 99.40% of super-scaffold level assembly into 15 clusters or pseudomolecules by Hi-C data with the size of clusters ranging from 303.18 Mb to 119.95 Mb (Fig. 2.1A). Repetitive or transposable elements in this TV-1 genome accounted for a total of 2.07 Gb covering 70.61% of the assembled genome. The orthologous genes were predicted by OrthoMCL between TV-1, *Camellia sinensis* var. *assamica* (CSA), *Camellia sinensis* var. *sinensis* (CSS), and Kiwi fruit (Fig. 2.1B). Further, the

single-copy gene-based genome evolution has confirmed that divergence of TV-1 occurred approximately 2 MYA and 5.5 MYA from CSA and CSS, respectively (Fig. 2.1C). Whole-genome duplication (WGD) study has revealed the independent origin of TV-1 when compared with Kiwi fruit, CSS, and CSA. A total of 30,039 high-confidence protein-coding genes were predicted in the genome.

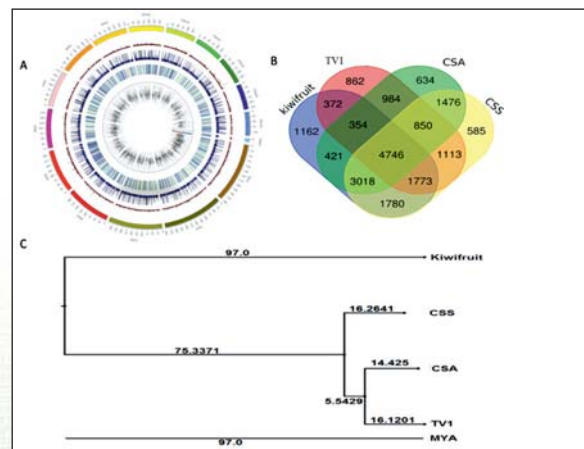


Fig. 2.1: Indian Tea Genome Study. (A) Chromosome-wise distribution of genes, SSRs, ncRNAs and InDels (compared to CSS genome), (B) Venn diagram showing orthologs and paralogs between TV1, CSS, CSA, and Kiwi fruit, (C) Single-copy genes based tree considering the evolution time between tea and KIWI fruit as 73-97 MYA (million years ago) (<http://www.timetree.org/>)

A first-generation Haplotype map (HapMap) of tea (*Camellia sinensis* L. O. Kuntz)

Nisha Singh, Hukam C. Rawal, Tilak Raj Sharma, Nagendra Kumar Singh, Tapan Kumar Mondal* [Singh *et al.* 2021, *Bioinformatics*]

Tea is a cross-pollinated woody perennial plant. Lack of genome-wide high-density SNP markers and genome-wide haplotype information has greatly hampered the utilization of tea genetic resources towards fast-track tea breeding programs. To address this challenge, a first-generation haplotype map of tea (Tea HapMap 1) was generated using standard methodology. Whole-genome re-sequencing data of 369 tea genotypes was used to generate 2,334,564 biallelic SNPs and 1,447,985 InDels (Fig. 2.2). Around 2928.04 million paired-end reads were generated with an average mapping depth of ~0.31X per accession. Identified polymorphic sites will be useful in mapping the genomic regions responsible for important traits of tea. These resources lay the foundation for future research to understand genetic

diversity within tea germplasm and effective germplasm utilization for breeding improved in tea varieties.

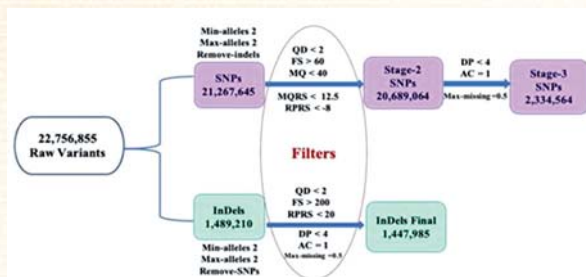


Fig. 2.2: Tea HapMap. Systematic representation of different filters applied to discover true sets of variants.

miRNA Prediction Methodology (miRPreM) and tRNA induced non-coding RNA Prediction Methodology (tiRPreM) for model and non-model organisms

Hukam C. Rawal, Tapan Kumar Mondal* [Rawal et al. 2021, *Brief. Bioinformatics*]

In recent years, microRNAs (miRNAs) and tRNA-derived RNA fragments (tRFs) have been reported extensively following different approaches of identification and analysis. We developed a benchmark methodology for the identification of miRNAs and tRFs each, termed as miRNA Prediction Methodology (miRPreM) and tRNA induced non-coding RNA Prediction Methodology (tiRPreM), respectively. The performance of these methodologies was

evaluated by using *Oryza coarctata*, a wild rice species as a case study for model and non-model organisms. With the organism-specific reference genome approach, 98 miRNAs and 60 tRFs were exclusively found. A high accuracy (13 out of 15) was observed when testing these genome-specific miRNAs in support of analyzing the data with respective organisms (Fig. 2.3). With such a strong impact of miRPreM, more than the double number of miRNAs (186) was predicted as

compared to the traditional approaches (79) and with tiRPreM, all known classes of tRFs were predicted within the same sRNA data. Moreover, the methodologies were presented in standard form to extend their applicability to different organisms rather than restricting them to plants. Both miRPreM and tiRPreM can fulfill the need for a comprehensive methodology for miRNA prediction and tRF identification, respectively for model and non-model organisms.

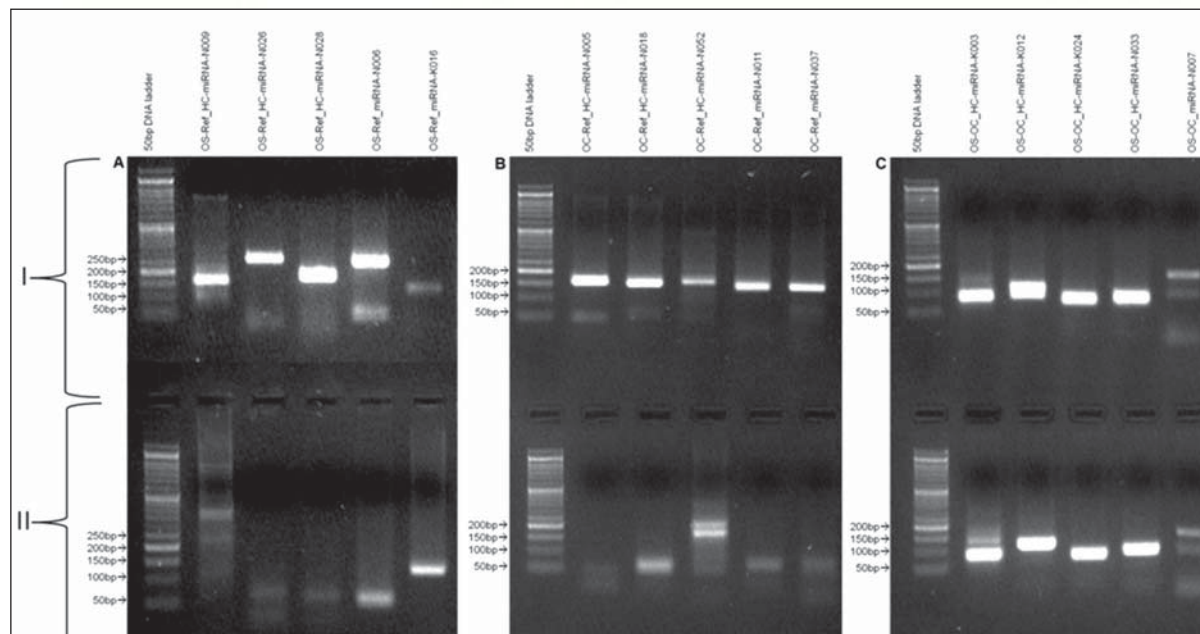


Fig. 2.3: Confirmation of organism-specific miRNAs in support of analyzing the data with the respective organism. (A) *O. sativa*-specific miRNAs for the presence in *O. sativa* DNA (upper panel; A-I) and absence in *O. coarctata* DNA (lower panel; A-II), (B) *O. coarctata*-specific miRNAs for the presence in *O. coarctata* DNA (upper panel; B-I) and absence in *O. sativa* DNA (lower panel; B-II) and (C) common miRNAs for the presence in both *O. sativa* DNA (upper panel; C-I) and *O. coarctata* DNA (lower panel; C-II).

Discovery of miRNA-lncRNA-mRNA module for stem elongation trait of deepwater rice

Alok Kumar Panda, Hukam C. Rawal, Jyoti Nishad, Soni Chowrasia, Tapan Kumar Mondal*

Deepwater rice (DWR) (*Oryza sativa* L.) variety is naturally grown in flood-prone harsh conditions and provides food to millions of people. This rice has a special adaptive character to elongate its internode rapidly to rescue from deep water flooding. The fast increase in elongation keeps the top portion of the leaf tip outside the water allowing the rice plants to efficiently photosynthesize and exchange gases for respiration. Micro RNAs (miRNAs) and Long non-coding RNAs (lncRNAs) are a class of non-coding RNAs that play an important role in gene regulation in

different biological processes. However, the role of miRNAs and lncRNAs in deepwater rice stem elongation has not been studied. To identify the miRNAs and lncRNAs in response to stem elongation, the uppermost internode and node tissues were sequenced after 14 days of deepwater stress treatment. A total of 291 and 58 miRNAs were found differentially expressed (DE) under deepwater treatment in internode and node tissues, respectively. Ten of these DE miRNAs were subsequently validated by stem-loop qRT-PCR. Further, a sum of 11335 and 9205 lncRNAs were identified from which 433 and 945 lncRNAs were found to DE in internode and node, respectively. Competitive endogenous RNA (ceRNA) study revealed that 69 and 27 lncRNAs were acting as endogenous target mimics (eTM) with the internode and node miRNAs, respectively. In the ceRNA study, miR1850.1, miR1848, and IN-nov-miR145 were up-regulated while miR159e

was down-regulated and their respective eTM lncRNAs and targets were found to have opposite regulation (Fig. 2.4A).

We have validated one of these modules (IN-nov-miR145-Cc-TCONS_00011544-Os11g36430.3) through transient expression in tobacco leaves. The tobacco leaves co-transformed with IN-nov-miR145-pSITE-4NB and ZFD-p1304-GUS showed a reduced GUS phenotype. However, the GUS expression again increased when co-transformed with TCONS_00011544-pEarlyGate201 and IN-nov-miR145-pSITE-4NB with ZFD-p1304-GUS confirming the possible role of lncRNA as a target mimic (Fig. 2.4B-C). In this novel attempt of identification and characterization of miRNAs and lncRNAs under deepwater stress in rice, we have provided a reference platform to study for the first time, interactions of these two non-coding RNAs with target genes through transient expression.

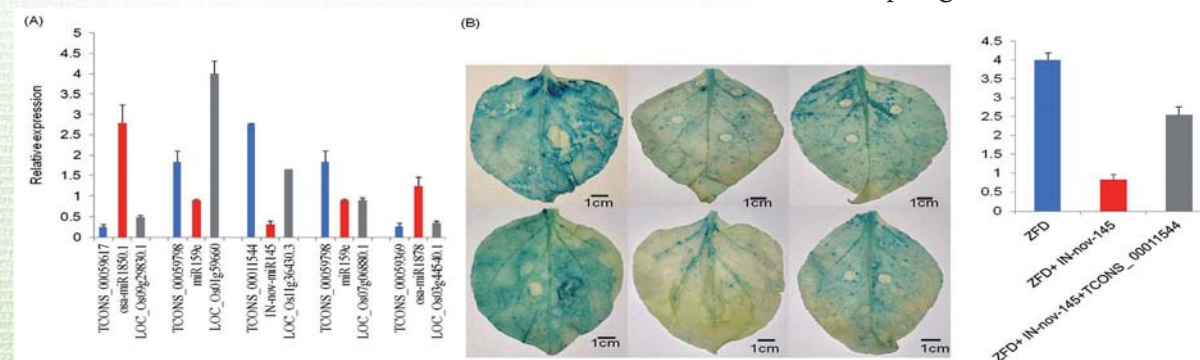


Fig. 2.4: A) qRT PCR of lncRNAs eTM of miRNAs and their corresponding target genes. B) The phenotype of GUS transient expression on tobacco leaves. C) RT-PCR of GUS genes of miRNA-lncRNA-mRNA module on tobacco leaves

Haplotype network of 94 diverse deep-water rice of Assam using eQTLs regulating transcript for internode elongation under deep water stress

Megha Rohilla, Nisha Singh, Tapan Kumar Mondal*

Several economically important genes of rice governed various major traits, but their allelic (haplotype) combinations remain undescribed.

Stem elongation and internode elongation underwater is a unique trait that is specific to deep-water rice. In this study, a haplotype network of 18 expression quantitative trait loci (eQTL) related to internode elongation and one anaerobic germination tolerance-related gene have been investigated among 94 deepwater rice landraces of Assam (Fig. 2.5A). This network analysis categorized all the 94 genotypes in different haplotype groups on basis of specific SNPs in eQTLs. Our investigation showed a maximum of 16 haplotypes for the eQTL named *SET protein 39*

with 42 SNP sites and the lowest number of 3 haplotypes for 5 different eQTLs (Fig. 2.5B). Maximum haplotype diversity in *SET protein 39* (*Os10g36250*) gene was found to be 0.8094 and lowest in *Annexin repeat, conserved site domain-containing protein* (*Os03g17634*) i.e., 0.0423 indicating that our genotypes are genetically much more diverse. Overall, this study guides the possibility for developing improved rice varieties with superior haplotype combinations of target genes enhancing rice yield under deep-water conditions through haplotype-based breeding.

Development of miniature inverted-repeat transposable elements and intron length polymorphic markers in tea

Megha Rohilla, Abhishek Mazumder, Shbana Begum, Tapan Kumar Mondal*

A large proportion of repeat sequences in the tea genome facilitates the discovery of numerous repeat-based DNA markers. We have employed a comparative genomics approach to mine tea genomes of *Camellia sinensis* var. *assamica* (CSA) and *Camellia sinensis* var. *sinensis* (CSS) to identify markers to differentiate tea genotypes. Forty-three and 60 *Camellia sinensis* Miniature Inverted-repeat Transposable Elements (CsMITE) families were identified which yielded 23170 and 37958 putative CsMITE sequences, respectively in these two different genomes. Further, we also reported 4912 non-redundant, *Camellia sinensis* Intron Length Polymorphic (CsILP) markers of which 85.8% were common in both the CSS and CSA tea genomes. To validate, a subset of randomly sampled 25 markers from CsMITEs and 33 from CsILP were employed for a polymorphism survey among the 36 highly diverse tea genotypes. Approximately, 45 percent of the CsMITE and CsILP markers were found to be highly polymorphic with clear and distinguishable alleles, which suggested the usefulness of the primers in germplasm characterization (Fig. 2.6a & b). These genome-

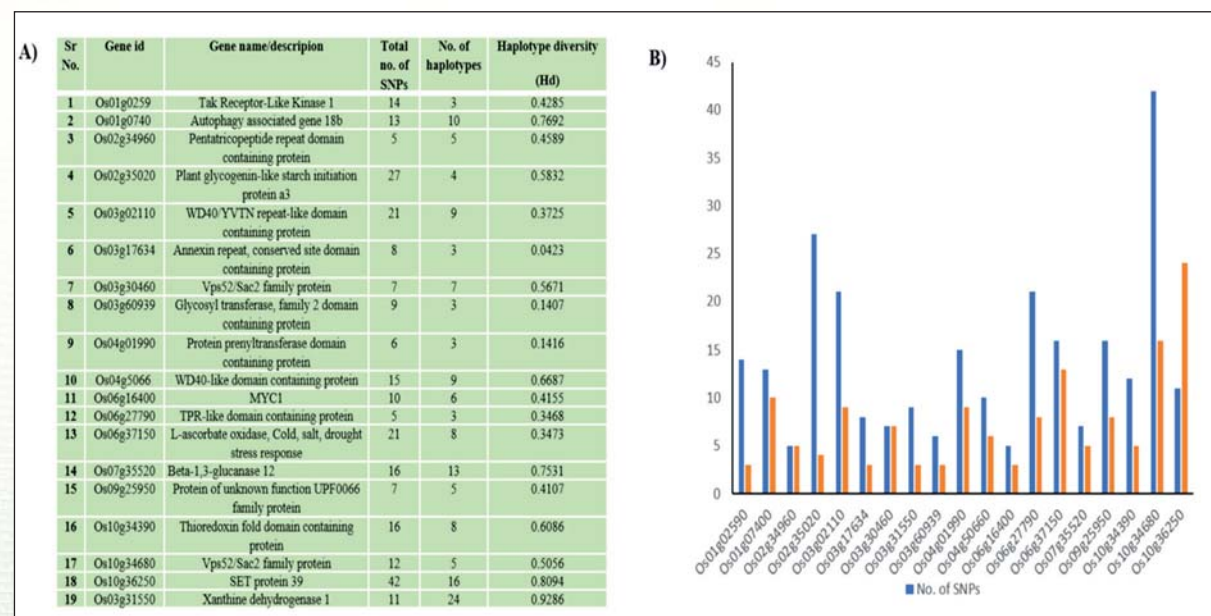


Fig. 2.5: A) Details of genes with number of SNPs and haplotypes used for haplotype network analysis, B) Gene wise distribution of the number of SNPs and number of haplotypes.

wide markers, which were reported for the first time in tea, will be a valuable resource for inclusion in a genetic diversity analysis as well as marker-assisted breeding of tea genotypes for quality improvement.

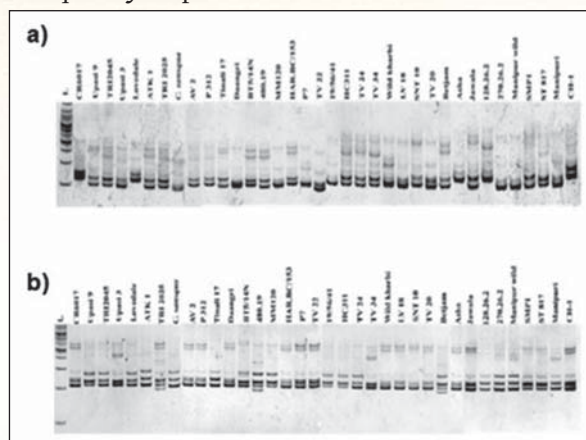


Fig. 2.6: a) PCR products amplified by CsMITE showing 4 alleles in 6% PAGE gel and b) PCR products amplified by CsILP marker showing 6 alleles in 6 % PAGE gel.

Down-Regulation of *Rhizoctonia solani* endogenous genes by Host-Delivered RNA interference to enhance Sheath Blight disease resistance in Rice

Ila Mukul Tiwari, Tapan Kumar Mondal*

Transgenic rice plants were developed using host-delivered RNA interference against sheath blight disease. Putative transgenic plants were screened by PCR using gene-specific and hygromycin-specific primers. For analyzing the tolerance of transgenic rice

against sheath blight disease, plants were inoculated with *R. solani* fungal strains using detached leaf assay. Observations were taken at 24, 48, and 72 h post-inoculation. Transgenic plants performed better as compared to wild type on the basis of parameters like lesion size, number of lesions, leaf area covered under lesion, and percentage of disease incidence (Fig. 2.7). Further analysis to check the relative expression of target pathogenicity gene post-inoculation will be carried out.

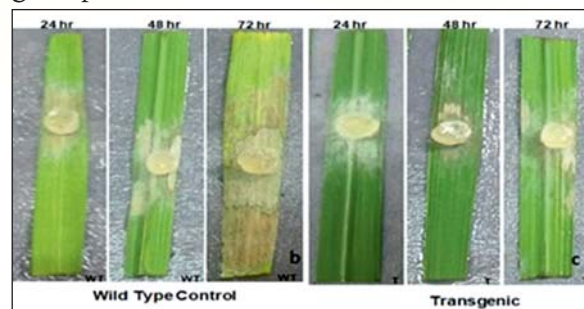


Fig. 2.7: Detached leaf assay showing the progression of sheath blight disease at different hours post-inoculation in control and transgenic plants

Investigation of the plant U-Box family of proteins (PUB) in rice (*Oryza sativa*) with emphasis on their roles in abiotic stress responses

Harmeet Kaur, Akansha Chaurasia, Tapan Kumar Mondal

A member of plant U Box ligases, PUB63 which is an early heat-inducible gene was

cloned into the plant expression vector pEG201 and sequenced. The construct was transferred into *Agrobacterium* and transgenic *Arabidopsis* plants were developed by floral dip method. Transgenic (T_2) seeds were given heat treatment at 50°C for 60 min in the water bath and then seeds were planted and grown at 22°C (Fig. 2.8 B); seeds grown without any treatment were used as controls (Fig. 2.8A). Seven-days-old seedlings grown on ½ MS were transferred to 45°C for 2h and then transferred back to 22°C for recovery. The transgenic lines showed better tolerance towards heat stress as compared to wild type Col-0 plants. For germination level heat treatment, the Col-0

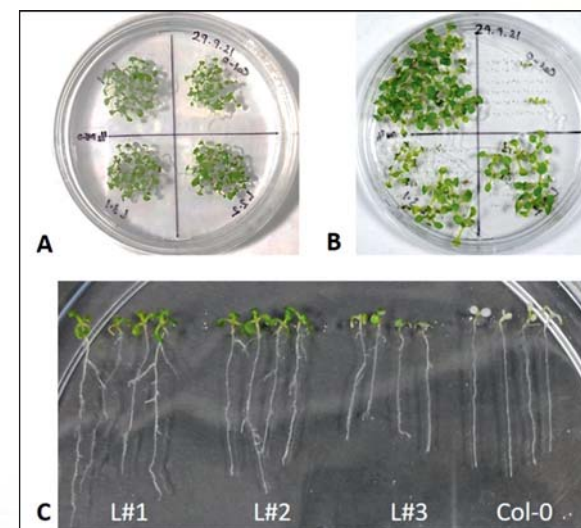


Fig. 2.8: PUB63 over expression improves heat stress tolerance in Arabidopsis transgenic lines at germination (A & B) and seedling stage (C).

seeds failed to germinate after heat stress whereas the transgenic lines over-expressing *PUB63* were able to grow well during the recovery (Fig. 2.8 C). Similarly, at the seedling stage also the Col-0 seedlings were not able to withstand the heat stress at 45°C and were completely bleached out whereas the transgenic lines displayed better growth and less chlorophyll bleaching during such stress.



Functional characterization of TALEffectors in Indian *Xanthomonas oryzae* pv. *oryzae* (Xoo) strain to decipher the outcome of their interaction with rice

Rhitu Rai*

Considering the determinative role of TAL effectors and their targets in rice-*Xanthomonas* interaction, the whole genome sequence of a xa13 compatible Indian Xoo strain RR2 was generated previously, to delineate complete sequences of all *tal* genes present in it. The sequence derived Repeat Variable Diresidues (RVDs) had revealed RR2 to harbor three major TALEffectors, one being an ortholog of *PthXo3* and the other two, orthologs of *PthXo2*. Interestingly, both the *PthXo2* orthologous genes harbor an aberrant repeat of 36 amino acids instead of the normal 34 amino acids.

Since aberrant repeats of 42 amino acids have been reported to shift the reading frame of TALEffector on the effector binding element on the host promoter region, we hypothesized that the *PthXo2* variants may also shift the reading frame and might induce the SWEET13 resistant allele of japonica rice. The three *tal* genes *tal2b*, *tal6b*, and *tal7* cloned in Xoo strain PXO99A_ME2 were characterized by gain of function assay. For checking their impact by loss, a *tal* mutant library with 10X coverage was generated and the library was screened for identifying individual mutants in the three major *tal* genes. Except for *tal6b*, the mutants for 16 of twenty *tal* genes were identified. The

$\Delta tal2b$ and $\Delta tal7$ mutants along with respective clones were tested for expression and function by cognate host gene induction profiling and infection assay on both indica and japonica rice varieties (Fig. 2.9). The *PthXo3* ortholog *tal2b* induced SWEET14; one of the two *PthXo2* orthologs harboring two aberrant repeats did induce SWEET13 in japonica and rendered it susceptible, whereas the other *PthXo2* ortholog functioned only to infect indica rice (Fig. 2.9). The results highlight fast evolution adapted by Xoo strains under selection pressure by accommodating more than one major TALEffectors targeting different targets in rice.

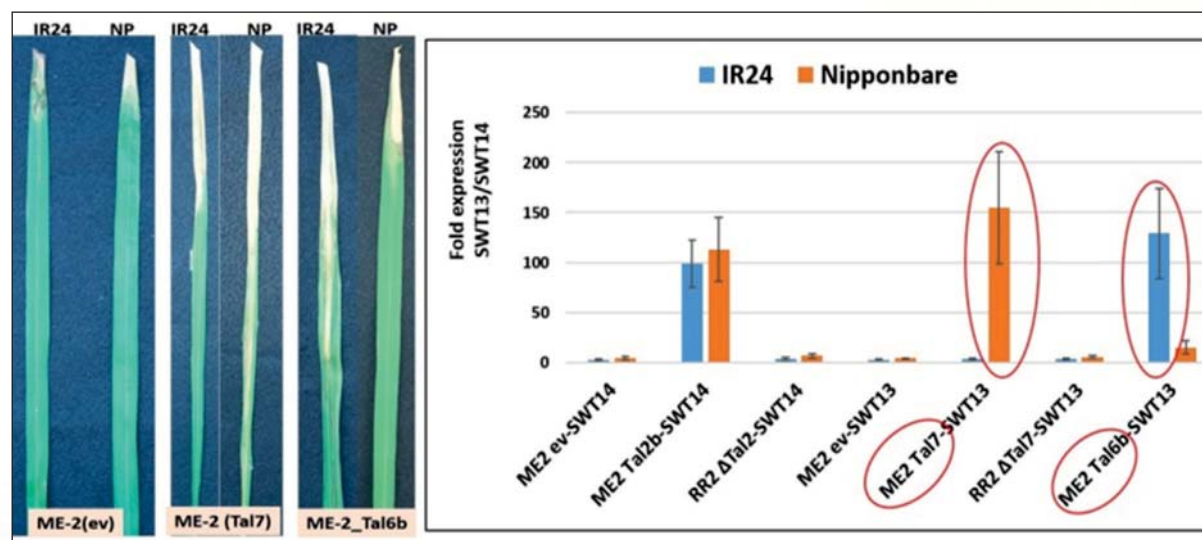


Fig. 2.9: Disease progression elicited by Xoo TAL effectors and expression profiling of cognate SWEET genes in rice.



Identification and cloning of genes for salinity tolerance in wheat

Kanika* and Ankita Vilasrao Chinche

Kharchia Local, a well-known salt tolerant Indian land race is a very useful genetic resource which can be used to improve our understanding on mechanisms of salinity tolerance. This year few RILs of mapping population derived from HD 2851 X Kharchia 65 were phenotyped under salt stress conditions in hydroponics. Various physio-biochemical, growth parameters of root traits were studied. It was found that reduction in relative water content and Membrane Stability Index was less in Kharchia 65 and RILs as compared to HD 2851. Salinity caused a decrease in primary root length, root surface area, root volume, and total root length in Kharchia 65 and most of the RILs as compared to HD 2851. The total dry weight of the Kharchia 65 and RILs decreased to less than that of HD 2851 following salt treatment. Dry matter partitioning to roots improved in Kharchia 65 and RILs but decreased in HD 2851. Compared to HD 2851, Kharchia 65 and RILs showed less reduction in average growth rate, relative growth rate,

and net assimilation rate. HD 2851 had a lower Na^+/K^+ ratio in roots than in leaves, whereas Kharchia 65 and RILs had a lesser Na^+/K^+ ratio in leaves than in the roots. The expression analysis of a few salts stress-responsive

genes was also carried out in selected RILs under salt stress conditions. Salt treatment induced upregulation of genes *TaNHX1* and *TaAVP1* was observed in root and shoot of one RIL-5 (Fig. 2.10).

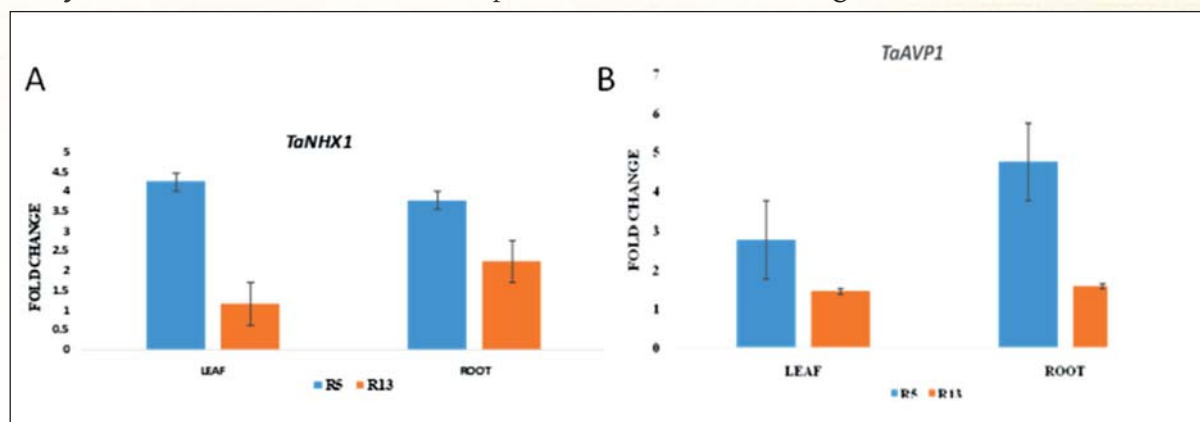


Fig. 2.10: Expression analysis of salinity-related genes A) Na^+/H^+ antiporter (*TaNHX1*) and B) Vacuolar pyrophosphatases like AVP1 (*TaAVP1*) genes from leaf and root samples of RIL-5 and RIL-13 after 14 days of treatment. The value represents mean $\pm$ SEs.



Genome-wide identification of MBF1 (Multi protein bridging factor1) from bread wheat

Mahammed Ashiq and Sharmistha Barthakur*

The Multiprotein Bridging Factor 1 (MBF1) proteins are transcription co-factors/activators, associated with TATA box-binding proteins and form a bridge between transcription factors and the basal

transcription machinery of stress-responsive genes. MBF1 genes are present in most Archaea and all eukaryotes and are involved in developmental processes and various stress responses. A genome wide analysis of MBF1 genes was carried out in Bread wheat (*Triticum aestivum*) which led to identification of 3 homologs. These three *TaMBF1* homologs (9 homeologs) were localized to chromosomes 2, 3, and 7, respectively. To understand the phylogenetic relationship among MBF1 proteins across plant species, a phylogenetic tree was constructed from selected 19 plant

species (Fig 2.11). The MBF1 proteins were grouped into 15 major clades. All the nine *TaMBF1* genes were found to be distributed in different clades of a phylogenetic tree. TraesCS2A02G389000, TraesCS2B02G407000, TraesCS2D02G386800 genes were grouped together in subclade of largest clade in the phylogenetic tree. TraesCS3A02G135300 and TraesCS3D02G151300 were grouped together

in one separate clade in the phylogenetic tree. TraesCS3B02G169100 gene present in the clade which contained TRIUR3_11900 gene from *Triticum Urartu*. TraesCS7A02G369500 TraesCS7D02G353800 genes were present in clade which contained genes from *Aegilops tauschii*. TraesCS7B02G259000 gene was present in a clade containing genes from *Hordeum vulgare*.

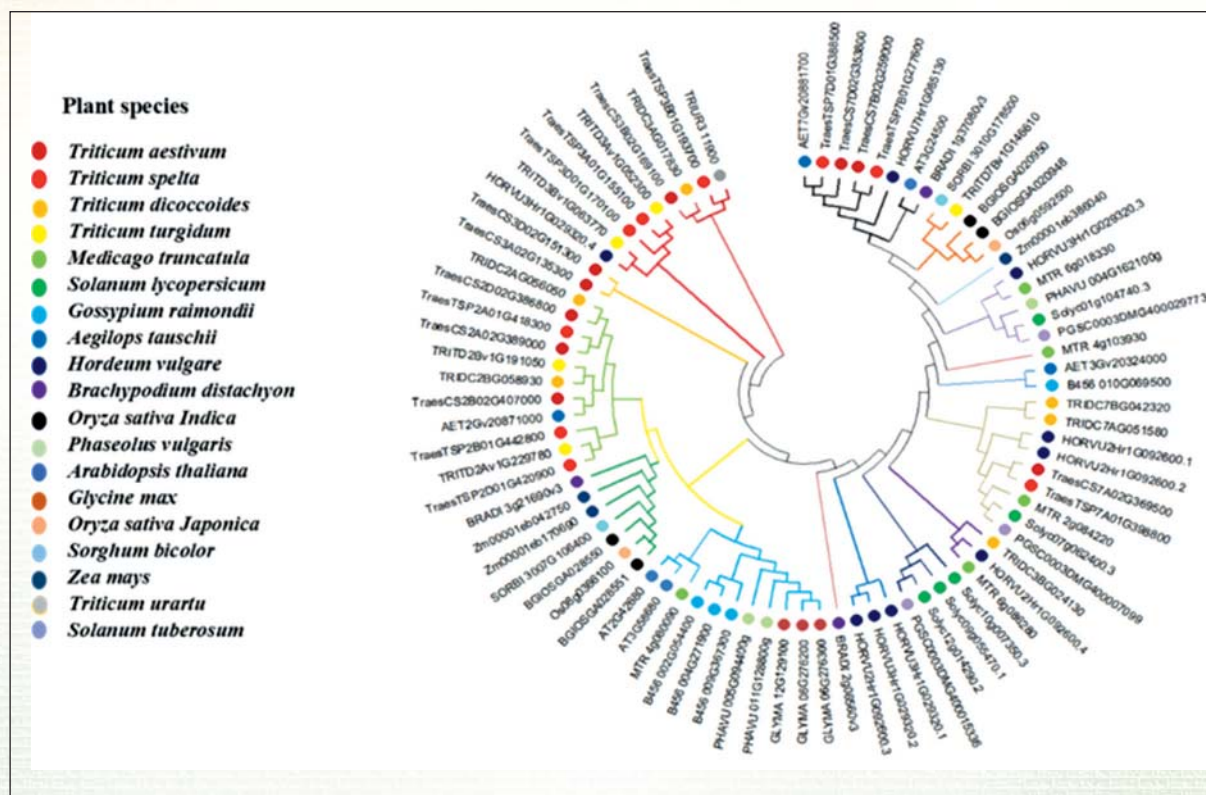


Fig. 2.11: Phylogenetic tree of the MBF1 genes from wheat and other plant species.



Genome-wide identification of Ran GTPase family genes from wheat (*T. aestivum*) and their expression profile during developmental stages and abiotic stress conditions

Soham Choudhury, Mansi, Senthilkumar K. Muthusamy, Jasdeep C. Padaria and Monika Dalal* [Choudhury *et al.* 2021. *Funct. Integr. Genomics*]

Maintenance of growth is important for sustainable yield under stress conditions. Hence identification of genes involved in cell division and growth under abiotic stress is of utmost importance. Ras-related nuclear protein (Ran) is a small GTPase required for nucleocytoplasmic transport, mitotic progression, and nuclear envelope assembly in plants. In the present study, two Ran GTPase genes *TaRAN1* and *TaRAN2* were identified through genome-wide analysis in wheat (*T. aestivum*). Comparative analysis of Ran GTPases from wheat, barley, rice, maize, sorghum, and *Arabidopsis* revealed similar gene structures within phylogenetic clades. In qRT-PCR analysis, *TaRAN1* showed significantly higher expression as compared to *TaRAN2* in shoot and root at seedling,

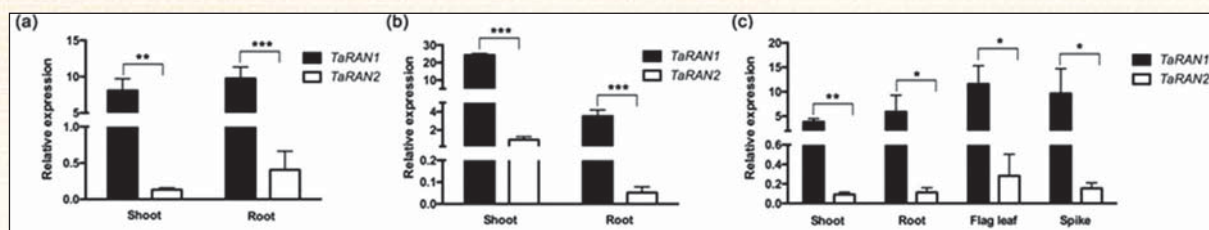


Fig. 2.12: Expression profile of *TaRAN1* and *TaRAN2* at different developmental stages in wheat (*Triticum aestivum*). Expression at (a) seedling (7-day-old seedlings); (b) vegetative (25-day-old plants), and (c) reproductive stage (full spike emergence). Error bars represent the SE of three biological replicates. Asterisks indicates significant difference between expression of *TaRAN1* and *TaRAN2* (* $P < 0.05$, ** $p < 0.001$, *** $p < 0.0001$).

vegetative and reproductive stages (Fig. 2.12). During progressive drought stress, the steady-state level of transcripts was maintained to that of control in roots under drought stress. The analysis suggests differential regulation of *TaRAN* genes under developmental stages and abiotic stresses. Delineating the molecular functions of Ran GTPases will help unravel the mechanism of stress-induced growth inhibition in wheat.



Transcriptional analysis of differentially induced proteins in *Alternaria* infected samples from *B. juncea*, *S. alba*, and *C. sativa*

G Prakash, Mahesh Rao, Rekha Kansal, Anita Grover*

A total of 40 differentially induced proteins were identified from *Alternaria* infected *B.*

juncea, *S. alba*, and *C. sativa* tissues through MALDI-TOF analysis. Among these 40

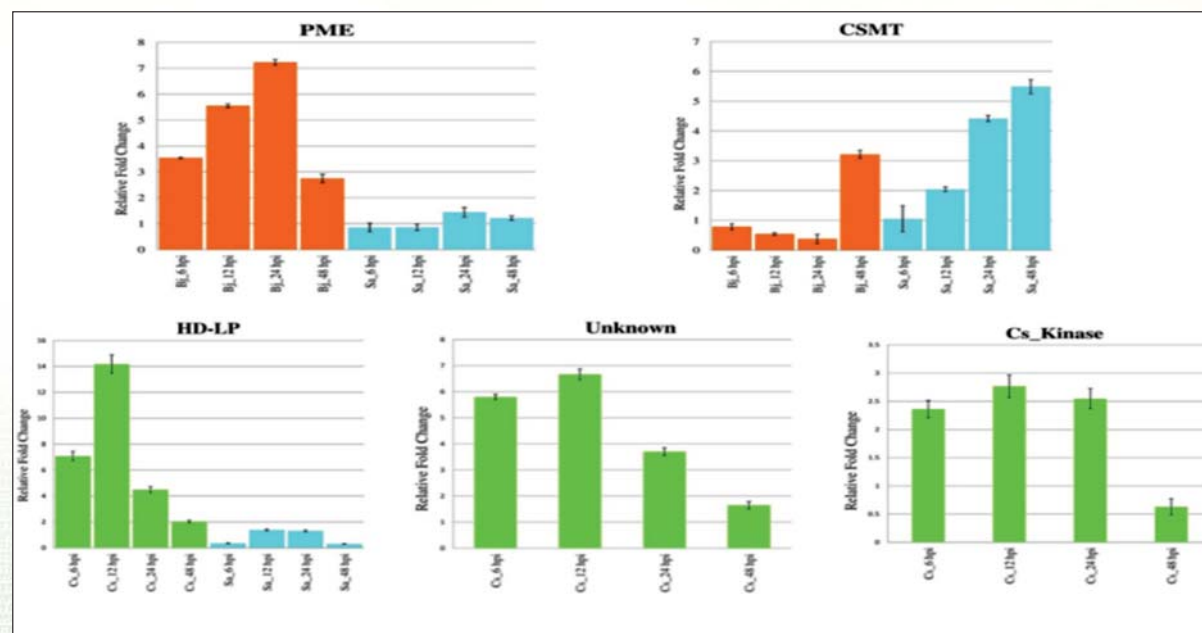


Fig. 2.13: Differential gene expression of *PME* and *CSMT* genes (in *B. juncea* and *S. alba*), *Homeodomain-like protein* (*HD-LP*, in *C. sativa* and *Sinapis alba*), *Unknown* and *Cs_Kinase* genes in the leaves of *C. sativa* after *A. brassicae* infection at 6, 12, 24, and 48 hpi. Values are mean of three replicates $\pm$ SD and indicate a significant difference between control and infected leaves at $p < 0.05$.

differentially expressed proteins, fifteen proteins showed significant up or down-regulation. Three of them have already been shown to be involved in disease resistance. Four of the proteins (*PME*, *CSMT*, *HDL*, one unknown gene, and *Cs-Kinase*) which are differentially regulated in *B. juncea*, *S. alba*, and *C. sativa* after *Alternaria* infection were studied at gene expression level (Fig. 2.13)



Use of insecticidal genes from legumes for developing aphid resistant mustard: Molecular analysis of F2 *B. juncea* with increased aphid resistance developed using gene pyramiding approach

Rekha Kansal*

Use of plant lectin and protease inhibitor genes to control aphids have been recommended as an alternative to pesticide splash and inbuilt resistance. Lectin and protease inhibitors have different modes of action. The ingested lectin bound to the midgut tract causes disturbance of the epithelial cells while protease inhibitors are polypeptides that quardary to insects' mid-gut proteolytic proteins, causing a decrease in the accessibility of amino acids, hindering the growth and development of insects. Pyramiding of these two genes will broaden the resistance spectrum of the crop. Towards this, transgenic *Brassica juncea* lines harbouring chickpea lectin gene was used as a female parent and crossed with transgenic *Brassica juncea* line with Urd bean protease inhibitor gene (male parent) and about 400 F1 seeds were collected. These seeds were sown and selfed to get the F2 seeds and were analyzed

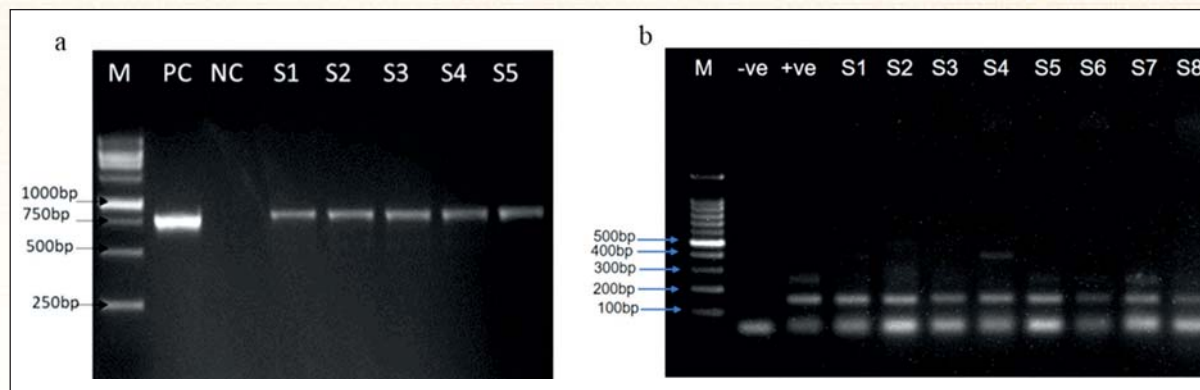


Fig. 2.14 Agarose gel electrophoresis of amplified (a) Chickpea lectin and (b) Urdbean protease inhibitor gene present in the *B. juncea* developed using the gene pyramiding approach.

for the presence of both the genes i.e Chickpea Lectin (CHPL) (Fig.2.14a) and Urdbean Protease Inhibitor (UPI) (Fig.2.14b). Total 15 plants were found positive for both the genes suggesting successful gene pyramiding.



Identification and utilization of genes and pathways for developing stem rot resistance in Indian mustard (*Brassica juncea*)

The Sclerotinia stem rot (white mold) disease of oilseed-*Brassica* caused by *Sclerotinia sclerotiorum* (Lib.) de Bary is one of the major biotic stress factors causing yield losses. This report summarizes the research-based findings of the *in vivo* expression of the predicted *S. sclerotiorum* effector candidates

(ECs) and identification of differentially expressed genes in susceptible and tolerant lines of *B. juncea* after *S. sclerotiorum* inoculation.

a) Expression analysis of the effector proteins predicted in *Sclerotinia sclerotiorum* ESR-01 isolate

Shaweta Arora, Sunita Yadav, Dwijesh C Mishra, Pankaj Sharma, Mahesh Rao, Navin C Gupta*

The genome-wide analysis of the secretory proteins in the whole genome of an Indian isolate 'ESR-01' of *S. sclerotiorum* revealed a total of 57 effector candidates from the 1006 secretory proteins. Out of the 57 effectors, 27 were annotated whereas the remaining 30 were unannotated. These effector candidate proteins were functionally diverse which included hydrolases, kinases, PgPi-

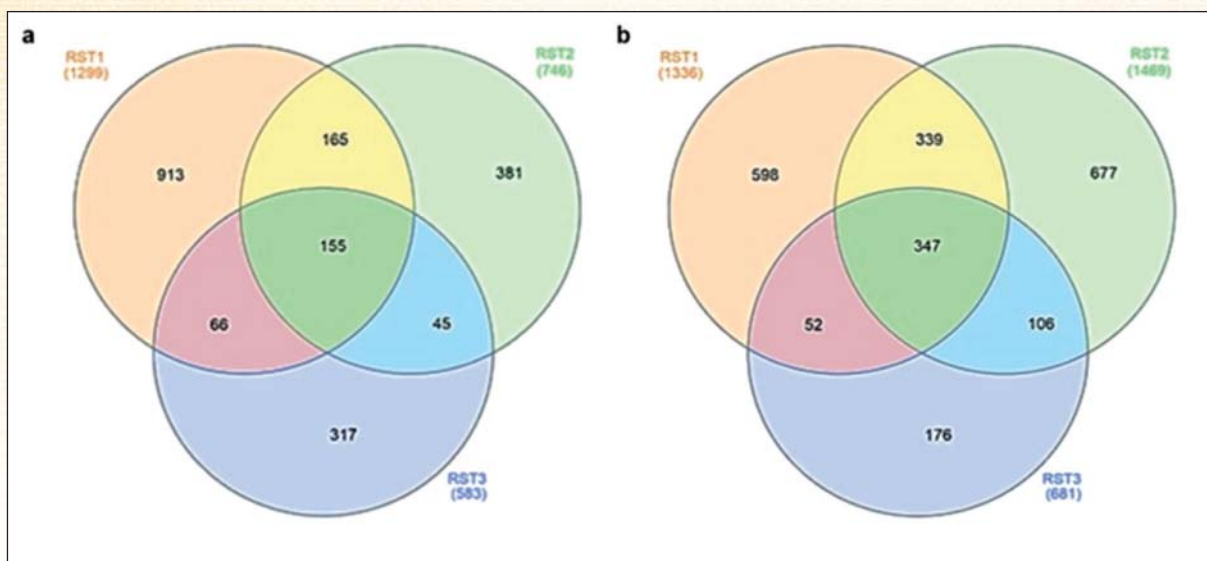


Fig. 2.16: The relative differentially expressed genes (RDEGs) of the susceptible and tolerant Brassica lines after 24, 48, and 96 hours of *S. sclerotiorum* inoculation. (a) Up-regulated, and (b) Down-regulated RDEGs (RST1 Vs RST2 Vs RST3).



Effect of drought stress on morphological, biochemical, physiological traits and expression analysis of microRNAs in drought-tolerant and sensitive genotypes of chickpea

Lalbahadur Singh, Deshika Kohli, Kishor Gaikwad, Rekha Kansal, Anil Dahuja, Vijay Paul, Chellapilla Bharadwaj, and Pradeep Kumar Jain*

The morphological and physiological characters of two chickpea genotypes Pusa362

(drought-tolerant) and SBD377 (drought susceptible) were analysed under drought stress. There was a severe reduction in biomass of chickpea plants subjected to drought (Fig. 2.17). However, there was not much difference in the root volume(s) suggesting a drought tolerance mechanism other than the root traits in Pusa 362 (drought-tolerant) as compared to SBD377 (drought-sensitive) genotype. The proline accumulation was significantly higher in Pusa 362 as compared to SBD377. miRNAs have been shown to regulate gene expression during drought stress conditions. Among the 16 validated miRNAs, expression of miR167, miR168, and miR171 showed

significant upregulation (>3-fold) in root tissue of drought-tolerant genotype (Fig. 2.18). These miRNAs target auxin response factors, WD-repeat, and scarecrow-like transcription factors, respectively which are known to play important roles in drought stress in plants suggesting a direct role of these miRNAs during drought tolerance in chickpea. miR390 and miR2118 were up-regulated in shoot samples in Pusa 362. Among the novel miRNAs, nov_miR8 in root and nov_miR2 in



Fig. 2.17: Effect of drought stress on the morphological characters of two chickpea genotypes: Pusa 362, drought-tolerant, and SBD377, drought susceptible. Forty five days old plants were subjected to drought stress of 45 days: A. Variation in plant height B. Variation in root length and C. Variation in leaf morphology (C: Control; S: Drought stress). Please note that SBD377 is a genotype with simple leaf.

shoot tissue showed maximum expression in Pusa 362. Nov_miR2 targets GMP synthase and nov_miR8 targets gene encoding laccase. GMP synthase is involved in the synthesis of purine nucleotides which among other functions play a key role as a secondary messenger in signal pathways activated during stress conditions. These drought-responsive miRNAs are likely to provide novel insights into post-transcriptional gene regulation under drought stress conditions in chickpea at the molecular level.

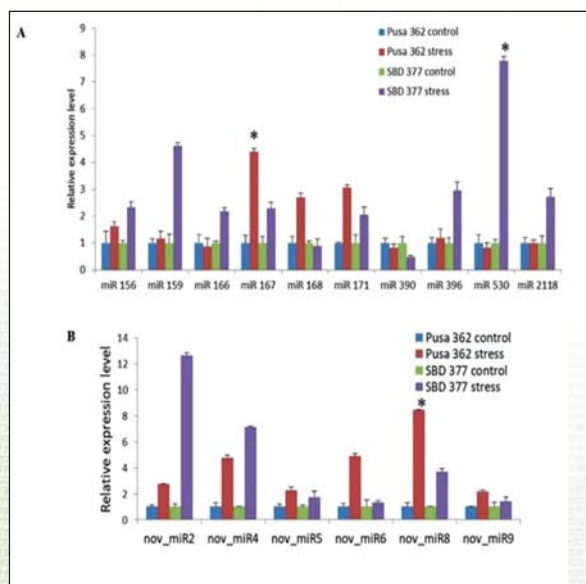


Fig. 2.18: Expression analysis of selected miRNAs in root tissues of chickpea genotypes- Pusa362, drought-tolerant and SBD377, drought susceptible: A) Conserved and B) Novel miRNAs. * Indicates statistically significant change ($p < 0.05$) in expression.

Gall-specific promoter, an alternative to CaMV35S, drives *Mi-msp2* gene and confers effective nematode resistance using HD-RNAi

Ila Joshi, Deshika Kohli, RC Bhattacharya, Ashok Chaudhury, Anil Sirohi, and Pradeep K Jain*

Root-knot nematodes (RKNs) are one of the major obligate plant parasites causing massive economic losses to crops. Targeting major nematode parasitism genes via HD-RNAi to confer silencing is established as one of the most effective approaches to curb nematode infection. Utilizing nematode-responsive root-specific (NRRS) promoters to design a dsRNA construct to hamper nematode parasitism. In this study, a previously validated peroxidase gall specific promoter, pAt2g18140, from *Arabidopsis* was employed to express the dsRNA construct of the nematode effector gene *Mi-msp2*. *Arabidopsis* RNAi lines of *CaMV35S::Mi-msp2* and *pAt2g18140::Mi-msp2* were compared with control plants to assess the

decrease in *Meloidogyne incognita* infection. The maximum reductions in the numbers of galls, females, and egg masses in the *CaMV35S::Mi-msp2* lines were 61%, 66%, and 95%, respectively, whereas, for the *pAt2g18140::Mi-msp2* lines, these were 63%, 68%, and 100%, respectively. The reduction in transcript level ranged from 79%-82% for *CaMV35S::Mi-msp2* and 72%-79% for the *pAt2g18140::Mi-msp2* RNAi lines. A reduction in female size and a subsequent reduction in next-generation fecundity (Fig. 2.19) demonstrate the efficacy and potential of the gall specific promoter pAt2g18140 for utilization in the development of HD-RNAi constructs against RKN, as an excellent alternative to the *CaMV35S* promoter.

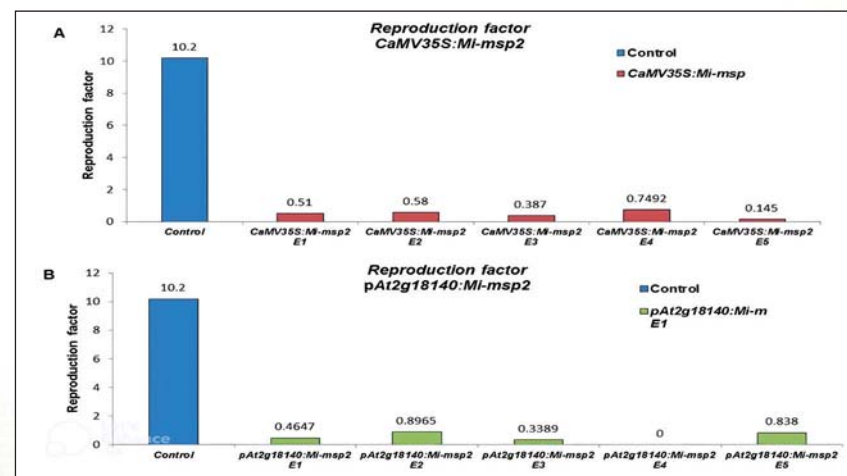


Fig. 2.19: Fecundity of *M. incognita*. A) Reproduction factors for the control and *CaMV35S::Mi-msp2*, B) Reproduction factors for the control and *pAt2g18140::Mi-msp2*. Each bar represents the mean $\pm$ SE. Double asterisks indicate statistically significant differences according to one-way ANOVA and Tukey's test ($p \leq 0.01$).



Development of GUI based Pipeline to extract the chloroplast genome from Illumina short reads

Shbana Begam*, Samarth Godara, Tapan Kumar Mondal

Chloroplast genomes (cpDNA) are relatively conserved among land plants in terms of size, structure, and gene content. The chloroplast genome harbours essential genes and these are widely used in biotechnology and studies of evolution and ecology. Assembling the chloroplast genomes can be challenging because of their structure that contains a pair of long inverted repeats (10–30 kb). Chloroplast genome extraction can be done with the available biotechnological kit which is costly and tedious. Hence, a computational approach has been attempted to extract the chloroplast genome from Illumina short reads data. A pipeline is designed to extract and reconstruct chloroplast genomes from whole-genome Illumina shorts reads data to reduce the cost and time (Fig. 2.20)

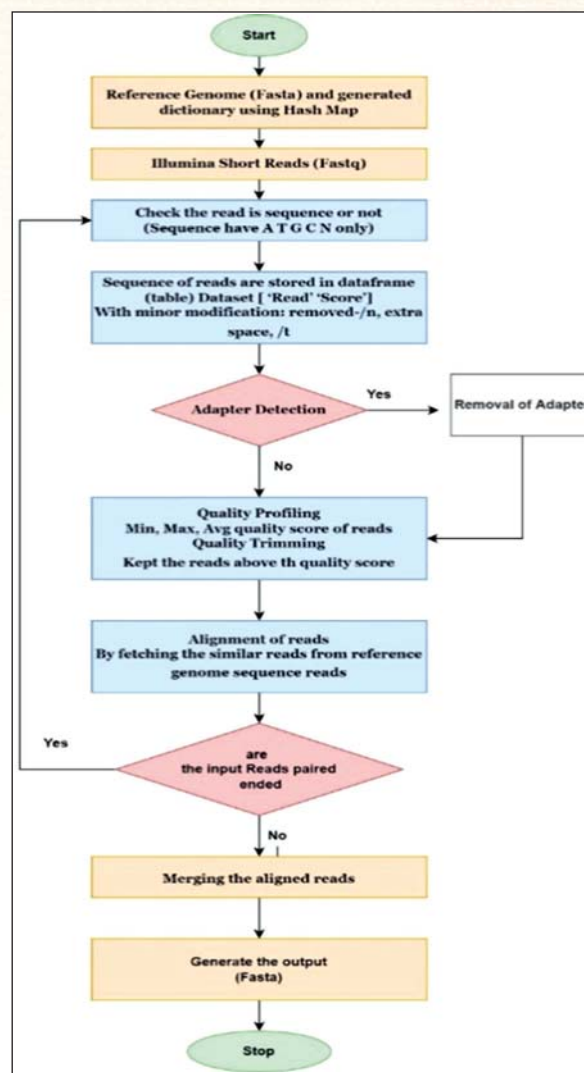


Fig. 2.20: Workflow for the chloroplast genome extraction pipeline

3. NOVEL GENES AND BIOMOLECULES FOR PLANT AND HUMAN NUTRITION



Comparative analysis of GS2 and Fd-GOGAT genes in Cultivated Wheat and their Progenitors under N stress

Gayatri, Karikalan Jayaraman, Subodh Kumar Sinha, Arivaradarajan Preeti and Pranab Kumar Mandal* [Gayatri et al. 2021, *Plant Mol Biol Rep*]

GS2 and Fd-GOGAT enzymes are core components involved in the assimilation of

primary nitrogen (N) in plants. We analyzed the extent of variation that occurred in these genes during evolution from the ancient diploid progenitors to the modern-day hexaploid bread wheat.

We compared the genomic sequences of both the genes from *T. aestivum* (hexaploid), *T. durum* (tetraploid), *Aegilops tauschii* (diploid DD), *A. speltoides* (diploid BB), and *T. urartu* (diploid AA). Both GS2 and Fd-GOGAT genes were located on chromosome 2 and were found

to comprise of 13 and 33 exons, respectively. Protein structure and phylogenetic analysis of these genes revealed their conserved nature and close relatedness during evolution (Fig. 3.1). Southern blot analysis revealed both these genes were present in three copies in bread wheat; two copies in durum wheat; and a single copy in their diploid progenitors. Genome species-specific enzyme activity, as well as differential gene expression suggested major differences in gene regulation, with insignificant differences in their protein structures. All growth parameters, except root length, decreased or remained unchanged with different degrees of plasticity in the genotypes under N stress. Combined PCA analysis of all parameters revealed that GS2/Fd-GOGAT enzyme activity, N accumulation, protein, chlorophyll, and carotenoid content were responsive to N stress. Our study confirmed the conserved nature of GS2 and Fd-GOGAT enzymes; however, their expression and subsequent effects were different in cultivated wheat and their progenitors.

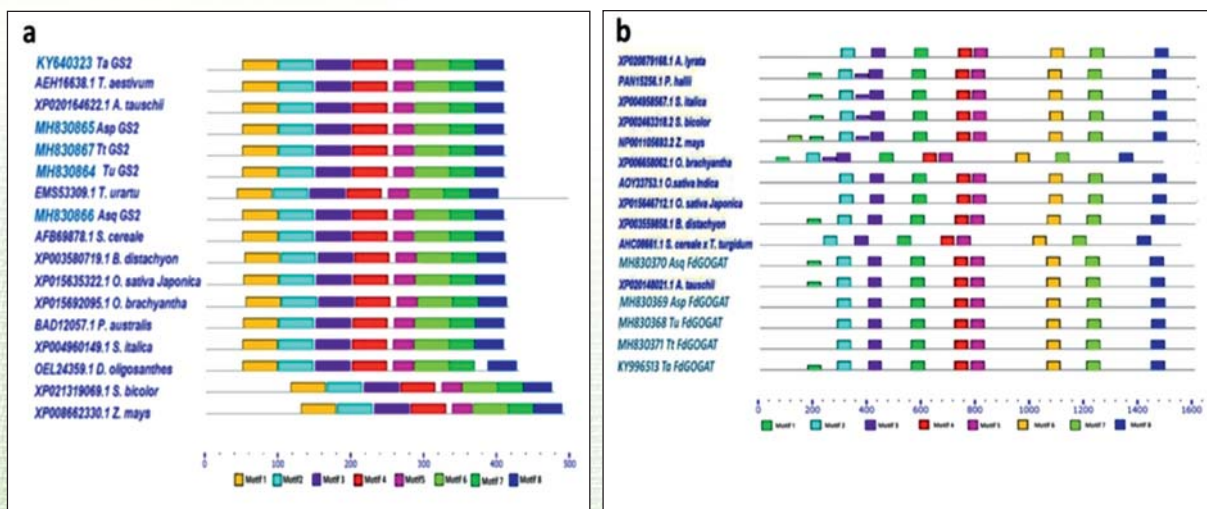


Fig. 3.1 Protein motif analysis for monocots was carried out using MEME. (a) GS2 amino acid sequences, (b) Fd-GOGAT amino acid sequences. Different colored boxes are assigned to the 8 motifs.

Molecular characterization of GS2 and Fd-GOGAT homeologues and their biased response to nitrogen stress in bread wheat (*Triticum aestivum* L.)

Gayatri, Karnam Venkatesh, Subodh Kumar Sinha, Pranita Roy and Pranab Kumar Mandal* (Gayatri et al. 2021, *Plant Growth Regul*)

A popular bread wheat variety HD-2976 was grown in hydroponics under N-optimal and N-stressed condition. Detailed characterization of Glutamine synthetase 2 (GS2) and Ferredoxin-dependent-glutamate synthase (Fd-GOGAT), two critical enzymes involved in GS/GOGAT cycle, have been carried out in bread wheat. GS2 and Fd-GOGAT genes were cloned from HD-2967 and their individual homeologues (A, B and D) were studied under different levels of N-stress. Homeologous cDNAs of GS2 and Fd-GOGAT showed > 93% and > 98% sequence similarity, respectively, in the coding region but were distinguishable at UTRs. The expression patterns and contribution of each homeologue of both the genes in HD-2967 under different N treatments showed downregulation in the shoot and upregulation in the root tissues (Fig. 3.2). However, the contribution of the individual homeologue varied with the tissue, time point and level of N-stress. Further, when the expression pattern and contribution of

each homeologue were studied in a diverse set of wheat genotypes, a generalized contribution pattern of each homeologue of wheat was observed. While the transcript ratio of A and D homeologues of both the genes demonstrated to be compensating each other, Fd-GOGAT-B showed the least contribution in all the conditions and at all stages of growth. Substantial variation in cis-regulatory elements was observed among all three homeologues of both the genes. Overall, characterization and spatio-temporal contribution of GS2 and Fd-GOGAT homeologues would provide applied value for further detailed studies as well as their exploration in primary N-assimilation process in wheat.

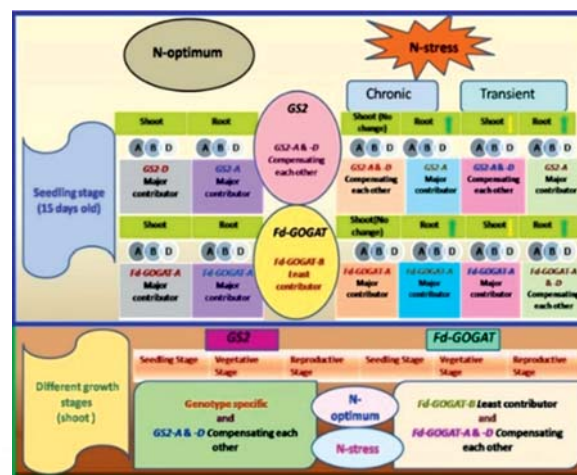


Fig. 3.2 Summary of expression pattern and homeologue specific contribution of GS2 and Fd-GOGAT genes at different growth stages in bread wheat

Integration of dual stress transcriptomes and co-localization of the major QTL with the transcriptome data of the parental genotypes under optimal and low N

Amitha Mithra Sevanthi, Subodh Kumar Sinha, Sureshkumar V, Manju Rani, Manish Ranjan Saini, Sapna Kumari, Megha Kaushik, Chandra Prakash, Venkatesh K, and Pranab Kumar Mandal* (Sevanthi et al. 2021, *Rice*)

We analyzed two hot spot regions on chromosome 6 and chromosome 1 between the flanking SNP markers, and spanning a length of 417 Kbp and 100 Kbp respectively for the differentially expressed genes under different N regimes between the parents. The QTL on chromosome 6 comprised of 31 genes of which two were DEGs in the transcriptome analysis including two UDP-glucuronosyl/UDP-glucosyltransferase family proteins (encoded by Os06g289200 and os06g289900) of which the former was downregulated in N22 under N- stress while the latter was upregulated under all stress conditions in N22. In addition, there was a zinc finger Dof-type family protein in the vicinity which was again downregulated in N22 (Fig. 3.3). The QTL on chromosome 1 harbored only two genes out of which Os01g166800 encoding ETG1 (E2F TARGET GENE 1) protein showed differential

expression only in IR64 shoots under N- and dual stress. These six genes have been picked by us for further functional validation.

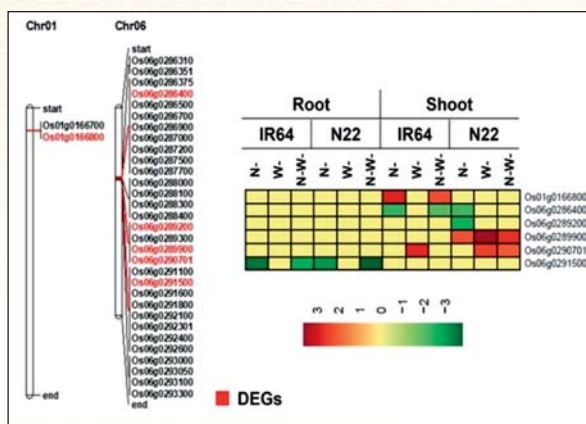


Fig. 3.3 Comparison of expression profiles of the differentially expressed genes present in the two major QTL regions identified for straw and seed N content on chromosomes 1 and 6



Responses of selected wheat genotypes under nitrate limiting condition for the expression of *Ta*NRT2 and *Ta*NAR2 genes and ^{15}N influx

Amresh Kumar, Pranab Kumar Mandal, Subodh Kumar Sinha*

We previously reported the group wise expression patterns of *Ta*NRT2 (Group I - XIII) and *Ta*NAR2 (Group I - III) genes under nitrate limiting conditions in two wheat genotypes

at seedling stage. In order to gain an insight into their stage and tissue specific expression pattern, we extended the experiment (qPCR analysis) at different physiological stages, viz., GS-19, GS-39 and GS-65, in root, shoot, flag leaf and ear tissues until anthesis. We also analyzed ^{15}N influx and N (%) translocation at these stages with four wheat genotypes. Our qPCR data suggests that *Ta*NRT2 genes of

different groups have growth stage and tissue specific preferential expression (Fig. 3.4). However, the expression patterns of *Ta*NAR2 genes showed constitutive expression with differential expression in different genotypes at different stages (Fig. 3.5). Interestingly, the selected genotypes also showed differences in ^{15}N influx in different tissues at different growth stages.

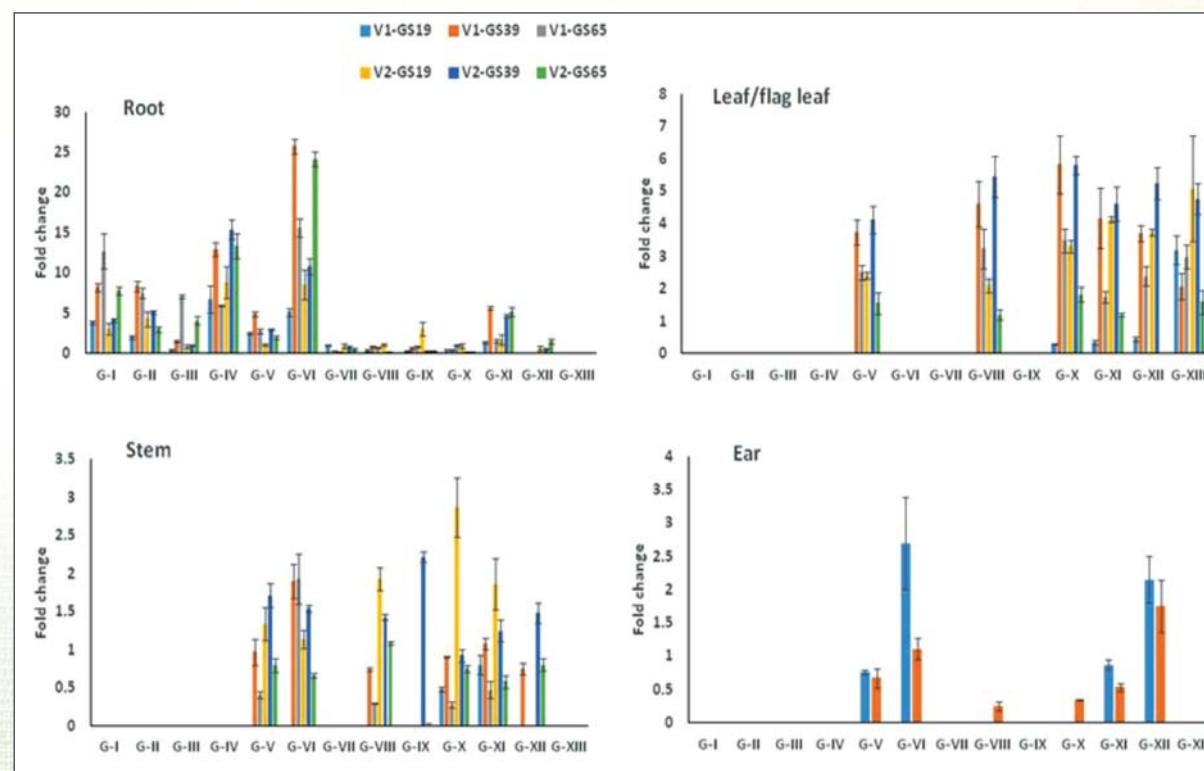


Fig. 3.4 qPCR analysis of *Ta*NRT2 genes (Group I-XIII) in different tissues of wheat genotypes at different growth stages (i.e., GS19, GS39 and GS65).

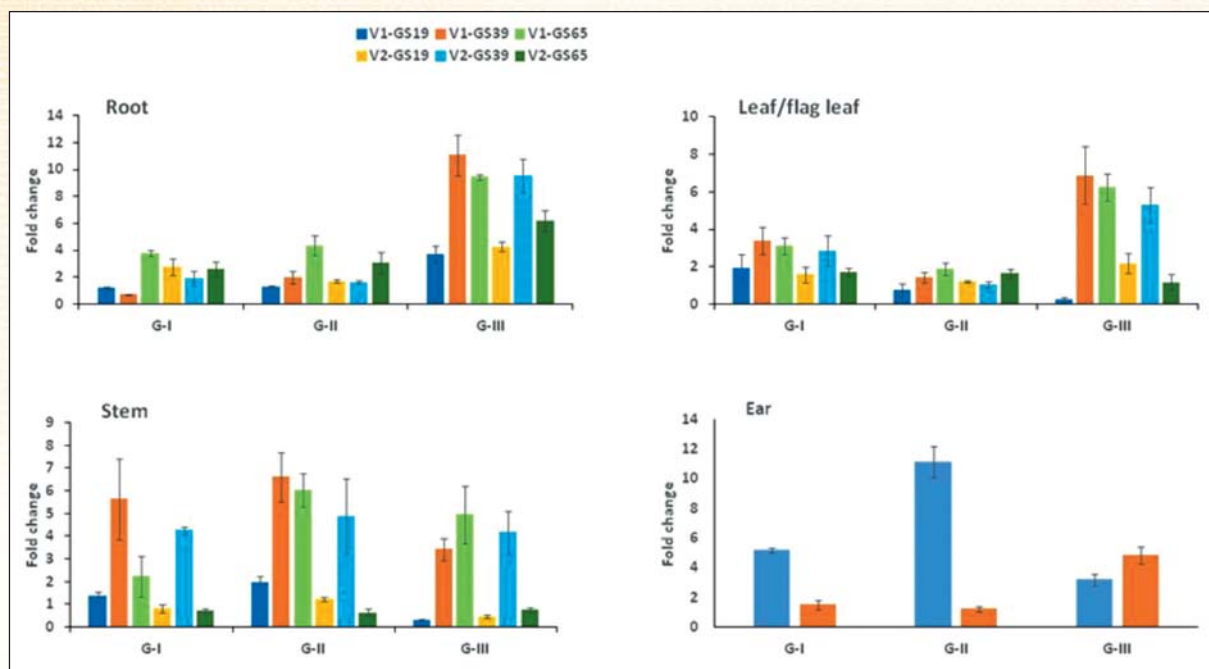


Fig. 3.5 qPCR analysis of *TaNAR2* genes (Group I-III) in different tissues of wheat genotypes at different growth stages (i.e., GS19, GS39 and GS65).

Construct generation and transformation of *Arabidopsis* mutant for function complementation of selected *TaNRT2* genes

Amresh Kumar, Pranab Kumar Mandal, Subodh Kumar Sinha*

In order to functionally characterize wheat nitratetransporter genes, three selected *TaNRT2* genes (previously cloned and sequenced) were cloned in a plant transformation vector and positive clones of each gene were mobilized into *Agrobacterium tumefaciens* (GV3101).

Confirmed *Agrobacterium* cultures were used for transformation of *Atnrt2.1* *Arabidopsis*

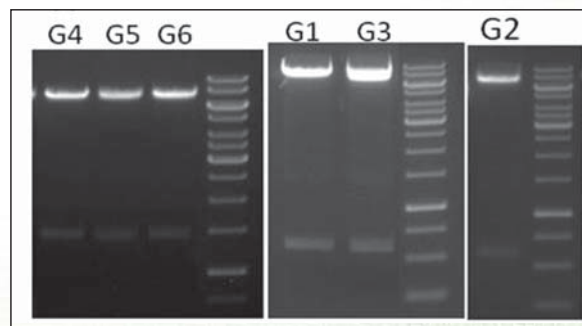


Fig. 3.6 Cloning of three *TaNRT2* genes in pORE-04 vector

mutant following floral dip method (Fig. 3.6). The transformants were selected on selection media and are being advanced for further characterization.



Characterization and in planta validation of a *CHI4* chitinase from *Cajanus platycarpus* (Benth.) Maesen for its efficacy against pod borer, *Helicoverpa armigera* (Hübner)

Maniraj Rathinam, Shaily Tyagi, Karthik Kesiraju, Rohini Sreevathsa* (Rathinam et al. 2021, Pest Manag Sci)

Pigeonpea is one of the economically important legume food crops and a major source of dietary proteins. Management of pod borer, *Helicoverpa armigera* has been prominent among crop improvement programs. Lack of resistance sources in the cultivated germplasm and crossing incompatibility with the pod borer-resistant wild relatives have prompted biotechnological interventions. Hence, identification and exploitation of genes from pigeonpea wild relatives in host plant resistance towards the pod borer assumes pertinence. Dynamic transcriptome analysis of the wild relative *vis a vis* cultivated pigeonpea identified a *CHI4* chitinase as one of

the putative insect resistance genes. The study presents variations in important amino acids in CHI4 chitinases from *Cajanus cajan* and *Cajanus platycarpus*. Comparative protein modeling and docking analysis of the two proteins demonstrated differences in substrate binding

efficacy of the chitinase from *C. platycarpus*, which resulted in a minimum binding energy of -8.7 Kcal/mol. Further, we successfully evaluated the insecticidal activity of the chitinase from *C. platycarpus* against *H. armigera* challenge through heterologous expression

in tobacco. Molecular characterization of transgenic plants confirmed that their efficacy against pod borer was due to the integration of CHI4 from *C. platycarpus*. Docking analysis demonstrated effective substrate interaction as a possible reason for efficacy against pod borer in the chitinase from *C. platycarpus*. This was authenticated by successful overexpression and bioefficacy assessment against *H. armigera* in tobacco (Fig. 3.7). CHI4 gene from *C. platycarpus* can be useful in the mitigation of *H. armigera* in pigeonpea as well as in other crops.



Genome-wide identification of secondary metabolite genes involved in various pod developmental stages in cluster bean

Sandhya Sharma* Harsha Srivastava, Kishor Gaikwad

Plants produce a large number of metabolites which can be utilized for drug development, pest resistance, countering abiotic stresses, etc. These secondary metabolites are involved in several biosynthetic pathways. In order to unravel their key roles in these pathways, the genome of cluster bean (*Cyamopsis tetragonoloba*) was analysed. Based on our analysis, we predicted 1053 secondary metabolite biosynthetic gene clusters in

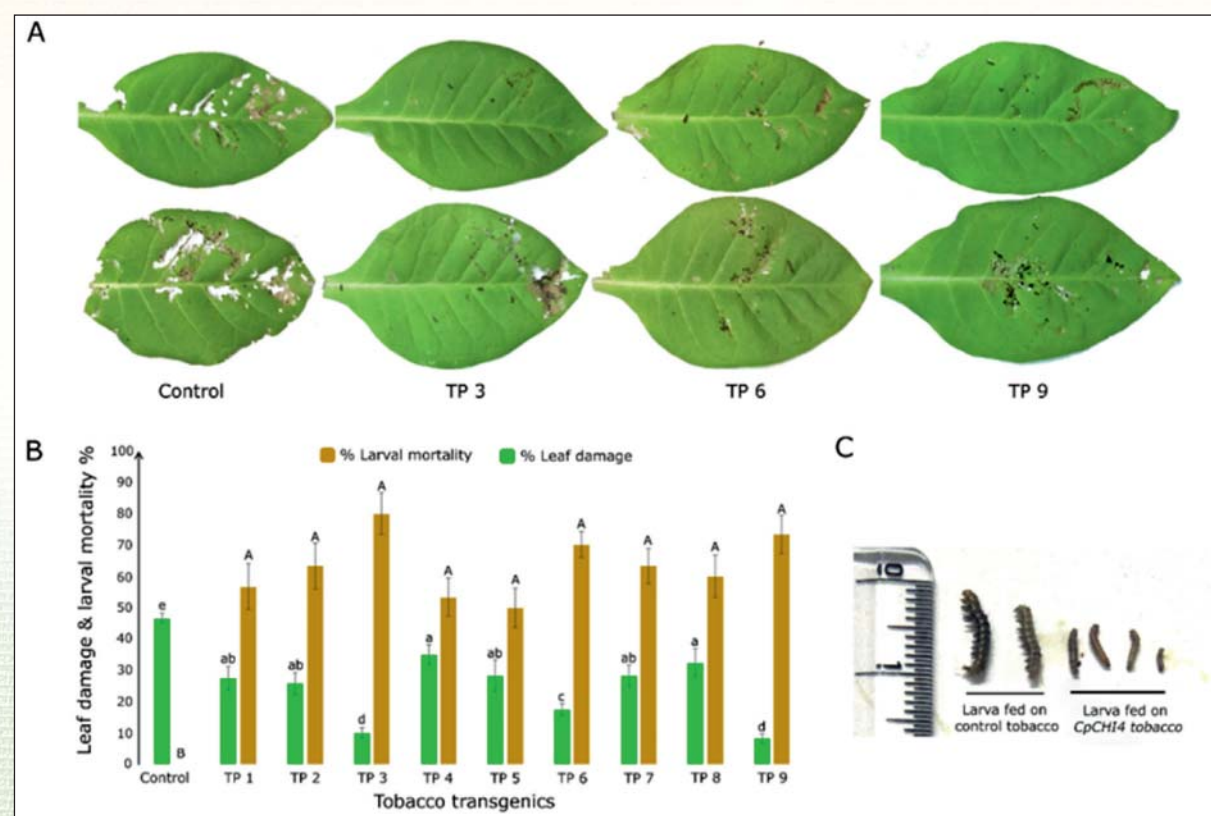


Fig. 3.7. *In planta* validation of *CpCHI4* in tobacco transgenics. (A) Illustration of the performance of selected *CpCHI4* transgenics in the *in vitro* detached leaf bioassay against deliberate challenging of *H. armigera*. (B) Graphical representation of the performance of individual *CpCHI4* transgenic plants and vector control plants to deliberate challenge by *H. armigera*. (C) Larval morphology at the end of the detached leaf insect bioassay. (a to e and A, B denotes significance at *p* (0.05) determined from one way ANOVA post hoc Tukey test).

the genome of cluster bean (*Cyamopsis tetragonoloba*) and analyzed their expression during various pod developmental stages using an existing transcriptome data set. A total of 27, 29 and 35 genes involved in phenylpropanoid biosynthesis, flavonoid biosynthesis and terpenoid biosynthesis were respectively identified in guar genome. *De novo* sequencing assembly led to the identification of a total of 209525, 375595 and 255401 unigenes at 25, 39 and 50 days after flowering respectively. Unigenes encoding enzymes of 'Phenylpropanoid biosynthesis pathway'

represented 249 transcripts, whereas the cluster of Unigenes encoding 'Flavonoid and terpenoid biosynthesis' represented 203 and 245 transcripts respectively. Secondary metabolite pathway genes were found to be distributed unevenly throughout the 7 chromosomes. Chromosome 1 and chromosome 5 contained the highest number of phenylpropanoid and flavonoid metabolic genes respectively. However, chromosome 2 and 1 contained the highest number of diterpenoid and monoterpenoid metabolic genes respectively (Fig. 3.8)

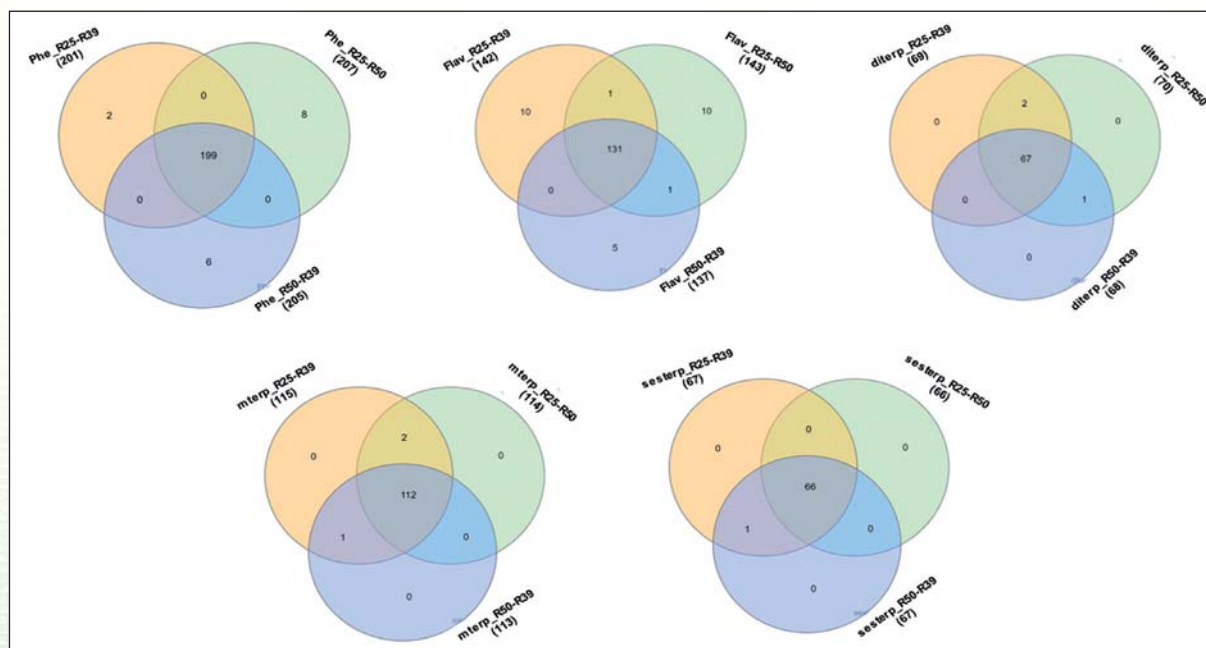


Fig. 3.8: Exclusive, inclusive and common differential unigenes involved in phenylpropanoid, flavonoid and terpenoid pathway at 3 different pod development stages in R936 genotype.



Identification of Compounds responsible for Aroma and Flavor in Mango

Joshitha Vijayan^{*1}, Pranab Kumar Mandal¹, Ratnasekhar CH², Ashish Yadav³, Anju Bajpai³ and Ajit Kumar Shasany¹ (¹ ICAR-NIPB, ² CSIR-CIMAP, ³ ICAR-CISH)

Mango fruit is widely appreciated for its flavor throughout the world. As it originated from the Indo-Sino-Myanmar region, India has a good variation in terms of germplasm resources in mango. Though a number of studies have characterized the aroma and flavor, most work has been limited to a few popular varieties. Moreover, very few studies have explored the genes for aroma and flavor production. In the present study, a GC-MS-HS profiling of volatiles was performed for 11 Indian mango varieties namely, Dashahari, Khasulkhas, Kishanbhog, Kesar, Chausa, Langra, Arunika, Mallika, Ambika, Fazri and Prabhasankar. Monoterpene and sesquiterpene compounds were identified in these 11 varieties. Fig. 3.9 represents the major compounds discovered in the aroma profile of the studied mango varieties. The bean plots in upper panels present in maximum abundance in the peel and the pulp of each of the varieties. The dot

plots in the lower panel represent the most abundant compounds in terms of percentage in the peel and the pulp of each of the varieties and also depicts the second most abundant compound too where the first compound is less than 75% of total identified compounds in terms of proportion.

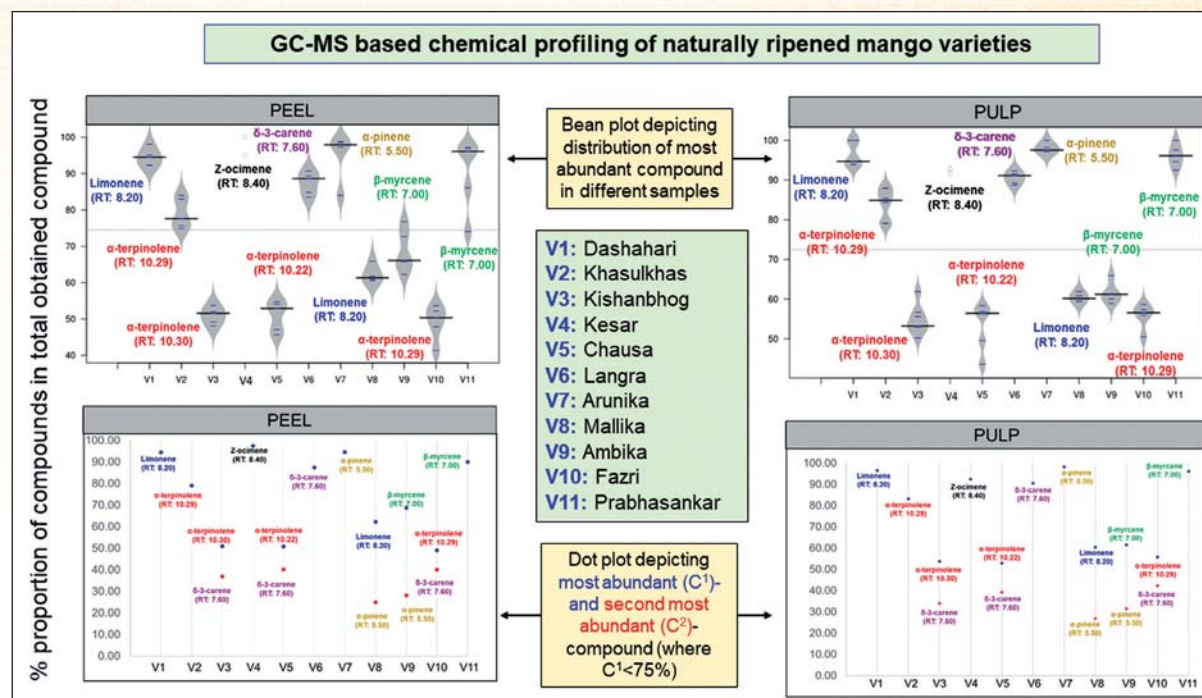


Fig. 3.9. Aroma profile of peel and pulp of different varieties of mango

4. GENETIC MODIFICATION OF CROP PLANTS FOR IMPORTANT TRAITS



CRISPR/Cas9 mediated genome editing in Indian mustard *Brassica juncea*

Muthuganeshan A, Satish Kumar, Smriti Singh, Anshul

Watts, R.C. Bhattacharya*

Indian mustard [*Brassica juncea* (L.) Czern.] is a major oilseed crop in our country which contributes 27.8% to the national oilseed economy. Because of the economic importance and substantial gap in the projected domestic demand and realized productivity, crop improvement research in mustard always assumed priority at our institute. Major objectives of Brassica improvement programme at our institute are: productivity enhancement, imparting resistance to biotic and abiotic stresses, and enhancement of oil- and meal-quality.

Recent developments in CRISPR/Cas system of genome editing offers precision in creating targeted mutation of genes. Therefore, for developing CRISPR/Cas9 mediated genome editing process in *B. juncea*, we have developed

a CRISPR/Cas9 vector for knockout of *PDS* gene in *B. juncea*. Bioinformatic analysis revealed the presence of at least three copies of the *PDS* gene in the allotetraploid genome of *B. juncea*. Specific sgRNAs targeting all the three *PDS* copies in the *B. juncea* genome, two from *B. rapa* (A-genome) and one from *B. nigra* (B-genome) were designed, synthesized, cloned and further subcloned in a plant binary vector. The cloning was confirmed through restriction digestion and sequencing. Using the binary construct, several transformed shoots were generated by hypocotyl transformation which were PCR screened for incorporation of the T-DNA cassette. Interestingly, severe retardation of chlorophyll content, a typical

symptom of *PDS* knock out was observed in these regenerated shoots. These shoots could be distinctly grouped into two categories: complete phenotype with complete loss of chlorophyll and partial phenotype with very low level of chlorophyll content (Fig. 4.1). Mutation in *PDS* gene could be detected when the PCR amplicons of *PDS* gene from such shoots were subjected to heteroduplex assay. Analysis of sequence data from these gene specific amplicons revealed targeted deletions in the *PDS* coding region (Fig. 4.2). The process of developing CRISPR/Cas mediated knockouts are being employed in validating novel target genes for enhancing yield and improving oil quality traits in mustard.

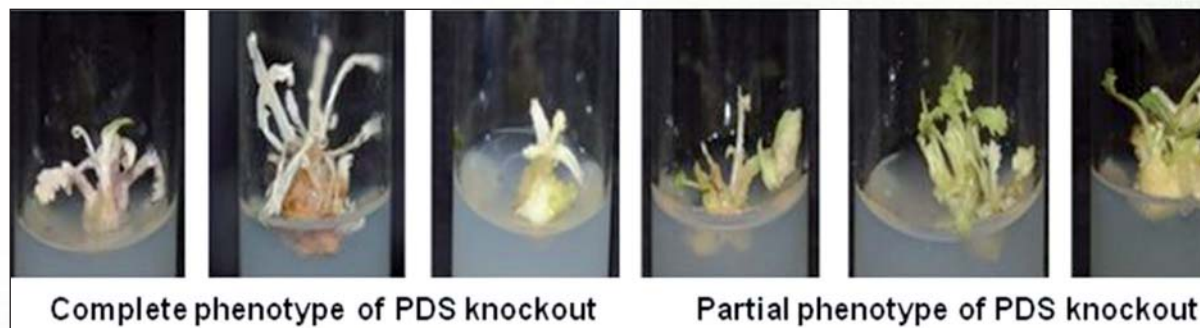


Fig. 4.1: Chlorotic phenotype of the *in vitro* regenerated shoots obtained in *Agrobacterium tumefaciens* mediated plant transformation of *B. juncea* using hypocotyl explants.



Fig. 4.2: Multiple sequence alignment of the PCR amplified PDS gene in Genome Edited lines of *B. juncea* showing regions of CRISPR/Cas9 mediated deletions in the targeted region.

Deciphering the genes and pathways associated with *Alternaria* resistance in introgression lines of Indian mustard

Sharani Choudhury, Mahesh Rao, Anamika Kashyap, Ashish Gupta, Navin C. Gupta, Anshul Watts, Rohit Chamola, R.C. Bhattacharya*

Alternaria leaf spot disease caused by *Alternaria brassicae* inflicts a significant annual yield loss in India and across the world. Genetic resistance is unavailable among the cultivated germplasm of Indian mustard. Based on exhaustive pathogen bioassay,

resistance has been demonstrated in a wild relative of Brassica, *Diplotaxis erucoides*. With the aim of introgressing the resistance genes in the cultivated gene pool of Indian mustard, distant hybridization was carried out using *Brassica rapa* as a bridge species. The F_1 progenies were backcrossed followed by several generations of selfing. The successive generation of progenies were screened for *Alternaria* resistance at multilocations including hot spots for *Alternaria* disease. For identification of *Diplotaxis*-derived resistant genes, pathways and markers associated with the traits in the stable resistant introgression lines (ILs) were compared with the susceptible

counterparts based on leaf transcriptome studies and GBS analysis.

A major group of genes that showed differential gene expression in resistant and susceptible introgression lines constituted transcription factors. Out of 35 transcription factors showing differential expression, 16 were upregulated whereas 19 were downregulated in resistant IL compared to susceptible (S) ILs. Among the unique TF transcripts, 56 were present in resistant ILs (composed of 27 upregulated and 29 downregulated genes in resistant IL) which were not detected in susceptible counterpart lines. Similarly, a set of 47 TF transcripts (composed of 20 upregulated and 27 downregulated genes) were uniquely detected in the susceptible ILs.

Plant secondary metabolites play a significant role in conferring quantitative resistance against plant pathogens. Among the total 75 secondary metabolite pathway related genes found significantly regulated in the transcriptome data, 32 and 25 transcripts were differentially upregulated and downregulated, respectively, in resistant ILs in contrary to 15 and 10 transcripts in the susceptible ILs. There was a significant regulation of the glucosinolate biosynthetic pathway genes when the plants of *D. erucoides* were challenged with the *A. brassicae* pathogen. A similar pattern of regulation of the transcripts related to glucosinolate

biosynthesis was also observed in some of the resistant ILs. A real time PCR based validation showed upregulation of the key glucosinolate biosynthetic genes in *D. eruroides* as compared to *B. juncea* (Fig. 4.3). Further, a global metabolome profile of Alternaria-challenged *B. juncea* and *D. eruroides* are being generated through LC-MS.

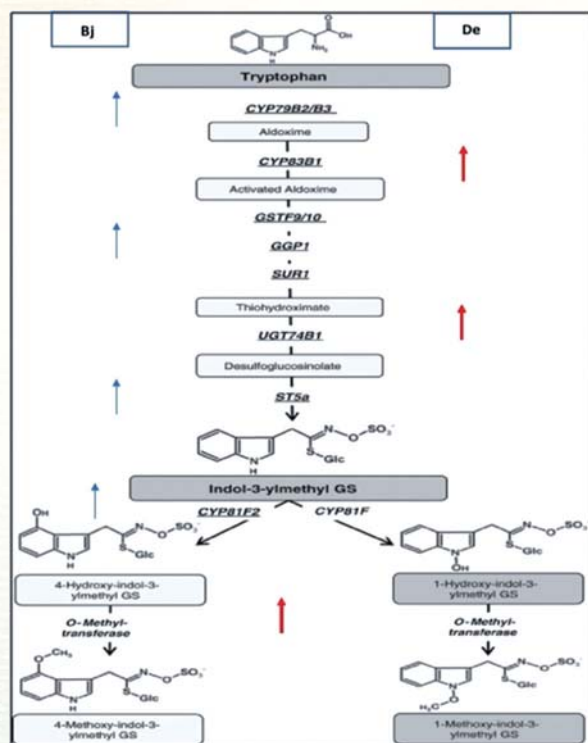


Fig. 4.3: Activation of biosynthetic genes under indole glucosinolate pathway in *B. juncea* and *D. eruroides*. (Blue arrows denotes the genes having higher upregulation in *B. juncea* and red arrows denote the genes having higher upregulation in *D. eruroides*).



Morphological and pathological characterization of Albugo candida isolates causing whiterust disease in rapeseed-mustard

Ashish Kumar*, Mahesh Rao, Jameel Akhtar, Rashmi Yadav, R. C. Bhattacharya

Eleven different isolates of *A. candida* collected from major mustard growing locations across the country were purified separately and maintained on seedlings of susceptible

cv. 'Varuna' under controlled conditions. Pathotyping of each isolate was done separately and incubation period for resulted infection varied with 8-12 days, 8-17 pinheads/100 mm² area, 1-6 mm pustule size and 10-90% disease severity was observed on the inoculated plants. Wild Brassica germplasm with the Brassica unit of the Institute, a total of 350 Brassica species obtained from NBPGR, 183 introgressed Brassica lines (BC₂F₁₀) and 90 Resynthesized Brassica (RBJ) accessions was tested under artificial conditions for infection with 5-6 isolates of the pathogen (Fig. 4.5). The results of the study are briefly presented in Table 4.1

Table 4.1: Disease response reaction of the wild germplasm and resynthesized lines of *Brassica juncea* to infection with *Albugo candida*

Genetic resources	Immune Response (PDI = 0)	Resistance reaction	Susceptible (PDI = 5-30)
Wild germplasm of Brassica	<i>Brassica oxyrrhina</i> , <i>Camelina sativa</i> , <i>D. eruroides</i> , <i>D. muralis</i> , <i>D. siettiana</i> , <i>Erucastrium lyratus</i> , <i>Erucastrium canariense</i> , two exotic accessions of <i>Crambe abyssinica</i> viz, EC694071 and EC694138 and three indigenous accessions of <i>Eruca sativa</i> viz., IC57706, I62597, IC62599	<i>Eruca sativa</i> IC508400, IC508401, IC508402, IC508404, IC310971 and IC369447 against 2-4 isolates.	<i>B. chinensis</i> , <i>B. fruticulosa</i> , <i>B. tournefortii</i> , <i>Diplotaxis assurgens</i> , <i>Capsella bursapastoris</i> , <i>Sinapis alba</i> , <i>C. abyssinica</i> (EC694159) and <i>Eruca sativa</i> (IC341907) to Ac-Pnt, Ac-Adhya, Ac-Bhrtr, Ac-Wlqtn and Ac-Samstr isolates
	<i>Lepidium</i> species: IC572843 immune to Ac-Pnt, Ac-Morna, Ac-Skn and Ac-Wlqtn		

Genetic resources	Immune Response (PDI = 0)	Resistance reaction	Susceptible (PDI = 5-30)
<i>Brassica</i> accessions/ advanced introgressed lines	EC766192 and EC766164 against 6-7 isolates of the pathogen at both cotyledonary and true leaf growth stages of the plants; <i>B. carinata</i> accessions: EC206642 and EC206654 against 5-6 isolates of the pathogen	EC766191, EC766193, EC313380 EC766272, IC414324, IC399312 found resistant against 4-5 isolates of the pathogen	-
Resynthesized <i>Brassica</i> (RBJ) lines	ERJ12, 15, 39, 40 and RBJ9, 18, 34 and 40 against 3-7 isolates	-	-

Field evaluation of *Brassica* germplasm

Ashish Kumar*, Mahesh Rao, Jameel Akhtar, Rashmi Yadav

Out of the 325 *Brassica* species accessions (IC-30 acc. & EC-295 acc.) evaluated under natural field conditions at Issapur and New area farm of ICAR-NBPGR, Delhi, two indigenous accessions, IC265495 and IC597932 and ten exotic accessions of *B. juncea*, viz., EC206651, EC766311, EC766313, EC766133, EC766134, EC766144, EC766145, EC766148, EC766164 and EC766193 and two accessions of *B. carinata* viz., EC206641 and EC206642 were identified as immune for white rust disease under natural field condition of Delhi. Moreover, these accessions were also found immune under artificial inoculated condition against Ac-Del isolate of *A. candida*.



Introgression of gene(s) for *Alternaria* tolerance from wild species into *Brassica juncea*

Anamika Kashyap, Pooja Garg, N C Gupta, Ashish Kumar, K K Singh, Usha Pant, Rohit Chamola, R C Bhattacharya and Mahesh Rao*

The ERJ lines were screened under natural conditions at GBPUAT, Pantnagar during 2020-21 (Fig. 4.4). On the basis of multi-year/multi-location data, both in-vitro and in-vivo, a total of 15 lines were identified which gave significant level of resistance consistently against *Alternaria brassicae* (Table 4.2). These lines can serve as pre-breeding material for the *Alternaria* resistance breeding.

Table 4.2: List of the selected resistant Introgression lines

S. No.	Introgression line No.	S. No.	Introgression line No.
1	ERJ-39	9	ERJ-164
2	ERJ-45	10	ERJ171
3	ERJ-106	11	ERJ172
4	ERJ-111	12	ERJ-186
5	ERJ 112	13	ERJ-191
6	ERJ-115	14	ERJ-194
7	ERJ-127	15	ERJ-201
8	ERJ-159		

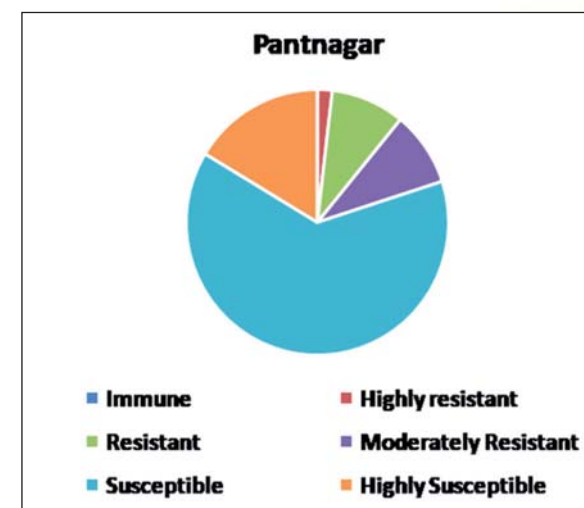


Fig. 4.4: Pie chart representing disease reactions of introgression lines under natural condition at GBPUAT, Pantnagar

Enlarging gene pool of *Brassica juncea* through resynthesis and generation of synthetic amphidiploids using wild species and *B. rapa*

Pooja Garg, Anamika Kashyap, N C Gupta, Rohit Chamola, Naveen Singh, R C Bhattacharya, KK Singh, Rashmi Yadav and Mahesh Rao*

The resynthesized *Brassica juncea* lines (RBJ1 to RBJ-92) along with their parents and *Brassica juncea* varieties were evaluated under multi-locations for agronomic performance. These lines were subjected to SNP genotyping and the data analysis is underway. A few lines were primarily identified for higher number of siliqua per plant and seeds per siliqua. The chromosomal pairing and chromosome number were analysed by cytogenetic tools (Fig.4.5) in a number of lines which

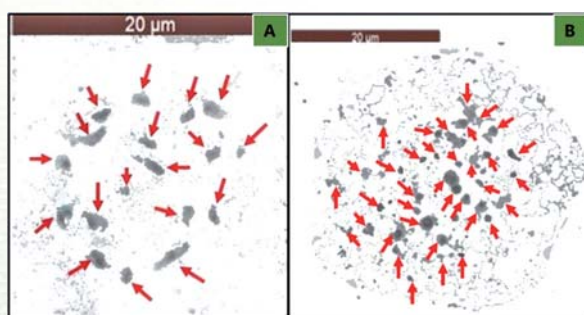


Fig. 4.5: Cytogenetic studies. A) Meiosis of F_1 amphihaploid showing 18I (univalent) chromosomes and B) Mitosis of advanced synthetic *B. juncea* showing 36 chromosomes.

represented 18 univalents during meiosis in the amphihaploid of *B. rapa* x *B. nigra* and 36 number of chromosomes in mitosis of synthetic *B. juncea* lines in most of the lines analysed.



Development of haploid inducer (HI) lines in *Brassica* sp.

Anshul Watts* and Ritesh Kumar Raipuria

An *in-vivo* HI line is needed in *Brassica* through which haploids can be produced at will by simple crossing. Therefore, CENH3 mediated HI strategy was employed for development of HI lines. CRISPR/Cas9 vector targeting both CENH3 paralogs of *B. juncea* and GFP-tailswap cassette were co-transformed in Varuna and RLM-198 variety of *B. juncea* using hypocotyl

mediated transformation approach. The transformants were selected on kanamycin and confirmed through PCR based analysis (Fig. 4.6A). Further to check any sequence variation in the CENH3 gene a T7 endonuclease digestion approach was used. *B. nigra* CENH3 paralog in some samples showed digestion using T7 endonuclease while most of the *B. rapa* samples showed no digestion (Fig. 4.6B).

Apart from CENH3 gene, pollen-specific phospholipase A and domain of unknown function protein 679 (DMP) genes showing haploid induction specific expression were identified. In order to identify the pollen specific phospholipase within *B. juncea*, first we screened the *B. juncea* genome sequence to identify various phospholipase A paralogs. We identified 39 paralogs of phospholipase A genes in *B. juncea*. Further various *in-silico* studies viz.

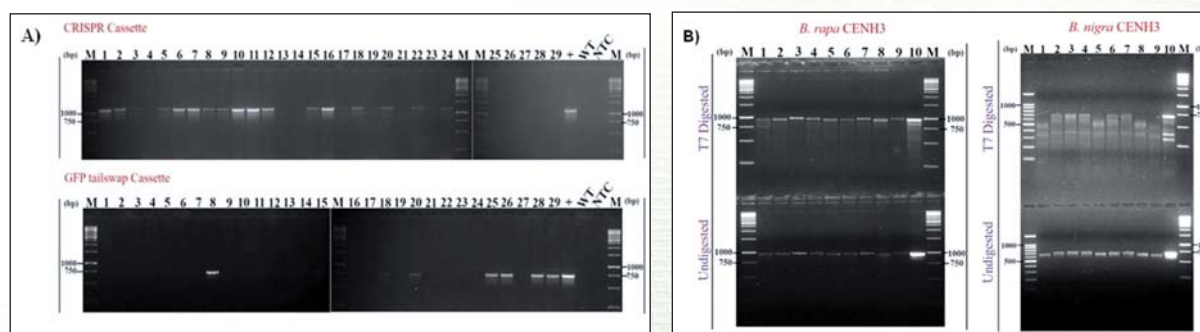


Fig. 4.6: Confirmation of putative transformants through PCR based analysis and T7 endonuclease assay A) CRISPR cassette and GFP-tailswap cassette was confirmed through use of one primer pair from each of the cassette B) T7 endonuclease assay was performed for both *B. rapa* and *B. nigra* CENH3 copies.

gene structure, gene and protein size, number of exons and introns, chromosome localization of various paralogs were also performed.

Understanding Brassica-Orobanchae interaction for developing Orobanchae resistance in *B. juncea*

Anshul Watts*, Ritesh Kumar Raipuria and RC Bhattacharya

Till now there is no *B. juncea* germplasm identified for tolerance/resistance to *Orobanchae* infestation. To identify such a germplasm we screened 216 *Brassica juncea* accessions comprising cultivated varieties, germplasm from ICAR-NBPGR and other *Brassica* accessions at *Orobanchae* hot spot in Haryana in 2020-21. Fig. 4.7 shows the frequency distribution of number of *Orobanchae* plants present and the number of accessions. *Lepidium didymium* which was earlier identified as a new host species in 2020 was again confirmed as the host species of *Orobanchae* in March 2021. The new host was submitted to ICAR-NBPGR NHCP herbarium and HS-24638 ID for the sample was provided by ICAR-NBPGR. An available molecular marker that can differentiate the *O. aegyptiaca* and *O. ramosa* was used in the samples collected from Haryana and Rajasthan. All the samples gave around 700 bp amplicon showing that these samples might be *O. ramosa* (Fig.

4.8). Interestingly, all these species possess hairy anthers which are characteristics of *O. aegyptiaca*.

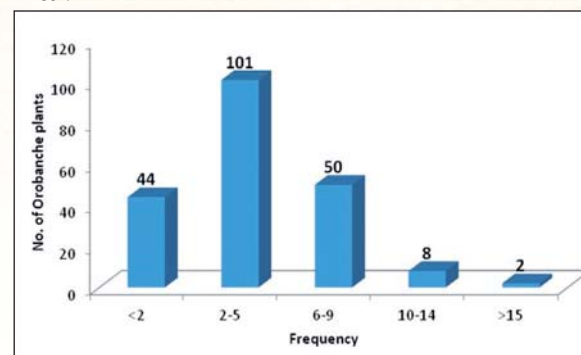


Fig. 4.7: Frequency distribution screening of the 216 accessions of *B. juncea* and the number of *Orobanchae* plants present along with each accession

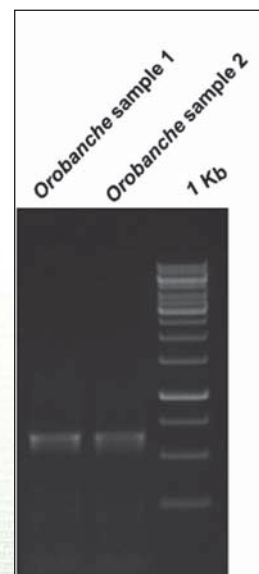


Fig. 4.8: PCR amplification in *Orobanchae* samples collected from Haryana and Rajasthan.

Further, to understand the gene expression of *Orobanchae* we carried out the transcriptome sequencing of different tissues of *Orobanchae* viz. Haustorium, flower bud, open flower and stem. We isolated the RNA from these tissues and mixed in equal amount. *De novo* assembly of transcriptome sequence was done and various genes present in different pathways of *Orobanchae* were identified (Figure 4.9). The details of various pathways are as below:

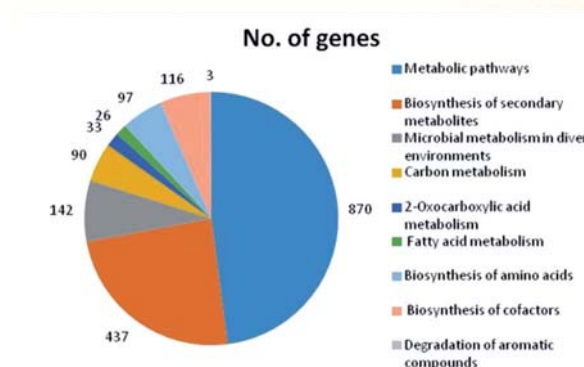


Fig. 4.9: Number of genes present in different pathways of *Orobanchae*



Time-course transcriptome analysis of rice-Magnaporthe oryzae interaction during panicle blast in rice

Amolkumar U Solanke*, Vishesh Kumar, Priyanka Jain, Suhas Gorakh Karkute, Amitha Mithra Sevanthi, Tilak Raj Sharma [Kumar et al. 2021, *Genes*]

Rice blast disease is a global threat to food security which results in more than 50% yield losses annually. Out of all blast diseases, panicle blast is a more severe form. The response of rice plant to leaf and panicle blast is distinct in different genotypes. To understand the specific response of rice in panicle blast, transcriptome analysis of blast resistant cultivar Tetep, and susceptible cultivar HP2216 was carried out using RNA-Seq approach after 48, 72 and 96 h of infection with *Magnaporthe oryzae*. Transcriptome data analysis of infected panicle tissues revealed that 3553 genes differentially expressed in HP2216 and 2491 genes in Tetep. The differentially expressed genes belonged to mainly defense pathways namely, hormonal regulation, synthesis of reactive oxygen species, secondary metabolites and cell wall modification (Fig. 4.10). In Tetep, cell wall strengthening pathway represented by PMR5, dirigent, tubulin, cell wall proteins, chitinases, and proteases was found to be specifically enriched. Additionally, many novel genes having DOMON, VWF, and PCaP1 domains which are specific to cell membrane were highly expressed only in Tetep post infection, suggesting their role in panicle blast resistance. This study helped us to identify valuable candidate genes that can unravel the mechanisms of panicle blast resistance and help in the rice blast breeding programme.

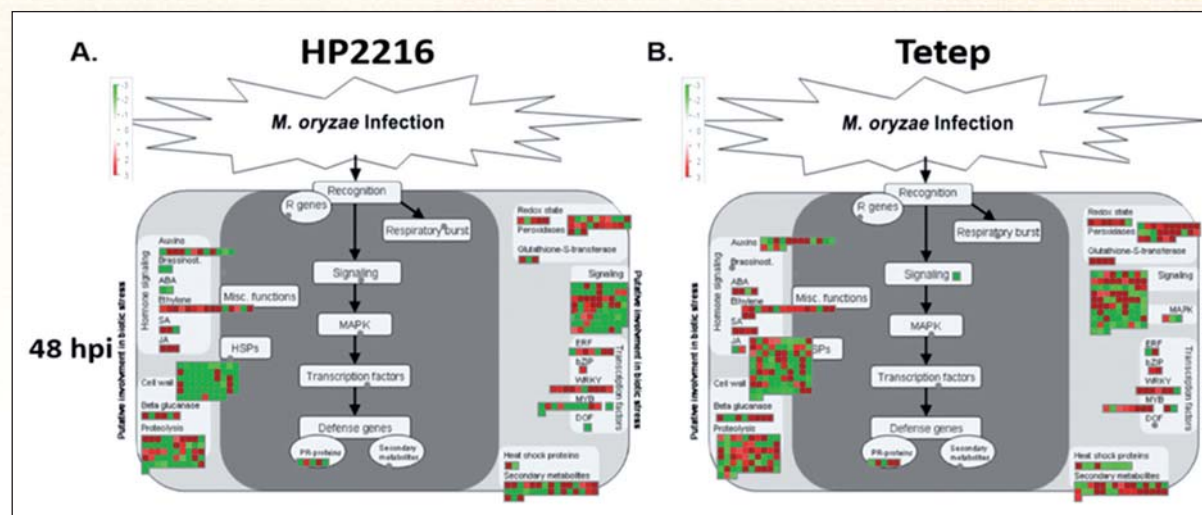


Fig. 4.10: MapMan overview of significant differentially expressed loci (SDEL) involved in biotic stress pathway, in A) HP2216 and B) Tetep, after 48 h of *M. oryzae* infection. Red color represents up-regulated loci and green represents down-regulated loci.

Transgenics of Finger millet *EcDREB2A* gene in tobacco shows enhanced tolerance to heat stress through ROS scavenging

Amolkumar U Solanke*, Sonam Singh, Ramakrishna Chopperla, Jasdeep C Padaria, TR Sharma [Singh et al., 2021, J Biotechnol]

High temperature stress is one of the primary constraints for plant development and productivity. Finger millet is a climate resilient crop and a potential source of novel stress tolerant genes. We generated subtractive cDNA library from finger millet cultivar MR1 ten-days-old seedlings subjected

to early heat stress (15, 30, 45 and 60 min) and late heat stress (2, 4, 8 and 24 h) at 42°C and sequenced them. Based on the annotation of the sequencing data, qPCR was carried out for 24 genes under heat stress to confirm differential expression. RACE-PCR was done to get complete CDS sequence of one of the selected heat responsive gene *EcDREB2A*. After sequence analysis and confirmation, the full-length gene was cloned into binary vector and mobilized into *Agrobacterium tumefaciens*. Tobacco cv. Petit Havana was transformed with *A. tumefaciens* strain EHA 105 harbouring the plasmids pBI121-35S:DREB2A. PCR positive transgenics were

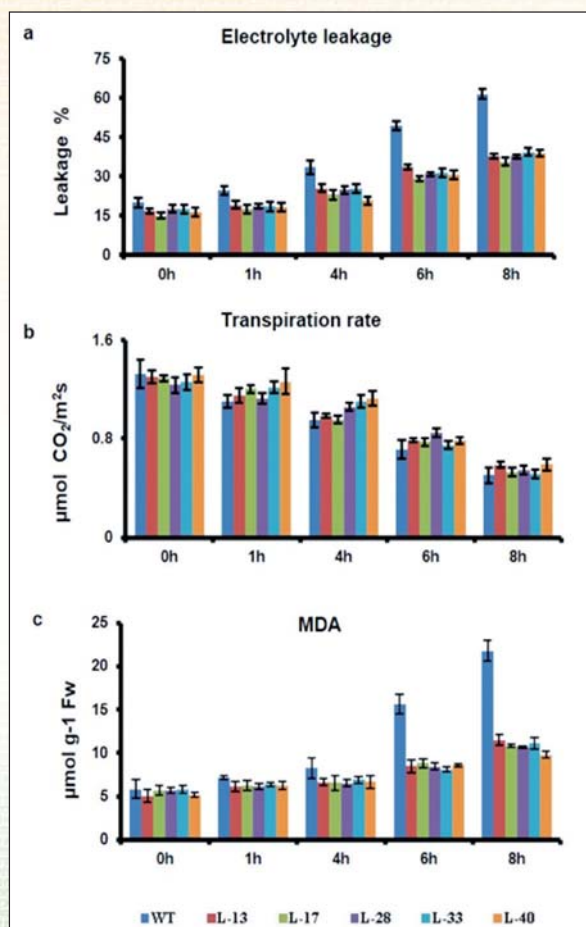


Fig. 4.11: Physiological analyses of transgenic lines after heat stress at 42°C. Estimation of (A) electrolyte leakage, (B) transpiration rate, and (C) MDA, from leaves of control plants (WT) and transgenic lines (L13, L17, L28, L33, and L40) under control and heat stress conditions. Bars represent means $\pm$ SE and values indicated by asterisks are significant at $p < 0.05$.

confirmed by southern analysis. Under heat stress of 42 °C for 7 days imposed on 14 days old seedlings, T1 transgenic lines performed better than control plants. Under heat stress conditions, increased seed germination with improved lateral roots, fresh and dry weight relative to wild type were observed. The *EcDREB2A* transgenics under heat stress showed improved rate of stomatal conductance, chlorophyll and carotenoids contents, and increased antioxidant enzyme activity compared to wild-type plants (Fig. 4.11). qRT-PCR analysis showed that all eight downstream genes were significantly upregulated in transgenic plants only after heat stress. Our data provided a clear demonstration of the positive impact of overexpression of *EcDREB2A* providing heat stress tolerance to plants.

Understanding molecular mechanism of Silicon mediated resistance against biotic stress using rice blast as a system (DST-WOS-A)

Gitanjali Jiwani and Amolkumar Solanke*

Silicon application in plants has been observed to provide protection against a broad range of pathogens, either by acting

as a physical barrier or by the soluble Si modifying the host resistance system. We treated rice plants (Tetep and HP2216) with calcium silicate and carried out differential gene expression analysis under infection by *Magnaporthe oryzae*, which causes rice blast disease. Rice leaves were harvested post 24h, 48h, 72h and 96h of infection. The defense related genes and some silicon pathway related genes were chosen for differential expression analysis. The transporters were down regulated in response to infection (Fig. 4.12). In response to calcium silicate treatment, some of the transporters were getting induced in cv. HP2216. Interestingly, calcium silicate treated plants that were infected, there is significant upregulation of *OsLsi1* and *OsLsi3*, indicating that in the presence of infection the plant does bring about changes in silicate levels of plant. The defense related genes were getting upregulated in response to infection in HP2216 while in case of Tetep these were down regulated. Calcium silicate treatment mostly led to upregulation of these genes. In case of infection in calcium silicate treated plants, these genes were highly upregulated, indicating the role of silicate in imparting resistance in rice.

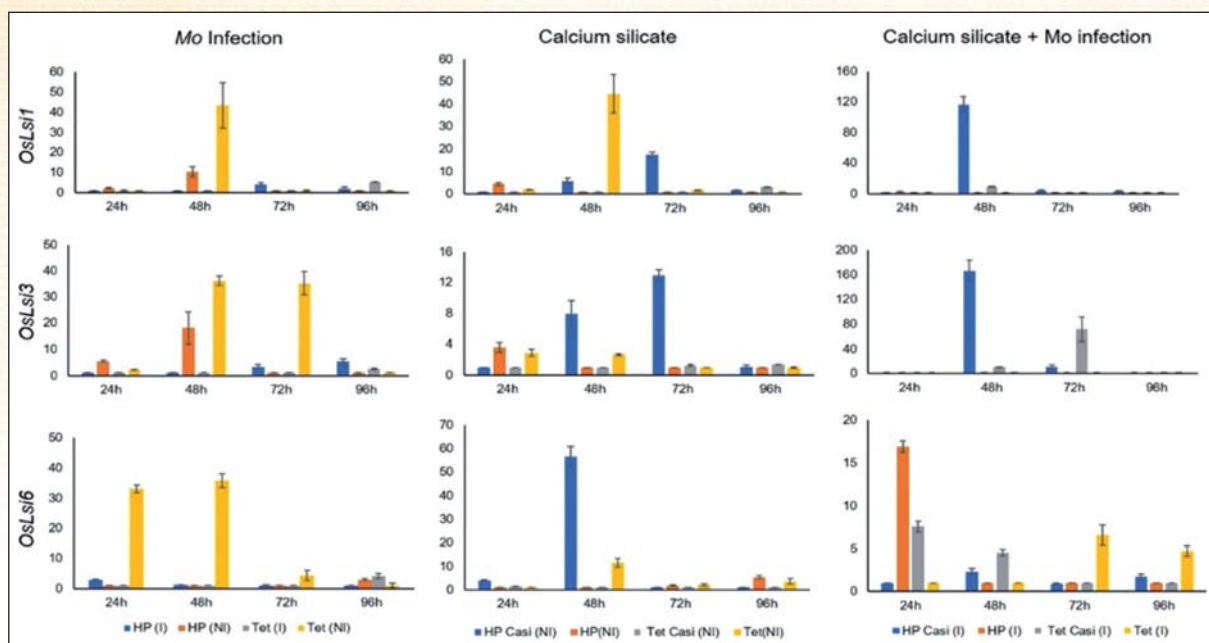


Fig. 4.12: Differential gene expression of silicon transporter genes post blast infection after calcium silicate amendment in HP2216 and Tetep cultivars of rice after 24, 48, 72 and 96 h of treatments.

Development of Standard Operating Protocols (SOP) for Genetic fidelity testing of Citrus plants

Amolkumar U. Solanke*, Amitha Mithra SV, Mohd. Tasleem, Bhuvnesh Sareen, Mayuri Mahalle, Thiagarajan T, Theivanai M, OP Awasthi

Under the National Certification System of Tissue culture Raised Plant (NCS-TCP), a program run by Govt. of India through

DBT, we are working as a Referral Centre for Genetic Fidelity Testing. Developing Standard Operating Protocols (SOP) for genetic fidelity testing of tissue culture raised plants which is one of the prime objectives of this program. In previous years, we reported SOPs for banana, potato, sugarcane, date palm, bamboo, guava and pomegranate. These protocols are transferred to DBT to distinguish somaclonal variations occurred during tissue culture process. This ensures

that genetically uniform and virus-free plants reach the farmers field, ensuring their income. This year, we developed SOP for genetic fidelity testing of citrus plants using ISSR markers. In India, mandarins are the leading group among citrus and cultivated in around 43% citrus growing area of the country. Five mandarin types namely, Kinnow mandarin, Murcott mandarin, Dancy tangerine, Nagpur mandarin and Cleopatra mandarin were collected from the germplasm resources of orchard, Division of Fruits and Horticultural technology, ICAR-Indian Agricultural Research Institute, New Delhi. We screened 100 ISSR markers in these mandarin cultivars, of which 8 primers showed polymorphism. When these markers were used to screen mother plants with their progeny, we found that two markers, UBC855 and UBC864, showed one distinct band each, which was present in mother plant but absent in five progeny plants. These bands were cloned and sequenced which confirmed their ISSR nature. Thus, ISSR primers, UBC807, UBC808, UBC812, UBC825, UBC827, UBC841, UBC855 and UBC864 can be used for genetic fidelity testing of citrus plants (Fig. 4.13). After refinement with larger varietal screening, these markers can act as a robust system for the genetic fidelity testing of citrus plants.

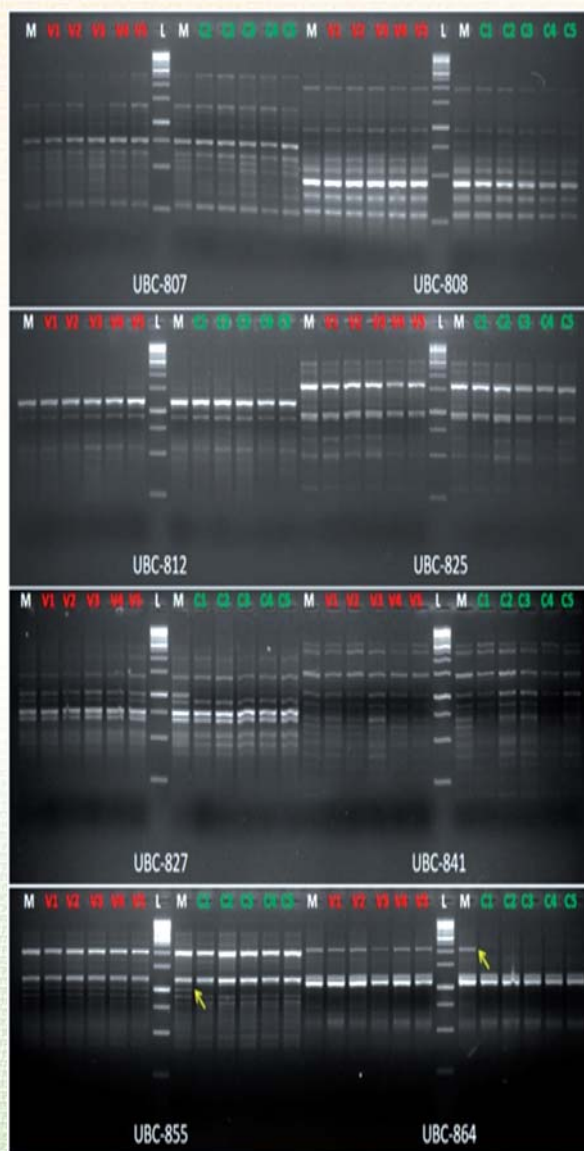


Fig. 4. 13: Genetic fidelity testing of citrus mother plant (M) with progeny plants (V and C) using ISSR markers.



Allele mining for a drought responsive gene *DRO1* determining root growth angle in donors of drought tolerance in rice (*Oryza sativa* L.)

Bablee Kumar Singh, Ramkumar MK, Monika Dalal, Amolkumar U Solanke, NK Singh and Amitha Mithra Sevanthi* (Singh et al. 2021 *Physiol Mol Biol Plants*)

Deeper Rooting 1 (*DRO1*) gene identified from a major QTL on chromosome 9 increases the root growth angle (RGA) and thus facilitates survival under drought. Twenty-four major Indian upland and lowland genotypes including the 'yield under drought' (DTY) QTL donors were subjected to allele mining of *DRO1* (3058 bp). Based on the nucleotide polymorphisms and the resulting amino acid substitutions, the 24 genotypes can be classified into 16 primary structure variants ranging from complete functional allele (Satti, IR36 and DTY 3.1 donor, IR81896-B-B-195) to truncated non-functional alleles in PMK2, IR64, IR20 and Swarna (Fig. 4.14). This information on structural variations in *DRO1* gene will be very useful for the breeders, especially in the light of recent breeding programmes on improving drought tolerance using several DTY donors and upland accessions.

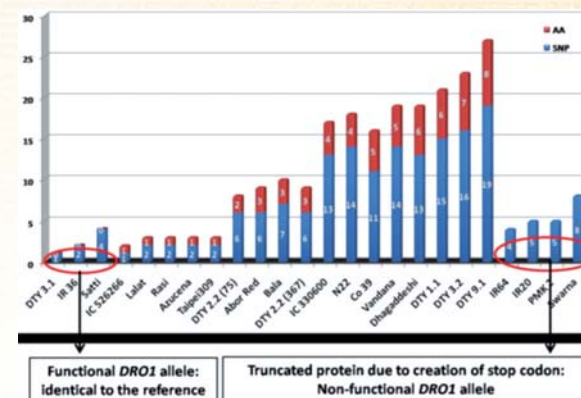


Fig. 4.14: Allelic variation in the *DRO1* gene in the rice accessions studied

Functional characterization of abiotic stress responsive genes from rice

Karikalan J, Amitha Mithra Sevanthi*, Venkat Raman. K, Pranab Kumar Mandal [Karikalan et al., 2021, *Environ Exp Bot*; Karikalan et al. 2021 *J Environ Biol*]

We identified a novel abiotic stress responsive gene LOC_Os06g10210 (*OsCHI2*) which showed upregulation in response to drought stress from the available global expression data. *OsCHI2* was isolated from a drought tolerant rice cv. Nagina 22 (N22) and transformed into a drought stress sensitive rice cv. Pusa Sugandh 2 (PS2) to express under the control of stress inducible *AtRd29A* promoter. Stable chromosomal integration of transgene was confirmed in the transgenic plants through molecular analysis.

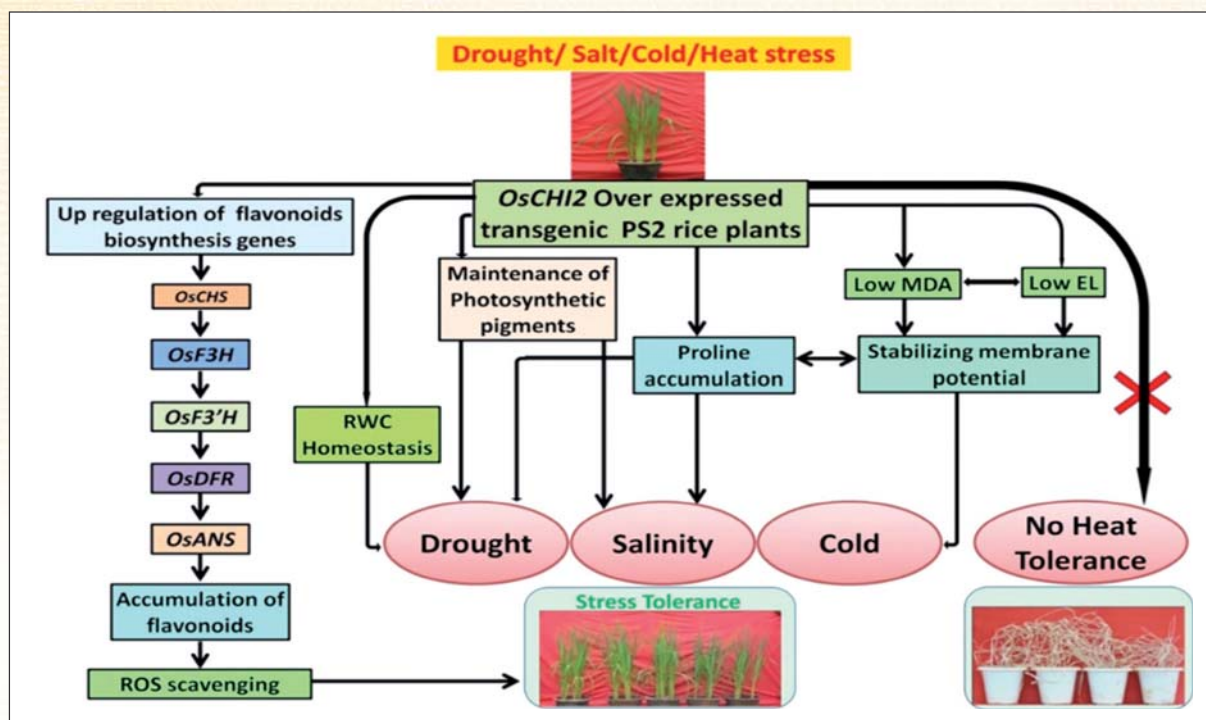


Fig. 4.15: Regulatory network model of the *OsCHI2* gene involved in drought, salt and cold stresses responses. Over expression of *OsCHI2* transgenic PS2 rice plants up-regulate the genes involved in the flavonoid biosynthesis pathway, which result in significant physiological and biochemical changes, including accumulation of proline and flavonoids, less reduction of relative water content (RWC), photosynthetic pigments and lower MDA content and electrolyte leakage which lead to the improved drought, salt and cold tolerance.

The expression level of the transgene in different transgenic lines was higher than wild type plants under abiotic stress conditions. Physio-biochemical studies revealed that the transgenic rice plants upheld higher relative water content, more photosynthetic pigments and proline with lower relative electrolyte leakage and less malondialdehyde content as compared to wildtype PS2 plants under drought,

cold and salinity stresses. *OsCHI2* transgenic lines had enhanced transcript abundance of structural genes of flavonoids biosynthesis pathway leading to higher production of flavonoids under abiotic stresses in the *OsCHI2* transgenic plants (Fig. 4.15). Subsequently, in the non-transformed N22 and PS2 plants, the transcript abundance of the flavonoid genes was studied under control and moisture deficit

stress conditions which clearly demonstrated the significantly higher transcript abundance of the six flavonoid biosynthesis genes in N22 compared to PS2 genes under moisture deficit stress. A linear pattern in the transcript abundance was also seen from the early to late pathway genes in descending order in N22 (Fig. 4.16).

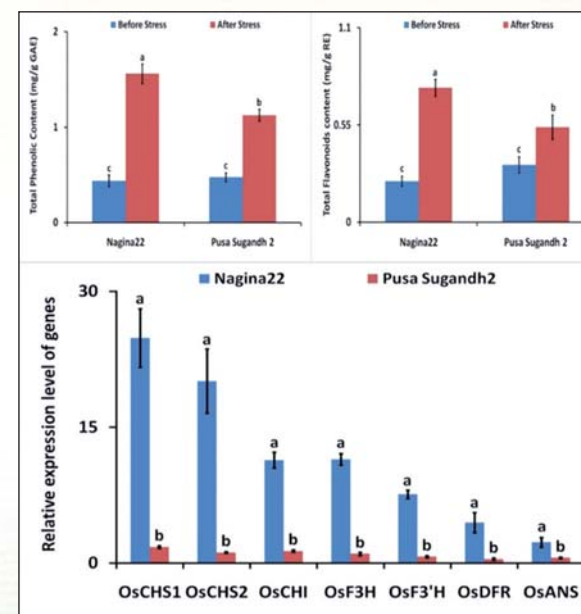


Fig. 4.16: Upper panel: Accumulation of the total contents of phenolics and flavonoids in contrasting rice genotypes (N22 and PS2) under water deficit stress. (A) Total phenolics content and (B) total flavonoids content of contrasting rice seedlings (N22 and PS2) subjected to water deficit stress by withholding water for a week. Lower panel: Relative expression analysis of flavonoid biosynthesis pathway genes (*CHS1*, *CHS2*, *CHI*, *F3H*, *F3'H*, *DFR*, and *ANS*) in contrasting rice genotypes of N22 and PS2 under water deficit stress.



Genetic Mapping of Anaerobic Germination-Associated QTL in Rice

Deepak Singh Bisht*, Balwant Singh, Vandana Rai, Nagendra Kumar Singh

Kumar Singh

For the success of the direct seeded rice it is important to develop rice varieties with potential to tolerate flooding during germination, also known as anaerobic germination (AG). We used

the wild rice germplasm resources available with us to identify accessions with promising anaerobic germination potential. The identified accessions were used to generate mapping population. Total 180 $F_{2:3}$ lines derived from Pusa1509 x NKSUR70 were phenotyped for seedling survival under the submergence stress (Fig. 4.17A). Using the genotyping and phenotyping data, three QTLs providing increased tolerance to anaerobic germination were identified: two on chromosome 7 and one on chromosome 3 with

LOD values ranging from 5 to 7.5. The major QTL on chromosome 7 explained about 27% of the phenotypic variation (Fig. 4.17B).

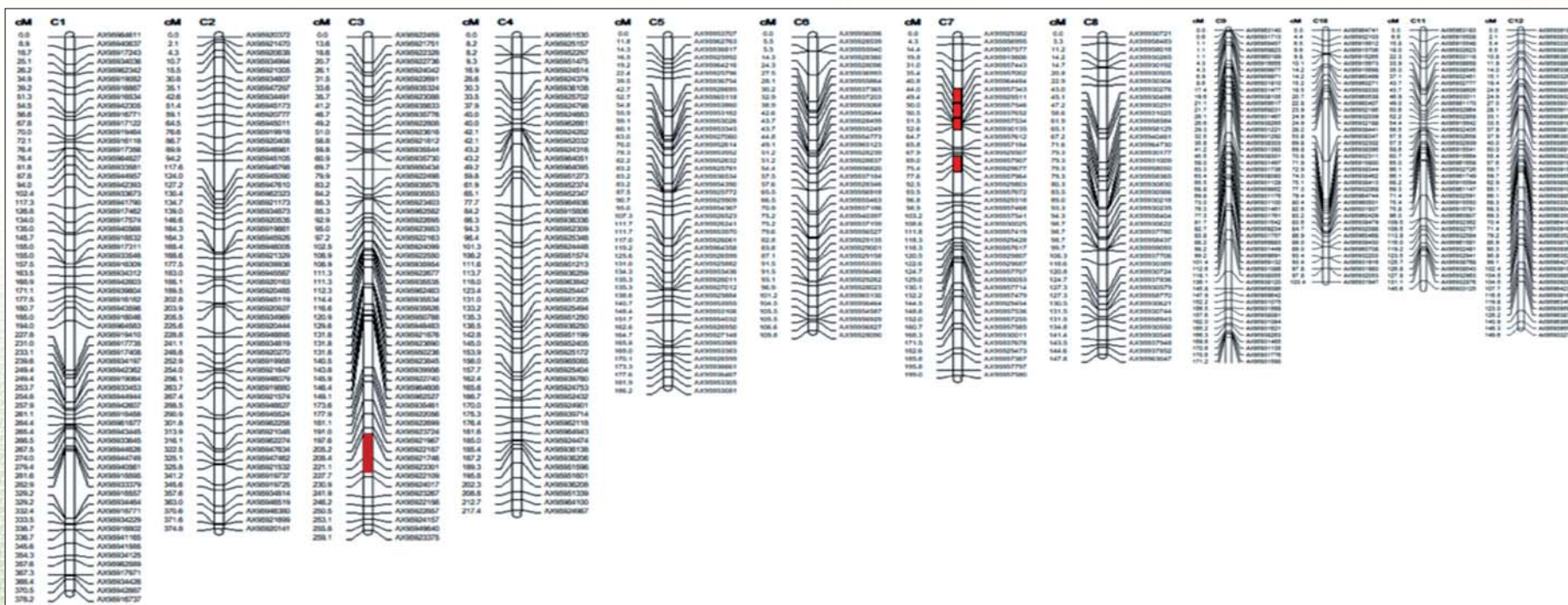
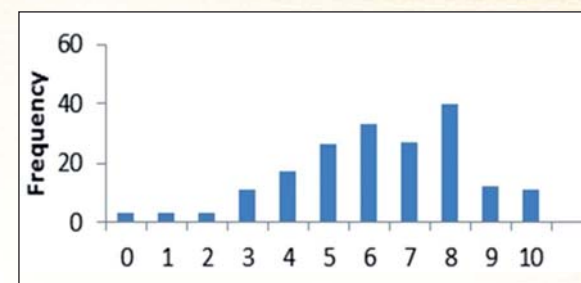


Fig. 4.17 (A) Phenotypic distribution of 184 $F_{2:3}$ progenies screened for anaerobic germination tolerance. (B) Location of anaerobic germination QTLs identified in the F_2 population. The position is shaded in the red colour.

5. National Professor- B. P. Pal Chair



Allele mining for agronomically important genes in wild rice germplasm and stress tolerant landraces of rice growing in the hot spots

Field evaluation of advance introgression lines for grain yield and tolerance to drought, flooding and salt stress in rice

Manish Kumar, Priyanka Kumari, Dharitree Phukan, Vandana Rai, Deepak Singh Bisht, Nagendra Kumar Singh*

Earlier, the $qGN4.1$ QTL was introgressed into the backgrounds of 12 different mega varieties of rice using marker-assisted backcross breeding approach. During Kharif 2021, field evaluation of rice introgression lines with yield QTL $qGN4.1$ was conducted. Amongst the twelve QTL- NILs, the Pusa Basmati- $GN4.1$, Pusa44- $GN4.1$, Samba Masuri- $GN4.1$ and Swarna- $GN4.1$ consistently showed significant yield superiority over the recipient parents. A representative figure of field evaluation of advanced breeding lines *Sambha Masuri- $qGN4.1$* is provided in Fig. 5.1. Additionally, QTL pyramiding of $qDTY3.2$ and $qGN4.1$ in Swarna and Samba Mahsuri genetic

backgrounds and $qGN4.1$ in Sarjoo 52 Sub-1 and MTU 1010 Saltol-1 was also performed. It is planned to nominate these lines for AICRIP trial in 2022.



Fig. 5.1: Field evaluation of Samba Mahsuri- $qGN_{4.1}$, during Kharif 2021

Evaluation, maintenance and utilization of wild rice germplasm collected from different agroclimatic zones of India

Deepak Singh Bisht, Amit Rathor, Vandana Rai, Nagendra Kumar Singh*

Rice is a food crop of global importance cultivated in diverse agro-climatic zones. However, in the process of domestication many beneficial alleles have been filtered out from the gene pool of cultivated rice that has eventually made the modern high-yielding varieties vulnerable to various stresses. In contrast, the wild rice, despite being agronomically inferior, has inherited a distinct potential of survival in a range of geographical habitats. These adaptations have enriched them with adaptability traits that are of immense agronomic value. At ICAR-NIPB under National Professor Project, a total of 802 accessions of wild rice have been collected from 11 agro-climatic zones of India. A set of 418 accessions have already been screened for tolerance to different abiotic and biotic stresses, including drought, salinity, flooding, sheath blight, BLB, blast, cereal cyst nematode, leap spot and morphological traits, identifying

promising accessions which are currently being utilized for mapping and trait introgression. During Kharif 2021, a total of 615 wild rice germplasm was multiplied and evaluated in the field (Fig. 5.2). Besides, three separate

expeditions were made to collect the wild rice germplasm from Terai region of Uttarakhand, Eastern Uttar Pradesh and regions bordering Andhra Pradesh, Odisha, Chhattisgarh and Telangana. These wild accessions will be

multiplied and evaluated during Kharif 2022. Further, introgression lines of wild rice QTLs from salt tolerance from NKSUR 173 and anaerobic germination from NKSUR 70 were grown at BC2 stage.

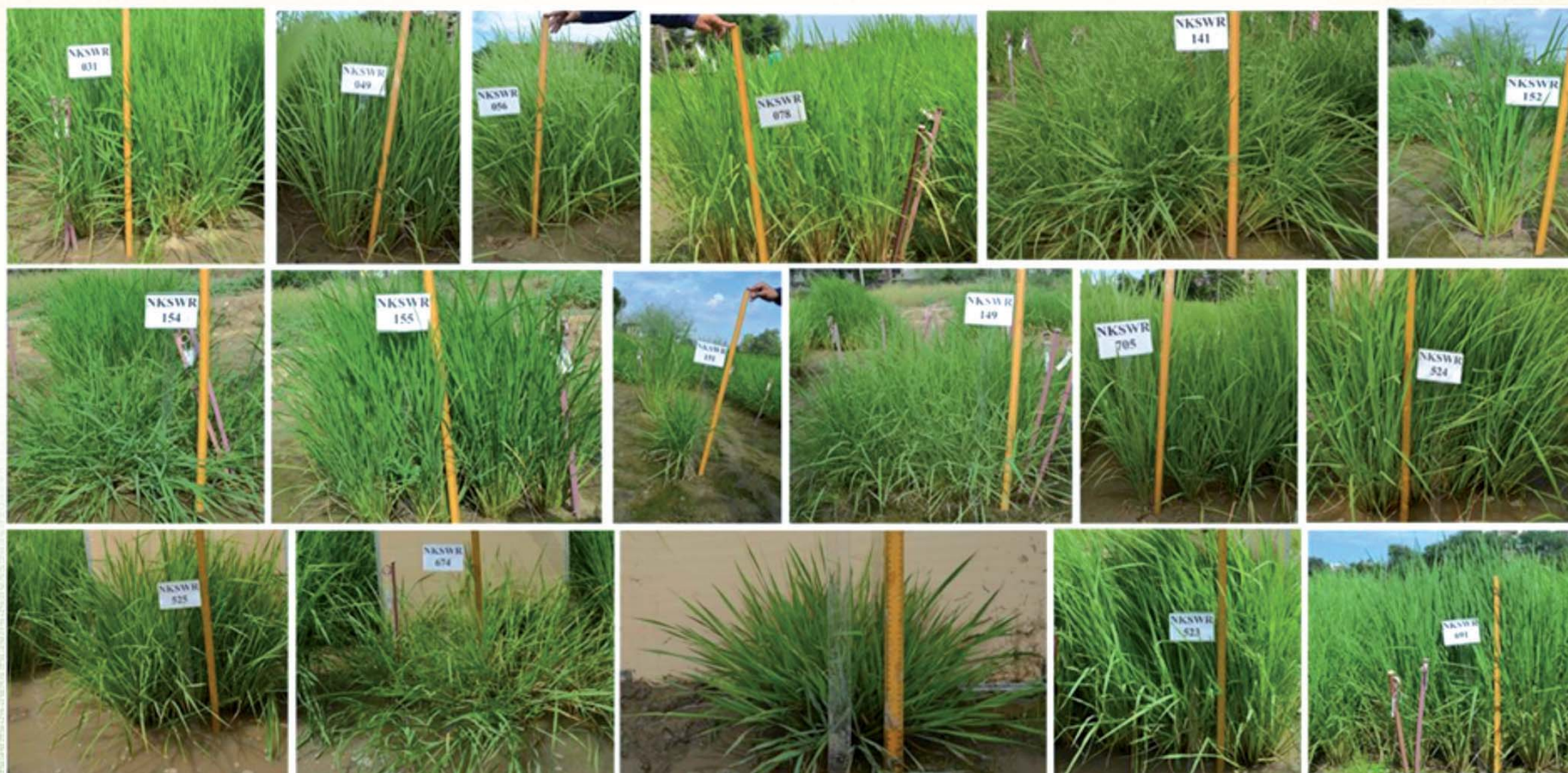


Fig. 5.2: Multiplication and field evaluation of wild rice accession during Kharif 2021

Human Resource Development

- POST-GRADUATE TEACHING PROGRAMME
- TRAINING AND CAPACITY BUILDING
- TRAINING PROGRAMME ORGANIZED

Post-graduate Teaching Programme

ICAR-National Institute for Plant Biotechnology (NIPB) has been actively engaged in Human Resource Development in the area of plant molecular biology and biotechnology since its inception. Currently 50 Ph.D. and 17 M.Sc. students are registered in the discipline of Molecular Biology and Biotechnology (MBB). In the year 2021, six Ph.D. and eight M.Sc. students were awarded with Doctoral and Master's degrees, respectively, in the 59th Convocation 2021.

Students on roll in the discipline of Molecular Biology and Biotechnology during the Academic Session 2021-22

Ph.D. Students

S No.	Name of the Student	Roll No.	Name of the Chairperson	S. No.	Name of the Student	Roll No.	Name of the Chairperson
1	Ms. Shruti Sinha	10497	Dr. Kishor Gaikwad	20	Mr. W. Sandesh Tulsiram	11370	Dr. Anita Grover
2	Mr. Sandeep Jaiswal	10639	Dr. Debasis Pattanayak	21	Mr. Krishnayan Paul	11546	Dr. Debasis Pattanayak
3	Ms. Priyanka Singh	10697	Prof. N. K. Singh	22	Mr. Muhammed Shamnas V.	11547	Dr. Subodh Kumar Sinha
4	Ms. Sharani Choudhury	10836	Dr. R. C. Bhattacharya	23	Mr. Naresh Kumar Samal	11548	Dr. R. C. Bhattacharya
5	Ms. Parichita Priyadarshini	10837	Dr. Pradeep Kumar Jain	24	Mr. Zaherul Islam	11549	Dr. Amolkumar U. Solanke
6	Ms. Jyotsana Tilgam	10838	Dr. Debasis Pattanayak	25	Mr. Deepesh Kumar	11550	Dr. S. V. A. C. R Mithra
7	Ms. Sreeshma N.	10839	Prof. N. K. Singh	26	Mr. Dhivyanandham K.	11551	Dr. Monika Dalal
8	Ms. Alka Bharati	10840	Dr. P. K. Mandal	27	Mr. Gopal	11552	Dr. Pradeep Kumar Jain
9	Mr. Kuldeep Kumar	10842	Dr. Kishor Gaikwad	28	Mr. Ankur Poudel	11646	Dr. Pranab Kumar Mandal
10	Mr. Deepanshu Jayaswal	10843	Dr. Rekha Kansal	29	Mr. Samar Deb	11816	Dr. S. V. A. C. R Mithra
11	Mr. Mahendra C.	10844	Dr. Kanika	30	Ms. Mareyam Mukhtar	11817	Dr. Amolkumar U. Solanke
12	Ms. L. Ashakiran Devi	11070	Dr. R. C. Bhattacharya	31	Mr. Jeet Roy	11818	Dr. Pranab Kumar Mandal
13	Ms. Bablee Kumari Singh	11071	Dr. Monika Dalal	32	Ms. Vibha Kamati	11820	Dr. Monika Dalal
14	Mr. Sachin	11074	Dr. Pranab Kumar Mandal	33	Ms. Priya	11821	Dr. R. C. Bhattacharya
15	Mr. Bipratip Dutta	11290	Dr. S.V.A.C.R. Mithra	34	Ms. Nitasana Rajkumari	11822	Dr. Tapan Kumar Mondal
16	Ms. Sheel Yadav	11291	Dr. Pradeep Kumar Jain	35	Mr. Anuj Kumar	11823	Dr. Subodh Kumar Sinha
17	Mr. Akash Paul	11293	Dr. Kishor Gaikwad	36	Ms. Renu Kumari	11824	Dr. Jasdeep C. Padaria
18	Mr. Saakre Manjesh	11294	Dr. Debasis Pattanayak	37	Ms. Priyanka Kumari	11928	Dr. Debasis Pattanayak
19	Ms. Shaziya Sultana	11367	Dr. Sharmistha Barthakur	38	Mr. Ahmed Mohammed Ismail	11934	Dr. Kanika

Ph D students (for academic session 2021-2022)

S. No.	Name of the Student	Roll No.	Name of the Chairperson
39	Mr. Ashfak Sirajmahammad Mujawar	12099	-
40	Mr. Machindra Sudhir Nirgude	12100	-
41	Ms. Rekha Mahato	12101	-
42	Ms. Meena S	12102	-
43	Ms. Mahi Baaniya	12103	-
44	Ms. Alvakonda Sheena Sabatina	12104	-
45	Ms. Sowmyapriya	12105	-
46	Ms. Shahina Perween	12106	-
47	Mr. Viraj Gangadhar Kamble	12107	-
48	Ms. Anindita	12108	-
49	Mr. Kumar Nupur Hrishikeshan	12109	-
50	Mr. Yaswant Kumar Pankaj	12217	-

M. Sc. Students

S.No.	Name of the Student	Roll No.	Name of the Chairperson
1	Mr. Gowtham T. P.	21315	Dr. P. K. Dash
2	Mr. Sagnik Chanda	21488	Dr. Jayanti M.
3	Mr. Adil Rahim Magray	21489	Dr. Sharmistha Barthakur
4	Ms. Rishika K S	21490	Dr. Sarvjeet Kaur
5	Ms. Pyla Bhuvaneswari	21491	Dr. Rhitu Rai
6	Mr. Yogesh Kumar S	21492	Dr. P. K. Dash
7	Ms. Nuzat Banu	21494	Dr. Jayanti M.
8	Mr. Ramesh R.	21495	Dr. Ajit Kumar Shasany
9	Mr. Tharun Kumar C J	21673	-
10	Mr. Ashutosh Diliprao Thakare	21674	-
11	Ms. Soumya Chakraborty	21675	-
12	Mr. Balaji B	21676	-
13	Mr. Sanjay T D	21677	-
14	Ms. Sonam Brijlal Ingle	21678	-
15	Mr. Subhash A	21679	-
16	Mr. Bhanu Kumar Tiwari	21680	-
17	Mr. Kiran Mahavir Magdum	21681	-

Degrees awarded in the discipline of Molecular Biology and Biotechnology (MBB) during the 59th Convocation held in February, 2021.

S No.	Student Name and Roll No.	Chairperson	Thesis Title
Ph.D. Students			
1.	Ms. Mamta Gupta (10217)	Dr. Sarvjeet Kaur	Isolation and characterization of vegetative insecticidal protein (vip3)-type gene(s) from native <i>Bacillus thuringiensis</i> isolates recovered from diverse habitats in India.
2.	Mr. Lianthanzuava (10313)	Dr. R. C. Bhattacharya	Understanding mustard-aphid interaction for development of aphid-resistance
3.	Mr. Manoj M. L. (10495)	Prof. N. K. Singh	Genetic analysis and identification of QTLs for anaerobic germination in tolerant wild rice accessions
4.	Mr. Kishor Panzade (10500)	Dr. Jasdeep C. Padaria	Transcriptomic profiling of high temperature responsive genes in <i>Ziziphus nummularia</i> (Burm. f.) Wight & Arn
5.	Mr. Pawan S. Mainkar (10549)	Dr. Rekha Kansal	Genome-wide identification and characterization of protease inhibitor genes from <i>Cajanus cajan</i>
6.	Ms. Sarita Kumari (10640)	Prof. N. K. Singh	Mapping and introgression of salt tolerance genes from wild rice into cultivated rice (<i>Oryza sativa</i> L)
MSc. Students			
1.	Mr. Megavath Ravi Naik (20998)	Dr. P. K. Mandal	Characterization of Nitrate Reductase gene homeologues in wheat (<i>Triticum aestivum</i> L.)
2.	Mr. Somya Gupta (21043)	Dr. Sarvjeet Kaur	Screening of cry52-type gene(s) from native <i>Bacillus thuringiensis</i> isolates recovered from diverse habitats in India.
3.	Mr. Ankan Chakraborty (21148)	Dr. Monika Dalal	Functional validation of PM19 promoter from wheat
4.	Mr. Samar Deb (21149)	Dr. Tapan Kumar Mondal	Indexing of allantoin accumulation under salinity stress in rice (<i>Oryza Sativa</i> L.)
5.	Ms. Priyanka Kumari (21150)	Dr. Subodh Kumar Sinha	Understanding self and non-self root-root interaction on root system architecture and expression of putative nitrate transceptor gene in bread wheat under nitrogen limited condition
6.	Mr. Jeet Roy (21151)	Dr. S.V.A.C.R Mithra	Growth and grain filling dynamics in selected EMS induced Nagina 22 mutants and mapping of the causal genes
7.	Ms. Ashika Debbarma (21152)	Dr. Sarvjeet Kaur	Screening of <i>Bt</i> isolates from different habitats of India for the presence of cry11-type
8.	Ms. Tamilarasi M. (21153)	Dr. Kanika	Molecular characterization of salt tolerance in wheat var. Kharchia-65

TRAINING AND CAPACITY BUILDING

Details of training by ICAR-NIPB staff during 2021

S.No.	Name	Subject Area	Duration	Host Institute
1	Dr.Sarvjeet Kaur	Massive Open Online Course (MOOC) on Theoretical Foundations of Educational Technology	1 - 30 June 2021	ICAR-National Academy of Agricultural Research Management (NAARM) Hyderabad
2	Dr. Sarvjeet Kaur	Generic Online Training in Cyber Security	17 Feb 2021	MEITY, GOI
3	Dr. Rhitu Rai	Training and Workshop for Vigilance Officers	16-18 Aug 2021	NAARM, Hyderabad
4	Dr. Ramawatar Nagar	<i>Agrobacterium</i> mediated transformation of chickpea (<i>Cicer arietinum</i>)	19- 1 Nov 2021	DBT Centre-Assam Agricultural University, Jorhat.
5	Dr. Nimmy MS	<i>Agrobacterium</i> mediated transformation of chickpea (<i>Cicer arietinum</i>)	19 Oct - 1 Nov 2021	DBT Centre-Assam Agricultural University, Jorhat.
6	Dr. SK Sinha	Basic Wheat Improvement Course-2021 (BWIC-2021)	18 Jan - May 31, 2021	The International Maize and Wheat Improvement Centre (CIMMYT), Mexico (in virtual mode)
7	Dr. Yuvaraj I	S2C2 online Workshop: Specimen Preparation and Data Collection Approaches	3 - 4 March 2021	Stanford –SLAC CryoEM center
8	Dr. Yuvaraj I	9 th Batch Generic Online Training Course in Cyber Security, by MeitY, Government of India	17 March 2021	Ministry of Electronics and Information Technology & C-DAC, Hyderabad
9	Dr. Yuvaraj I	Online Training Program on “SNP Mining GWAS & Genomic Selection	16-21 Dec 2021	ICAR-Indian Agricultural Statistics Research Institute
10	Dr. Yuvaraj I	Online Industrial Training Program on Artificial Intelligence & Machine Learning Using Python	13 - 24 Dec 2021	Finland Labs

TRAINING PROGRAMME ORGANIZED

DBT-BCIL sponsored Training program on 'Genetic Fidelity Testing and Virus Indexing of Tissue Culture-Raised Plants'

A DBT-BCIL sponsored online training program was conducted at ICAR-National Institute for Plant Biotechnology, New Delhi, India from 8th to 12th March, 2021 on "Genetic Fidelity

Testing and Virus Indexing of Tissue Culture-Raised Plants". Dr. Amolkumar Solanke and Dr. Amitha Mithra Sevanthi were Course Directors for the training. All the contract scientists and technical assistants from all the ATLs (GKVK, Bangalore; ICAR-IISR Lucknow; VSI Pune and ICAR-CPRI, Shimla) participated in the online training. Prof. Anupam Varma, Adjunct Professor, IARI, New Delhi inaugurated the

training program and interacted with all participants. Dr. Ajit Kumar Shasany, Director, ICAR-NIPB; Dr Manoj Kumar Modi and Dr Chandrabhan Dhruw, DBT; Dr Shiv Kant Shukla, BCIL; Prof. V.K. Baranwal, Dr G.P. Rao, Dr R. P. Pant, Division of Plant Pathology, IARI, New Delhi and all PIs from ATLs delivered lectures and interacted with participants.



EXTENSION AND OUTREACH ACTIVITIES

SCSP programme

ICAR-National Institute for Plant Biotechnology organized extension and outreach activities through Schedule Caste Sub-Plan (SCSP) programme for socio-economic development

of Schedule cast communities (SCs) in the country. In the year 2021, ICAR-NIPB conducted 14 SCSP programmes in six states of India (Uttar Pradesh, Uttarakhand, Madhya Pradesh, Rajasthan, Bihar and Odisha). More than 2500 farmers benefited with these programmes. The

farmers' families were provided with different farm equipments like sprayers, spades, fawda, sickles, khurpi, vegetable seeds, saplings and planting material as per requirement and availability. The detailed list of the programmes conducted is given here:

Sl. No.	Date	Organizers	in collaboration with	Beneficiary farmers
1	06/03/2021	Dr. Amolkumar Solanke, Dr. Amitha Mithra Sevanthi, Mr. Anshul Kumar Verma	KVK, Datia, Madhya Pradesh	180
2	06/03/2021	Dr. Rohini Sreevathsa, Dr. Prasanta Dash	Balanga, Puri, Odisha	320
3	15/03/2021	Dr. Amitha Mithra Sevanthi, Dr. Amolkumar Solanke, Dr. Pranab Kumar Mandal	KVK, Ujjain, Madhya Pradesh	175
4	16/03/2021	Dr. Sandhya Sharma, Ms. Rita	Buxa, KVK Jaunpur, UP	201
5	19/03/2021	Dr. Sandhya Sharma, Ms. Rita	Kerakat, KVK Jaunpur, UP	
6	20/03/2021	Dr. Sanjay Singh, Dr. Ramawatar Nagar, Dr. Mahesh Rao	Shiv Shakti Foundation, from Jhiran Village Bijnor, Rajasthan	200
7	23/03/2021	Dr. Sanjay Singh, Dr. Mahesh Rao, Dr. Ramawatar Nagar, Dr. NimmyMS, Dr. Pankaj, Dr. Joshitha Vijayan	Mahayogi Gorakhnath Krishi Vigyan Kendra, Gorakhpur, UP	200
8	23/03/2021	Dr. A K Shasany, Dr. Deepak Singh Bisht, Dr. Rekha Kansal, Dr. RC Bhattacharya, Dr. PK Jain, Dr. Rohit Chamola, Ms. Shabana Begum	Satyon village, Tehri Garhwal district, Uttarakhand	100
9	25/03/2021	Dr. Nimmy MS, Dr. Mahesh Rao, Dr. Ramawatar Nagar; Dr. Joshitha Vijayan	Muradih village, KVK Sipaya, Gopalganj District, Bihar	236
10	27/03/2021	Dr Naveen Gupta, Mr Deepak Rathore, Dr Prasanta Dash	Janakideipur village, Chandanpur, Puri, Odisha	350
11	27/11/2021	Dr. Navin C Gupta, Dr. Prasanta Dash, Mr. Mittravesh Chaoudhary	Gunta, Bansur KVK, ICAR-DRMR, Bharatpur, Rajasthan	100
12	01/12/2021	Dr. Anshul Watts, Dr. Mahesh Rao, Dr. Joshitha Vijayan	Bhukarka village, KVK, Nohar, Hanumangarh district Rajasthan	250
13	03/12/2021	Dr. Ajit Kumar Shasany, Dr Prasanta Dash, Dr. Pranab Kumar Mandal	Talabeda Village, Talcher, Angul District, Odisha	150
14	10/12/2021	Dr. Navin C Gupta, Dr. Rhitu Rai, Dr. Amolkumar U Solanke, and Mr. Kunal Mann	Nekpur, Bulanshahr, UP	200

GLIMPSES OF SCSP PROGRAM CONDUCTED BY ICAR-NIPB DURING 2021



KRISHI MELA AND FARMERS-SCIENTIST INTERACTION

The products and technologies developed by ICAR-NIPB were showcased with appropriate charts and posters in Pusa Krishi Mela held from 25-27 February 2021 at the IARI Krishi Mela ground. Scientists and technical staff of NIPB explained the work and attended the queries of the farmers and other visitors during this period.



A Farmers-Scientist interaction was held at NIPB on February 27, 2021. The Director, ICAR-NIPB Dr. A. K. Shasany welcomed farmers of adopted villages by NIPB under MGMG programme and ensured to extend all support to farmers and showed NIPB's commitment towards increasing their income and nation building. Dr. Sanjay Singh, Nodal Officer (MGMG) coordinated the programme. Farmers were introduced about the current research work undertaken at NIPB by our scientists. Farmers shared their problems and discussed with NIPB Scientists. Dr. Shasany also discussed the importance of medicinal and aromatic plants in raising their farm income and appealed them to incorporate such high value crops in their agricultural practices on their barren lands without affecting their arable lands.



‘POSHAN VATIKA MAHAABHIYAN’ AND ‘TREE PLANTATION ABHIYAN’

The Institute organized “Poshan Vatika Mahaabhiyan” and “Tree Plantation Abhiyan” under Curtainer Raiser of “International Year of Millets 2023” on 17 September 2021 at Agota Village, district Bulandsahar, U.P. In this program a group of 200 farmers from this village participated in this abhiyan. On this occasion, scientists of NIPB explained different technologies/research initiatives of institute and also discussed the importance of nutri-cereals and their role on human health. During the function, fruit plant saplings were distributed among farmers in order to support sustainable agriculture and rural livelihood. Director of the institute, Dr. A. K. Shasany, and staff planted tree saplings in the institute campus. The millet-based food packets were distributed among 71 girls of a primary school located in the Pusa Campus to create awareness about the benefits of millets in our daily food intake.



PM'S INTERACTION WITH FARMERS AND SCIENTISTS

On 28th September 2021, the scientists of the institute and about one hundred farmers including females of Bulandshar district of Uttar Pradesh participated in the interaction program of Hon'ble Prime Minister Shri Narendra Modi with farmers and agricultural scientists. After the live program, the seeds of improved varieties of mustard were distributed to all the farmers by the Director of the institute, Dr. Ajit Kumar Shasany and scientists of the institute briefed all participants about NIPB technologies.



Other Institutional Activities

- INSTITUTIONAL PROJECTS
- EXTERNALLY FUNDED PROJECTS
- TECHNOLOGY COMMERCIALIZATION AND IPR
- VARIETIES/DATABASES/SOFTWARE DEVELOPED AND GERMPLASM REGISTRATION
- AWARDS AND HONOURS
- RECRUITMENTS/PROMOTIONS/RETIREMENTS
- OTHER ACTIVITIES
- LIST OF PUBLICATIONS
- IMPORTANT COMMITTEES

INSTITUTIONAL PROJECTS

Project Title	Date of start	Date of Completion	Principal Investigator	Name of Associates
Genome analysis in crop plants	April 2021	March 2026	Dr. Kishor Gaikwad	Dr. Jasdeep C Padaria Dr. PK Dash Dr. Anil K Singh Dr. Ramavatar Nagar Dr. Yuvaraj I.
Bioprospecting agriculturally important utilizable genes	April 2021	March 2026	Dr. Tapan Kumar Mondal	Dr. Anita Grover Dr. Rekha Kansal Dr. Sarvjeet Kaur Dr. Pradeep Kumar Jain Dr. Sharmishta Barthakur Dr. Kanika Kumar Dr. Monika Dalal Dr. Rhitu Rai Dr. Vandna Rai Dr. Navin Chandra Gupta Dr. Nimmy MS Dr. Shbana Begam
Novel genes and biomolecules for plant and human nutrition	April 2021	March 2026	Dr. Pranab Kumar Mandal	Dr. Ajit Kumar Shasany Dr. Subodh Kumar Sinha Dr. Rohini Sreevathsa Dr. Sandhya Sharma Dr. Joshitha Vijayan
Genetic modification of crop plants for important traits	April 2021	March 2026	Dr. Ramcharan Bhattacharya	Dr. Debasis Pattanayak Dr. Ashish Kumar Dr. SV Amitha Mithra Dr. Amolkumar Solanke Dr. Mahesh Rao Dr. Deepak Singh Bisht Dr. Anshul Watts

EXTERNALLY FUNDED PROJECTS

S.No.	Title of the project	Funding agency	Budget (Rs. Lakhs)	Principal Investigator
1.	Exploiting alien genetic resources for developing climate resilient wheat and understanding mechanism of heat tolerance	ICAR	25.60	Dr. Jasdeep Chatrath Padaria
2.	Decoding the genomes of guar and black gram for the identification of important genes	ICAR-CRP on Genomics	145.00	Dr. Kishor Gaikwad
3.	Molecular mapping and identification of candidate gene(s) responsible for the cleistogamous trait in pigeonpea <i>Cajanus cajan</i> Millsp	DST-SERB	36.63	Dr. Kishor Gaikwad
4.	Genetic improvement of rice for yield, abiotic and biotic stress tolerance through RNA guided genome editing (CRISPR-Cas9)	ICAR-NASF	78.10	Dr. Prasanta K Dash
5.	Identification of QTLs for sub component of WUE through strategic utilization of whole genome sequences and accurate phenotyping rice.	ICAR-NASF	28.72	Dr. Prasanta K Dash
6.	Genome wide association mapping for identification of novel genes/QTLs of Bao rice population of Assam for deep water tolerance and nutritional quality trait	DBT	82.00	Dr. Tapan Kumar Mondal
7.	High throughput approach to discover noble salt tolerant genes from wild halophytic species <i>Oryza coarctata</i>	DST-SERB	42.59	Dr. Tapan Kumar Mondal
8.	Development of transgenic wheat for heat stress tolerance	NPFGGM	35.00	Dr. Sharmistha Barthakur
9.	Deciphering the transcriptional regulator(s) of hydrotropism in wheat	DST-SERB	40.70	Dr. Monika Dalal
10.	Identification and functional characterization of the key resistance/susceptible determinants for Sclerotinia stem rot disease in oilseed Brassica	DST-SERB	32.004	Dr. Navin C Gupta
11.	Identification of effector candidates from necrotrophic plant pathogen, <i>Sclerotinia sclerotiorum</i> and its target proteins in Indian mustard (<i>Brassica juncea</i>)	DBT	38.16	Dr. Navin C Gupta
12.	Meta-Analysis and computational approach for biomolecular interactions profiling of <i>Sclerotinia sclerotiorum</i> -Brassica pathosystem	ICAR-NPABCB (CABin)	50.00	Dr. Navin C Gupta
13.	Study on methylation patterns under drought stress for identification of key genes and microRNAs involved in drought stress tolerance in chickpea	DST-SERB	38.00	Dr. PK Jain
14.	Genetic modifications to improve biological nitrogen fixation for augmenting nitrogen needs of cereals	ICAR-Incentivizing Research in Agriculture	96.80	Dr. Pranab Kumar Mandal
15.	Development of Low Immunogenic Wheat	ICAR-CRP on Biofortification	29.55	Dr. Pranab Kumar Mandal

S.No.	Title of the project	Funding agency	Budget (Rs. Lakhs)	Principal Investigator
16.	Identification and functional validation of partner proteins of two-component high affinity nitrate transporter of wheat	DST-SERB	36.32	Dr. Subodh Kumar Sinha
17.	Functionality of Methionine sulfoxide reductase gene(s) in redox homeostasis-mediated resistance to <i>Helicoverpa armigera</i> in pigeonpea wild relative <i>C. platycarpus</i>	DST-SERB	42.00	Dr. Rohini Sreevathsa
18.	Amelioration of cold-induced sweetening in potato	ICAR (NPTC)	10.00	Dr. Debasis Pattanayak
19.	Genomics-led improvement of biotic and abiotic stress tolerance in mustard rape for economic and environmental sustainability	DBT	108.00	Dr. Ramcharan Bhattacharya
20.	Development of haploid inducer line, and enhancement of seed-meal quality in Brassica juncea through CRISPR/Cas mediated genome editing	DBT	50.00	Dr. Ramcharan Bhattacharya
21.	A way forward to developing aphid-resistance in Indian mustard (Brassica juncea): Host- and virus-mediated gene silencing of parthenogenetic genes in mustard aphid	DST-SERB	38.00	Dr. Ramcharan Bhattacharya
22.	An integrated omics approach to develop white rust (Albugo candida) resistance in Brassica juncea	DST-SERB	35.24	Dr. Ashish Kumar
23.	Maintenance, Characterization and use of EMS Mutants of upland variety Nagina 22 for Functional Genomics in Rice- Phase II	DBT	258.98	Dr. SV Amitha Mithra /NK Singh
24.	CRISPR-Cas9 based genome editing of multiple negative regulators for blast resistance in rice	DST-SERB	31.93	Dr. Amolkumar Solanke
25.	Referral Centre for Genetic Fidelity testing of Tissue culture raised plants (NCS-TCP)	DBT	160.33	Dr. Amolkumar Solanke
26.	RiceMetaSys: Understanding of rice gene network for biotic and abiotic stress management through system biology approach	CABin Scheme- ICAR	15.0	Dr. Amolkumar Solanke
27.	Broadening genetic diversity of Indian mustard (Brassica juncea) through resynthesis of amphidiploids genome by crossing parental diploid species	DST-SERB	33.64	Dr. Mahesh Rao
28.	Induced mutation of synthetic B. juncea haploids for isolation of climate resilient genotypes	BRNS	31.46	Dr. Mahesh Rao
29.	Strengthening cultivar diversity of rapeseed mustard to manage climate related risks and foster productivity in stress prone areas of Central and Western India	Bioversity International	69.58	Dr. Mahesh Rao
30.	In-vivo maternal haploid induction system in Brassica juncea through CRISPR/Cas9 mediated genome editing of pollen specific DMP gene	DST-SERB	30.27	Dr. Anshul Watts

TECHNOLOGY COMMERCIALIZATION AND IPR

The mandate of the Institute Technology Management Unit relates to registration of patents, facilitation of contract research projects and commercialization of IPR enabled technologies of the institute through Public- Private Partnership. The following activities were undertaken by the ITMU during the year January 2021 to December 2021.

I. Patent Granted:

S. No.	Title of patent application	Patent Application number & date of filling	Date of Grant	Granted Patent Number
1.	Promoter from <i>Gossypium hirsutum</i> L for enhanced expression of foreign genes in late Boll development stages of cotton.	2977/DEL/2015 21.09.2015	26.11.2021	382816

II. Patent Filed:

S. No.	Title of patent application	Patent Application Number	Date of filing
1.	A novel DUF740 Polynucleotide associated with multiple stress tolerance from rice.	202111038665	26.08.2021
2.	A rice blast disease resistance governing gene containing a von Willebrand Factor Domain.	202111056668	07.12.2021

III. Commercialization:

Memorandum of Understanding (MOU) has been signed with Imperial life science Pvt. Ltd Gurugram, Haryana and Agrinnovate India Ltd. NASC, New Delhi on 23.11.2021 for licensing single copy gene based 50K 'OsSNPnks' rice chip. The license fee of Rs 10 lakhs has been paid by the company.

IV. MTA/MoUs signed:

1. A Material Transfer Agreement (MTA) has been signed for transfer of synthetic codon optimized *cry1Ac34* and *vip3Aa44* genes to Forest Research Institute, Dehradun on 8.07.2021 for research purpose.
2. A Material Transfer Agreement (MTA) has been signed for transfer of *Brassica rapa* to ICAR-IARI, Pusa New Delhi on 07.12.2021 for research purpose.
3. Memorandum of Understanding (MOU) has been signed with Lovely Professional University, Phagwara Punjab on 28.10.2021 for long term collaboration for promotion of students training and post graduate research in biotechnology.
4. Memorandum of Understanding (MOU) has been signed with The International crops Research Institute for the Semi - Arid Tropics (ICRISAT), Patancheru, Telangana on 1.10.2021 for promotion of research, training and to create trained manpower in the area of plant biotechnology and genetic engineering.

VARIETIES/DATABASE/SOFTWARE DEVELOPED AND GERMPLASM REGISTRATION

Release of drought-tolerant chickpea variety in collaboration with IARI

Contributors: Shailesh Tripathy, VS Hegde, C Bharadwaj, and Pradeep K Jain

A chickpea variety BG 4005 (Pusa Chickpea 4005) was released by IARI, with ICAR-NIPB as a Collaborator. It is a drought-tolerant introgression line of Pusa 362, developed through Marker-assisted backcross breeding. It has an overall weighted mean yield of 1618 kg/ha and has a yield potential of 1940 kg/ha under drought conditions in the northwestern plain zone (NWPZ). It matures in 131 days in NWPZ. The MAS-derived IL, BG 4005, has drought tolerance mainly because of improved root architecture with 11% yield superiority over recurrent parent Pusa 362. Annually, a huge amount of yield loss is incurred due to occurrence of water stress at maturity in chickpea. The proposed variety BG 4005, being drought tolerant will minimize the risk of crop failure under water stress conditions.

Databases developed

1. Mondal TK (2021). TeMID: an exclusive database for Tea SSR markers.
2. Mondal TK (2021). Tea HapMap-1: a database for of tea haplotype mapping.

Software developed

1. Mondal TK (2021). TENGEXA one software developed. This is a R package-based tool for tissue enrichment and gene expression analysis. It performs the analysis irrespective of organism, number of input genes or tissues at various FPKM thresholds. It has got copy right from Indian patent office.

Germplasm registered

1. Mondal TK (2021). Kalajoha rice for seedling stage salinity tolerance (NBPGR acc. INGR21014)
2. Mondal TK (2021). Negheri Bao for anerobic germination tolerance (NBPGR acc INGR21015)

3. Mondal TK (2021). Saragpahla-2 anerobic germination tolerance (NBPGR acc INGR 21016)
4. Mondal TK (2021). UPASI-9 (NBPGR acc. INGR21086) for high yield
5. Mondal TK (2021). ATK-1 (NBPGR acc. INGR21088) for drought tolerance
6. Mondal TK (2021). CR-6017 (NBPGR acc. INGR21085) for better quality
7. Mondal TK (2021). UPASI-3 (NBPGR acc. INGR21087) for high drought tolerance

AWARDS AND RECOGNISION

- Dr Amolkumar Solanke joined as an Expert for 'Refinement of Plant Protection and Agricultural Botany Course Curricula for the Implementation of the Post Graduate Diploma in Climate Smart Agriculture', for Mahatma Phule Krishi Vidyapeeth, Rahuri, MS.
- Dr. NC Gupta (2021) received 'Distinguished Scientist Award' from Agro-Environmental Development Society (AEDS). In: 5th International conference on advances in agriculture, environmental and biosciences for sustainable development, pp 169, held on 05-07 August 2021, Virtual Mode (Zoom).
- Dr. NC Gupta (2021) received Oral presentation award. In: 5th International conference on advances in agriculture, environmental and biosciences for sustainable development, pp 169, held on 05-07 August 2021, Virtual Mode (Zoom).
- Dr. Sandhya Sharma recieved Best paper presentation award during 4th International Conference on Global Approaches in Natural Resource Management for Climate Smart Agriculture (GNRSA-2021)
- Dr. Sandhya Sharma recieved Best poster presentation award in Agriculture Science Congress held at BHU Varanasi, India 2021
- Dr. SK Sinha was selected as Member, Institute Biosafety Committee, Sardar Vallabh Bhai Patel University of Agriculture and Technology, Meerut, U.P.
- Dr. TK. Mondal Fellow- Indian Society of Genetics and Plant Breeding-2021
- Dr. TK. Mondal Fellow- International Society of Environmental Botany-2021
- Dr. TK. Mondal Fellow-Indian Botanical Society-2020

PROMOTIONS / RECRUITMENTS / RETIREMENTS/ TRANSFERS

Promotions

1. Dr. S.V. Amitha Mitra, Dr. Amolkumar U. Solanke and Dr. N.C. Gupta were promoted from the post of Scientist (S.S) to Senior Scientist.
2. Dr. Mahesh Rao, Dr. Nimmy M.S, Dr. Ramavatar Nagar, Dr. Anshul Watts, and Dr. Deepak Singh Bisht were promoted from the post of Scientist to Scientist (S.S).
3. Sh. Rahul Kumar, F&AO was promoted to the post of Sr. F&AO.
4. Sh. Vipin Kumar, Assistant was promoted to the post of Assistant Administrative Officer.
5. Smt. Rekha Chauhan, UDC was promoted to the post of Assistant.

Recruitments: NIL

Retirements

1. Dr. Sandhya Rawat, Ex-CTO was superannuated from ICAR services w.e.f. 31th Oct, 2021.

Transfers

1. Dr. Anil Kumar Singh, Principal Scientist joined ICAR-NIPB on transfer from ICAR-IIAB, Ranchi.
2. Sh. Sumit Singh, SAO joined ICAR-NIPB on transfer from ICAR-IARI, New Delhi.
3. Sh. Harshit Agarwal, SAO, ICAR-NIPB on transfer joined ICAR-IARI, New Delhi.

Visit Abroad: NIL

OTHER ACTIVITIES

Hindi Chetna Maas

To celebrate Hindi chetna maas (14th Sept. to 13th Oct. 2021) competitions such as Hindi essay writing, debate competition, poetry recitation, noting and drafting were conducted for the staff of NIPB.

Hindi workshops

Four Hindi workshops were conducted in online mode during the year 2021.

S. No.	Quarter & Date	Guest Speaker	Topic
1	Jan-March 2021 (24-03-2021)	Shri Dheeraj Sharma, Deputy Director-Rajbhasha ICAR-National Dairy Research Institute, Karnal	“राजभाषा हिंदी का वैज्ञानिक कार्यान्वयन”
2	April-June 2021 (22-06-2021)	Dr. Chandra Mani Brahmhatt, Director Indraprastha Literature Festival, New Delhi	“हिंदी भाषा का बदलता रूप व स्वरूप”
3	July- Sept 2021 (15-09-2021)	Dr. Ram Kathin Singh, Director Nand Education Foundation for Rural Development, (NEFORD) Lucknow, UP	“कृषि शिक्षा में हिंदी भाषा का योगदान एवं दिक्कतें”
4	Oct-Dec 2021 (29-12-2021)	Ms Sunita Kumari, Assistant Director-Rajbhasha ICAR-NIPB & ICAR-IARI	“कार्यालयों में हिंदी के कार्यान्वयन संबंधी नीति नियमावली”

Waste to wealth program

NIPB in collaboration with KVK Shikopur, Gurugram organized ‘Waste to Wealth’ programme in village Tatarpur, Gurugram on 12th October 2021 under the cleanliness campaign run by the Government of India. About 100 farmers participated in this programme. Under the programme Dr. Bharat Kumar, Scientist of KVK briefed about the methods of using the waste. Dr Pargat Singh talked about the benefits of making vermicompost adding it to the soil. Scientists of NIPB, Dr. Sanjay Singh informed the farmers about all the programs being run by the Government of India and Dr. Ashish Kumar, informed the farmers about the diseases of crops and their management.



INTERNATIONAL WOMEN'S DAY CELEBRATION

ICAR-NIPB celebrated International Women's Day on March 8, 2021. Mrs. Kalpana Mohapatra, President Pusa Institute Ladies Association (PILA) and first lady of ICAR graced the occasion as chief guest. Mrs. Seema Malhotra, Director SCOPE PLUS, a NGO from South Delhi was the guest speaker. The programme was coordinated by Dr. Anita Grover (Chairperson, Women Cell, NIPB). The chief guest, Mrs. Mohapatra talked about Indian way of life and ethos as is prevailing since years. Dr. Ajit Shasany, Director, NIPB also addressed the staff on this occasion. A newly designed atrium of NIPB was also inaugurated by the Chief Guest.



SWACHH BHARAT ABHIYAN

Two cleanliness drives namely, Special National Swachhata Campaign (23-24 October, 2021) and Swachhta Pakhwada (16-31 December, 2021) were organized by ICAR-NIPB. Under 'Special National Swachhta Campaign' a cleanliness drive was organized in Village Dusgharah and Janak vihar.

Under the aegis of Swachh Bharat Abhiyan, the Swachhta Pakhwada was organized at ICAR-NIPB during 16-31 December, 2021. During the two-week long programme, many events were organized at the institute, nearby residential areas and market places. A workshop was organized on the topic "Waste to wealth, Safe disposal of all kinds of wastes". Drawing and poetry competitions were also conducted for children of NIPB staff on the theme of Cleanliness and Health.



LIST OF PUBLICATIONS

Research Articles

1. Abhay G, Namita D, Raju VS, Ramkumar MK, Singh MK, Bhupinder S, Gopala Krishnan S, Panwar S, Mithra SVA (2021) Genetic diversity analysis of wild and cultivated Rosa species of India using microsatellite markers and their comparison with morphology based diversity. Journal of Plant Biochemistry and Biotechnology <https://doi.org/10.1007/s13562-021-00655-3>.
2. Ambika N, Venkatraman H, Nimmy M S, Bharadwaj C, Shailesh T, Rajesh KS, Rajendra K (2021) Unraveling genetics of semi-determinacy and identification of markers for indeterminate stem growth habit in chickpea (*Cicer Arietinum* L.). Scientific Reports 11:21837.
3. Anil K, Jaiwal R, Sreevathsa R, Chaudhary D, Jaiwal PK (2021) Transgenic cowpea plants expressing *Bacillus thuringiensis* Cry2Aa insecticidal protein imparts resistance to *Maruca vitrata* legume pod borer. Plant Cell Reports 40(3): 583-594.
4. Anshuman S, Tyagi S, Saratale GD, Guzik U, Hu A, Sreevathsa R, Reddy VD, Rai V, Mulla SI (2021) A comprehensive review on the influence of light on signaling cross-talk and molecular communication against phyto-microbiome interactions. Critical Review on Biotechnology, 41(3): 370-393.
5. Bashyal BM, Gupta AK, Singh D, Choudhary R, Aggarwal R (2021) First report of *Sclerotium hydrophilum* Sacc causing stem rot disease of rice in North Eastern Plain Zone of India. Plant Disease 105(03):700.
6. Begam S, Jain R, Arora A, Marwaha S (2021) Particle swarm optimization based multi-objective optimization for crop planning: A case study of Bundelkhand. Journal of the Indian Society of Agricultural Statistics 75(1), 47-57.
7. Chakraborty K, Ray S, Vijayan J, Molla KA, Nagar R, Jena P, Mondal S, Panda BB, Shaw BP, Swain P, Chattopadhyay K, Sarkar RK (2021) Preformed aerenchyma determines the differential tolerance response under partial submergence imposed by fresh and saline water flooding in rice. Physiologia Plantarum 173(4): 1597-1615.
8. Chatterjee M, Yadav J, Rathinam M, Mandal A, Chowdhary G, Sreevathsa R, Rao U (2021) Exogenous administration of dsRNA for the demonstration of RNAi in *Maruca vitrata* (Lepidoptera: Crambidae). 3 Biotech 11(4): 1-12.
9. Choudhury S, Muthusamy SK, Padaria JC, Dalal M (2021) Genome-wide identification of Ran GTPase family genes from wheat (*T. aestivum*) and their expression profile during developmental stages and abiotic stress conditions Functional & Integrative Genomics 21 (2): 239-250.
10. Chowrasia S, Nishad J, Pandey R, Mondal TK (2021) *Oryza coarctata* is a triploid plant with initial events of Cd photosynthesis evolution. Plant Science 308:110878.
11. Chungada AS, Gahukar SJ, Akhare AA, Behere GT, Solanke AU and Patil SR (2021) DNA fingerprinting in citrus for species identification. The Pharma Innovation Journal 10(5): 306-309.
12. Dalai D, Chakraborti M, Mondal TK (2021) The core set of sequence-tagged microsatellite sites markers between halophytic wild rice *Oryza coarctata* and *Oryza sativa* complex. Euphytica 217:4- 57.

13. Danakumar T, Kumari J, Singh AK, Sinha SK, Pradhan AK, Sharma S, Jha SK, Bansal R, Kumar S, Jha GK, Yadav MC, Vara Prasad PV (2021) Genetic dissection of seedling root system architectural traits in a diverse panel of hexaploid wheat through multi-locus Genome-Wide Association Mapping for improving drought tolerance. *International Journal of Molecular Sciences* 22(13): 7188.
14. Dharmateja P, Kumar M, Pandey R, Mandal PK, Babu P, Bainsla NK, Gaikwad KB, Tomar V, Kranthi KK, Dhar N, Ansari R (2021) Deciphering the change in root system architectural traits under limiting and non-limiting phosphorus in Indian bread wheat germplasm. *PloS one* 16 (10), p.e0255840.
15. Dhatwalia D, Aminedi R, Kalia V, Veena PV, Bhattacharya R (2021) Host-mediated attenuation of gut sucrose in mustard aphid *Lipaphis erysimi* impaired its parthenogenetic reproduction on Indian mustard *Brassica juncea*. *Pest Management Science* 78(2):803-811.
16. Gayatri JK, Sinha SK, Roy P, Mandal PK (2021) Comparative analysis of GS2 and Fd-GOGAT genes in cultivated wheat and their progenitors under N stress. *Plant Molecular Biology Reporter* 39:520-545.
17. Gayatri VK, Sinha SK, Roy P, Mandal PK (2021) Molecular characterization of GS2 and Fd-GOGAT homeologues and their biased response to nitrogen stress in bread wheat (*Triticum aestivum* L.). *Journal of Plant Growth Regulation* 1-15.
18. Gupta P, Rai R, Vasudev S, Yadava DK, Dash PK (2021) Ex-foliar application of glycine betaine and its impact on protein, carbohydrates and induction of ROS scavenging system during drought stress in flax (*Linum usitatissimum*). *Journal of Biotechnology* 20; 337:80-89.
19. Hada A, Patil BL, Akansha B, Karthik K, Savithramma DK, Bheema P, Sreevathsa R, Rao U (2021) Micro RNA-induced gene silencing strategy for the delivery of siRNAs targeting *Meloidogyne incognita* in a model plant *Nicotiana benthamiana*. *Pest Management Science*. 77: 3396-3405.
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- stress tolerance to rice. *Environmental and Experimental Botany* 190:104582.
26. Karthik K, Jyotsana N, Maniraj R, Navinder S, Rohini S (2021) Exploitation of novel Bt ICPs for the management of *Helicoverpa armigera* (Hübner) in cotton (*Gossypium hirsutum* L.): A transgenic approach. *Frontiers in Microbiology* 12: 661212.
 27. Karthik K, Tyagi S, Mukherji S, Rai R, Singh NK, Sreevathsa R and Dash PK (2021) An apical meristem-targeted in planta transformation method for the development of transgenics in Flax (*Linum usitatissimum*): optimization and validation. *Frontiers in Plant Science* 11:562056.
 28. Kashyap AS, Manzar N, Rajawat MVS, Kesharwani AK, Singh RP, Dubey SC, Pattanayak D, Dhar S, Lal SK, Singh D (2021) Screening and biocontrol potential of Rhizobacteria native to Gangetic plains and hilly regions to induce systemic resistance and promote plant growth in chilli against bacterial wilt disease. *Plants* 10: 2125.
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